BREAK THE LIGHT: A POP-UP PLATFORM TO MINIMIZE BLUE LIGHT EFFECTS FOR STUDENT WELLNESS

A Research Paper

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Of the Requirements

For the Course

Inquiries, Investigations, and Immersion

By

SERAPHIM TANGLAO MANUEL

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Abstract

The research seeks to promote student wellness by developing a Pop-Up Break Platform for managing digital habits in order to address the negative effects of blue light exposure on sleep quality, anxiety, and overall well-being. The materials were developed following an experimental method combined with the Agile development process and are in the form of customizable pop-up reminders, alarms, and digital notepads designed to interrupt prolonged screen use. Generally, the materials are interactive, engaging, and student-centered, reflecting literary and design characteristics of playfulness, personalization, and accessibility. The materials are grounded in the Circadian Rhythm Disruption Theory, Cognitive Load Theory, and Digital Intervention principles, which emphasize the need for both physiological and cognitive protection in the digital learning environment. The materials are expected to be highly acceptable in terms of usability, vocabulary, visual design, and functionality, as evaluated by experts and student participants. Ultimately, the study aims to demonstrate that well-designed digital interventions can encourage healthier screen practices, reduce anxiety, and improve sleep hygiene among adolescents.

Approval Sheet

Endorsed for approval of the Faculty of the Senior High School Unit of the Living Stone International School, the research entitled **Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness** by Seraphim Tanglao Manuel.

IRENE LINAG
Research Adviser

Recommended for approval of the Faculty of the Senior High School Unit of the Living Stone International School for having successfully passed the oral examination of the research entitled **Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness.**

Abigail del Rosario
Chairman

Neil Jason Generoso
Critique

Michael John Cunanan
Member

For having successfully completed the research entitled **Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness** in partial fulfillment of the requirements for the course Inquiries, Investigation and Immersion, is hereby approved.

SERAPHIM TANGLAO MNUEL
Principal, High School Unit
Living Stone International School

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Table of Contents

Title Page i

Approval Page ii

Abstract iii

Acknowledgment iv

Table of Contents.....vi

List of Tables vi

List of Figures x

List of Appendices xi

CHAPTER I: The Problem and Its Background 1

Introduction 12

Significance of the Study 11

Theoretical Framework 10

Conceptual Framework 10

Statement of the Problem 11

Hypothesis of the Study 12

Scope and Delimitation 13

Definition of Terms 19

CHAPTER II: Review of Related Literature 21

Relevant Theories 21

Blue Light Exposure and Sleep Quality 22

Melatonin Suppression and Sleep Onset 29

Circadian Rhythm Disruption 36

Sleep Quality and Mental Health 44

Blue Light and Cognitive Overload 51

Cognitive Load Theory in Digital Learning 51

Impact on Academic Performance 51

Anxiety and Emotional Regulation 52

Digital Interventions for Screen Time Management 53

StayFree and Digital Wellbeing Features 53

Pomodoro and Time-Boxing Strategies 53

Synthesis 54

CHAPTER III: Research Methodology 56

Methods and Techniques of the Study 56

Population and Sample 58

Instruments of the Study 59

Data Gathering Procedure 60

Data Processing and Statistical Treatment 61

CHAPTER IV: Presentation, Analysis, and Interpretation of Data 67

CHAPTER V: Summary, Conclusions, and Recommendations 71

References 89

Appendices 92

List of Tables

No.	Table Title	Page
1	Summary of findings on LSIS, AUF, and NU comparison (Manuel, 2025)	13
2	Screen Time vs. Melatonin Levels (Grzelak, 2024)	19
3	Average GAD-7 Scores among Grade 11 Students (LSIS, NU, AUF)	70
4	Average Awareness Scores among Grade 11 Students (LSIS, NU, AUF)	72
5	Average PSQI Scores among Grade 11 Students (LSIS, NU, AUF)	77
6	Average Coping Strategy Awareness among Grade 11 Students (LSIS, NU, AUF)	74
7	Distribution of Respondents by Grade Level	68
8	Distribution of Respondents by Device Used	69
9	Mean Sleep Quality Scores (Pre-test vs Post-test)	95
10	Mean Anxiety Scores (Pre-test vs Post-test)	97

List of Figures

No.	Figure Title	Page
1	Conceptual Framework for Blue Light Exposure	12
2	Cognitive Load Interaction Model based on CLT	26
3	StayFree In-App Blocking Dashboard	33
4	Pomofocus Pomodoro Landing Page	34
5	Pomofocus Instructions Page	35
6	Pomofocus Short Break Screen	36
7	Pomofocus Long Break Screen	37
8	Notion Dashboard – School Setup	41
9	Notion Dashboard – Class Schedule	42
10	Notion Dashboard – College Final Plan	43
11	Coffeelings Mood Log (“Amazing/Okay”)	44
12	Coffeelings Mood Log (“Tired/Sad”)	45
13	Coffeelings Mood Calendar	46
14	Conceptual Framework of Blue Light Exposure (Adapted from Manuel 2024)	47
15	Agile Sprint Cycle	56
16	Data Gathering Procedure Diagram	61
17	Frontend Development (UI)	84
18	Backend Development (Database)	85

List of Appendices

	Appendix	Page
A	Letter of Request	92
B	Letter of Acceptance	93
C	Letter of Endorsement	94
D	Proposed Instrument of the Study	95
E	Validation Reports	96
F	Reliability Testing Results	97
G	Transcripts of Interview	98
H	Transcripts of Focus Group Discussion	99
I	Curriculum Vitae	104

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

The effects of blue light exposure on adolescent health have become more evident in today's digital age, where screen time is deeply embedded in daily routines. Behar and Radrizzani (2022) conducted a systematic review revealing that exposure to blue light in the evening delays sleep onset, shortens sleep duration, and suppresses melatonin production. These disruptions can impair cognitive performance, increase fatigue, and raise the risk of anxiety and depression, especially in adolescents, who are still developing emotional regulation and are highly engaged with technology. Their findings highlight the importance of applying practical strategies like limiting screen time before bed and using blue light filters to support healthier sleep hygiene, which is closely tied to academic performance and mental well-being.

Tonon et al. (2022) further support these concerns by examining how nighttime exposure to artificial light (ALAN), particularly blue light, is associated with disrupted circadian rhythms and a higher risk of major depressive disorder in adolescents. Their study shows that sleep irregularities caused by blue light exposure can intensify emotional distress and long-term mental health challenges. Similarly, Kocevska et al. (2021) conducted a large-scale meta-analysis involving over one

million individuals across the Netherlands, the United Kingdom, and the United States, confirming that blue light-related sleep disturbances are widespread, especially among younger populations. These studies collectively provide strong evidence for this research's focus on promoting awareness of the psychological and physiological effects of blue light and the need to encourage healthier digital habits both at school and at home.

This study builds upon the researcher's prior qualitative research titled *"The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students aged 15 to 16"* (Manuel, 2024). That study utilized semi-structured interviews and open-ended surveys with five students to explore the impact of screen time on mental health. It found high anxiety levels and poor sleep quality among respondents, recommending broader sampling and comparative analysis across academic strands.

The subsequent quantitative study, *"Comparative Study on Blue Light Exposure in Grade 11 Students from Livingstone International School, National University Clark, and Angeles University Foundation"* (Manuel, 2025), surveyed 66 students across various strands (STEM, ABM, HUMSS, and GAS). Utilizing tools such as the GAD-7 and PSQI, the study found statistically significant differences between schools. Students from Livingstone reported the highest anxiety levels, while those from AUF showed the poorest sleep quality. Data analysis involved One-way ANOVA, Post Hoc Tests, and the Kruskal-Wallis Test. These findings emphasized the

effect of academic environments on screen time behaviors, circadian rhythms, and students' well-being.

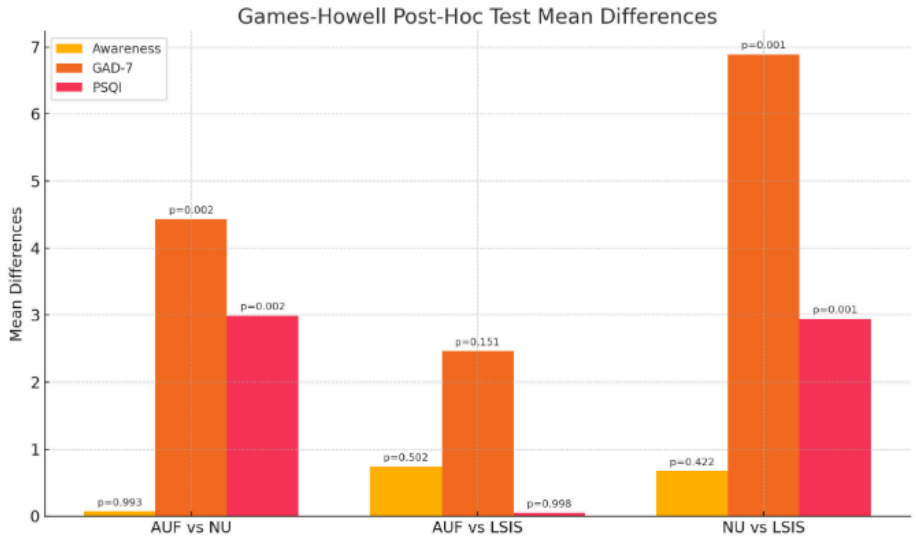


Table 1.1: Summary of findings on the study comparison between LSIS, AUF, and NU (Manuel, 2025)

The urgency of the present study is further underscored by a pre-intervention assessment conducted on February 8, 2026. Data revealed that while many students are aware of the physical strain and health risks associated with prolonged screen time, they often feel cognitively and emotionally "paralyzed" by academic workloads. As a result, excessive device usage becomes a coping mechanism rather than a conscious habit. Early evaluations of the Break The Light platform demonstrated an immediate positive shift in users' reported mood upon interaction. This suggests that digital interventions may function not only as tools for managing physical health but

also as immediate affective resets that help students disrupt cycles of cognitive overload and emotional fatigue caused by sustained digital exposure.

Building upon these findings, the present study focuses on the development of the Break The Light platform, a pop-up break intervention designed to mitigate the adverse effects of blue light exposure and continuous screen use. By incorporating alarms, pop-up reminders, and notepad features, the platform aims to promote healthier digital habits, improved emotional regulation, and greater awareness of screen time behavior among students.

Significance of the Study

This study will benefit the following groups:

- **Students**: As the primary population of interest, students benefit from a practical digital tool that provides immediate emotional and mental relief, particularly through mood improvement and reduced cognitive strain. The platform also supports the development of healthier long-term digital habits by encouraging regular breaks, improved time management, and mindful screen usage.

- **Parents**: For parents, especially those of younger or elementary-aged children, this research offers a framework for promoting healthy digital boundaries at an early stage. Interest expressed by parents during the evaluation phase

highlights the platform's potential as a preventive tool to manage excessive screen exposure before unhealthy digital habits become deeply ingrained.

- **Teachers and workers:** Teachers and professionals who spend extended hours using digital devices may also benefit from the platform's customizable break reminders and alerts, supporting digital wellness, reduced eye strain, and improved productivity.
- **Researchers:** This study contributes to digital wellness and educational technology research by serving as a baseline analysis of user engagement and "user stickiness" in wellness applications. It identifies factors that influenced students' willingness to continue using the platform despite initial technical challenges, such as login issues, thereby offering insights into sustained adoption of digital interventions.

The tools provided will include options such as:

- Pop-up break reminders (e.g., a chick hatching or clown jumpscare)
- Custom alarms
- A notepad for tasks or reminders
- Puzzle game

Theoretical Framework

This study is anchored in the Circadian Rhythm Disruption Theory, which explains how external cues—particularly light exposure—affect the body's internal clock responsible for regulating sleep, hormone production, and emotional regulation (Czeisler et al., 1999). Exposure to blue light, especially during nighttime hours, suppresses melatonin production, resulting in delayed sleep onset, irregular sleep patterns, and increased vulnerability to anxiety and depressive symptoms. This theory supports the need for interventions that interrupt prolonged screen exposure and encourage healthier sleep-wake cycles.

The study is further supported by Cognitive Load Theory (CLT) proposed by Sweller (1998), which explains how excessive information and continuous stimuli overload working memory. Prolonged screen use contributes to extraneous cognitive load, impairing focus, learning efficiency, and emotional regulation. The Break The Light platform functions as an external regulatory tool that reduces cognitive overload by prompting breaks and encouraging mindful digital behavior.

Conceptual Framework

This study is built on two main concepts: the circadian rhythm disruption caused by blue light exposure, and cognitive overload from continuous screen use

without breaks. These concepts explain how students experience poor sleep and academic fatigue, leading to anxiety, low focus, and burnout.

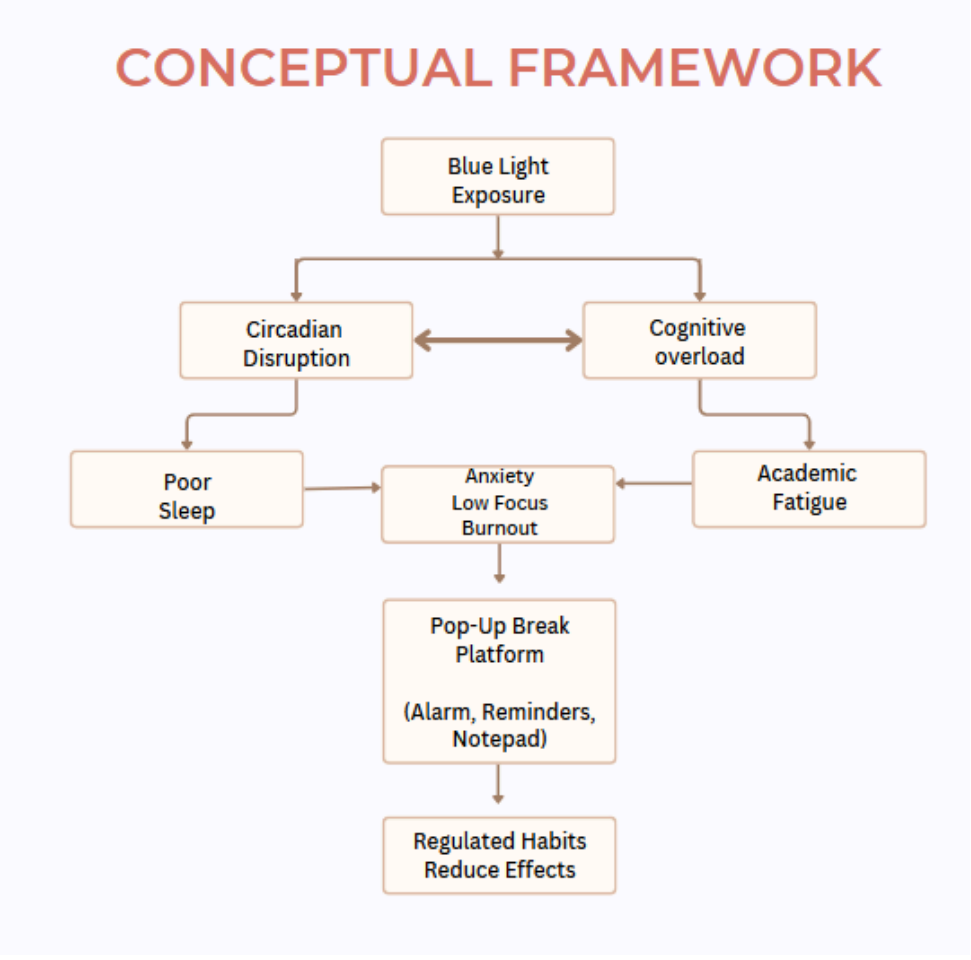


Figure 1.1: Conceptual Framework for Blue Light Exposure

- ***Blue Light Exposure*** starts the cycle. It affects both the body clock and the brain's ability to rest and focus.

- ***Circadian Disruption*** refers to how blue light at night delays melatonin release, causing poor sleep and emotional instability.
- ***Cognitive Overload*** happens when the brain is overworked due to nonstop screen time and lack of mental breaks, lowering focus and performance.
- These lead to ***Poor Sleep and Academic Fatigue***, which increase the risk of ***Anxiety, Low Focus, and Burnout*** among students.
- To address this, the ***Pop-Up Break Platform*** acts as the solution. With features like *alarms, cute pop-ups, and a notepad,* it encourages breaks and time management.
- The goal is to create ***Regulated Habits and Reduce the Negative Effects*** of blue light on student wellness.

Statement of the Problem

This study seeks to answer the following main research question:

"How can a pop-up break platform be developed to reduce the negative effects of blue light exposure on sleep quality, anxiety levels, and digital habits among junior and senior high school students from different academic strands and schools?"

Specifically, it aims to answer the following questions:

1. What features should be included in the platform to effectively support students in managing screen time?

2. Which specific functions do students find most helpful in reducing blue light exposure and its related effects?
3. How do users perceive the immediate impact of the Break The Light platform on their mood and level of digital mindfulness during short-term interaction?

Scope and Delimitation of the Study

Scope of the Study

The primary focus of this research was the development, deployment, and user-centered evaluation of "Break The Light," a cross-platform web intervention designed to mitigate the psychological and physiological impacts of blue light exposure.

- Demographic Diversity: While the study originated within Living Stone International School (LSIS), the scope organically expanded to a global scale. The participant pool (n=35 for pre-intervention; n=24 for evaluation) included a heterogeneous group ranging from Grade 7 to University level. This spectrum allowed the researcher to capture data across various developmental stages—from early adolescence to young adulthood.
- Geographic and Institutional Reach: The study's reach extended beyond local boundaries, incorporating respondents from prestigious local and international institutions, including De La Salle University (DLSU) Manila, Keio

University (Japan), Brent International School (Subic), and student respondents from India and the Brunei/Malaysia region. This geographic diversity strengthens the study by proving that "digital paralysis" and academic stress are universal phenomena transcending cultural and regional borders.

- Technical Infrastructure: To facilitate this global reach, the platform was deployed using Firebase Hosting and Authentication. This allowed for real-time, 24/7 accessibility and secure user data management, ensuring that the intervention was available to participants regardless of their time zone or device type.
- Data Metrics: The study utilized standardized psychological tools, specifically the GAD-7 (Generalized Anxiety Disorder-7) and PSQI (Pittsburgh Sleep Quality Index), to establish a rigorous baseline for student wellness.

Delimitation of the Study

Despite its broad reach, the study is bound by certain constraints:

- Self-Reported Data: The findings rely on the subjective honesty of participants. Variables such as "mood improvement" and "sleep quality" were measured via self-reporting, which may be subject to response bias.

- Duration: The evaluation phase focused on short-term interaction (February 2026). Consequently, the study does not claim to track permanent physiological changes or long-term behavioral shifts over several months.
- Technical Constraints: While the platform was accessible globally, user experience may have been influenced by local internet speeds or the "login error" identified during the testing phase.
- Focus Area: The study specifically targets "student wellness" in the context of academic-related digital strain and does not account for clinical psychological conditions or external lifestyle variables (e.g., diet, pre-existing medical conditions).

Definition of Terms

ABM students *– Students pursuing the academic strand of Accountancy, Business, and Management.*

Angeles University Foundation (AUF) *– A private Roman Catholic university in Angeles City, Philippines, serving as one of the secondary research sites.*

Blue Light *– A type of visible light with a wavelength between 380 to 500 nanometers (nm). It contains the shortest wavelength and the highest energy of all visible light. It affects human circadian rhythms and melatonin levels, potentially causing twice as much damage to the human retina as green light. Sources include sunlight and artificial sources like LED TVs, computer monitors, and smartphones.*

Brent International School Subic *– An international co-educational day school located in the Subic Bay Freeport Zone, Philippines, included in the diverse respondent pool.*

Circadian Rhythm *– The natural internal process that regulates the sleep-wake cycle and repeats roughly every 24 hours. Often called the "body clock," it controls sleep, temperature, hormones, and appetite.*

Claro M. Recto ICT High School *– A specialized public high school in Angeles City focusing on Information and Communication Technology.*

De La Salle University (DLSU) Manila *– A private Catholic research university in Taft Avenue, Manila, included in the university-level respondent demographic.*

Digital Intervention *– A technology-based tool (such as "Break The Light") designed to interrupt excessive screen time and promote healthier physical, emotional, and cognitive outcomes.*

Firebase *– A backend-as-a-service (BaaS) platform developed by Google, used in this study to host the website and manage secure user authentication.*

Flutter and Dart *– The open-source UI framework (Flutter) and programming language (Dart) used to develop the cross-platform interface and functionality of the Break The Light platform.*

Francisco G. Nepomuceno Memorial High School *– A public high school in Angeles City, Philippines, represented in the evaluation phase.*

GAS students *– Students pursuing the General Academic Strand in Senior High School.*

Generalized Anxiety Disorder (GAD-7) *– A seven-item, self-report scale used for screening and severity assessment of anxiety.*

Holy Angel University (HAU) *– A private Catholic university in Angeles City, Philippines, represented in the student respondent pool.*

HUMSS students *– Students pursuing the Humanities and Social Sciences academic strand.*

India (International Respondents) *– Refers to participants located in India who accessed the platform, representing the study's international reach.*

Keio University *– A private research university in Tokyo, Japan, representing the international university-level demographic of the study.*

Living Stone International School (LSIS) *– The primary research site and institution where the study was initiated.*

Mean *– The average value of a set of numbers, calculated by dividing the sum of all values by the total count.*

National University (NU) Clark *– An extension of National University located in Clark Freeport Zone, serving as one of the contributing research sites.*

P-value *– A statistical measure used to determine the probability that a result happened by chance; values below 0.05 are typically considered statistically significant.*

Pampanga High School *– A public high school in San Fernando, Pampanga, included in the diverse participant demographic.*

Pittsburgh Sleep Quality Index (PSQI) *– A self-report questionnaire that assesses sleep quality over a one-month interval across seven components to produce a global score.*

Pomodoro Technique *– A time management method developed by Francesco Cirillo that uses a timer to break work into 25-minute intervals (called "Pomodoros") separated by short breaks.*

Pre-intervention Assessment *– The initial data collection phase (including GAD-7 and PSQI) conducted prior to platform interaction to establish baseline wellness levels.*

San Roque National High School *– A public secondary school included in the geographic scope of the respondents.*

Significant Difference *– A statistically verified change between two groups, indicating that the observed difference is unlikely to have occurred by chance.*

Sleep Quality *– A clinical measurement of sleep excellence based on duration, sleep latency, and the feeling of alertness upon waking.*

St. Peter Inter College *– An educational institution represented in the diverse respondent pool.*

STEM/STEAM students *– Students pursuing the fields of Science, Technology, Engineering, and Mathematics, often including the Arts (STEAM).*

CHAPTER II

REVIEW OF RELATED LITERATURE

In today's fast-paced digital world, where screens are practically glued to our hands, the impact of blue light exposure on adolescent health has become a pressing concern. It's as if we're constantly bathed in this digital glow, and for teenagers who are always online for school, social media, or gaming, it's a significant issue. Studies have shown that excessive exposure to blue light, especially in the evening, disrupts our body clocks, making it difficult to fall asleep and even affecting our mood and cognitive function (Alicja Grzelak, 2024).

This chapter examines the profound impact of blue light exposure on adolescent health, delving into the theoretical underpinnings of circadian rhythm disruption and cognitive load. It discusses relevant literature concerning these phenomena and introduces the concept of pop-up interventions as a promising strategy for digital wellness, concluding with a synthesis of the reviewed literature to inform the research methodology.

Classification of Relevant Theories

Visual Representation of Melatonin Suppression

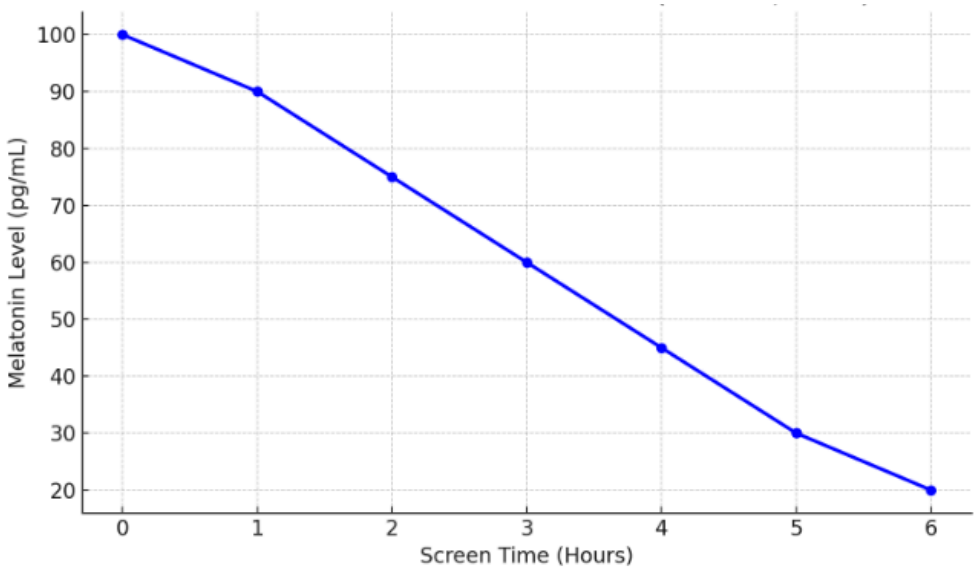


Table 2.1: Screen Time vs. Melatonin levels (Grzelak, 2024)

A visual representation adapted from the work of Grzelak (2024) demonstrates the negative correlation between screen time and melatonin production, the hormone responsible for regulating sleep-wake cycles. In her systematic review titled "Managing Blue Light Exposure and Its Impacts on Sleep Quality and Circadian Health," Grzelak examined physiological data from controlled evening-exposure experiments, particularly on adolescents and young adults. Her findings indicate that with every additional hour of screen time, melatonin levels significantly decline, resulting in poorer sleep onset and quality.

This physiological trend is supported by data Grzelak gathered across multiple studies measuring melatonin levels in pg/mL against screen exposure in hours, which found that melatonin levels may drop by more than 75% after just 4–6 hours of continuous screen time in the evening. Her analysis emphasized that melatonin suppression does not require extreme screen exposure; even 1–2 hours of use before bedtime can lead to measurable hormonal disruption.

This graph reinforces the Circadian Rhythm Disruption Theory and illustrates how blue light emitted from screens functions as a powerful external stimulus, delaying melatonin release. It also provides a compelling reason to design pop-up interventions that help reduce exposure through timed breaks or bedtime cutoffs.

By visualizing this data, students and educators can better grasp the steep physiological cost of evening device use. This further validates the current study's rationale for creating behavioral interventions, such as the Pop-Up Break Platform, to protect adolescent health during their most vulnerable developmental years.

Theory I: *The Circadian Rhythm Disruption Theory: When Your Body Clock Goes Haywire*

The Circadian Rhythm Disruption Theory explains how light, particularly blue light, functions as a powerful zeitgeber for the human internal body clock, regulating

wakefulness and the production of melatonin, the sleep-inducing hormone. Back in 1989, Czeisler and his team demonstrated that even dim blue light could effectively reset human circadian rhythms (How to Write a Literature Review | Guide, Examples, & Templates, 2023), and subsequent research confirmed that blue light from digital devices interferes with melatonin secretion, thereby disrupting the natural sleep-wake cycle. The crucial feature this study considers from this theory is how mistimed light exposure disrupts the circadian rhythm, leading to various health issues. This is highly relevant for this research, as previous studies (Manuel, 2024; 2025) found a clear link between high blue light exposure and poor sleep and anxiety in Grade 11 students.

A systematic review by Marcia Ines Silvani, Robert Werder, and Claudio Perret (2022) emphasized how evening blue light exposure negatively affects sleep, shortens its duration, and suppresses melatonin. These disruptions extend beyond mere tiredness, significantly impairing cognitive performance and increasing the risk of anxiety and depression. This systematic review highlights how crucial it is to address the physiological disturbances caused by blue light, as they contribute to a decline in mental health among adolescents. It also indicates that strategies such as blue light filters and reduced screen time before bed could prove beneficial. The detail of this literature that the present paper will consider in writing the research methodology is the evidence for blue light's physiological impact on sleep and melatonin suppression.

Building on that, Prather and Hurst (2022) explored the complex, two-way relationship between sleep and circadian rhythms, showing how disruptions in one lead to problems in the other. Their work reinforces how screen time, especially before bed, leads to less and poorer quality sleep, which in turn amplifies the risk of anxiety and depression. This study emphasizes the interconnectedness of sleep, circadian cycles, and mental health, making it clear that interventions must consider the holistic impact. This study's findings, similar to the work by Silvani, Werder, and Perret (2022), consistently demonstrate the detrimental effects of evening blue light on sleep patterns and mental well-being. However, Prather and Hurst (2022) uniquely underscore the bidirectional nature of the sleep-circadian rhythm relationship, providing a more comprehensive understanding of how disturbances cascade. The specific detail of the literature that the present paper will consider is the interconnectedness of sleep, circadian cycles, and mental health as a holistic framework for intervention.

Similarly, a broad meta-analysis by Kocevska et al. (2021), which encompassed 1.1 million individuals across various countries, confirmed the widespread nature of blue light-related sleep problems, particularly among young people. Their findings align with previous research, indicating that disrupted sleep patterns constitute a significant public health concern affecting diverse age groups,

not solely adolescents. This global perspective reinforces that this is a widespread problem requiring attention. Kocevska et al. (2021) further substantiate the findings of Silvani, Werder, and Perret (2022) and Prather and Hurst (2022) by providing extensive, population-level evidence for blue light-induced sleep disturbances. While Silvani, Werder, and Perret (2022) focused on specific physiological mechanisms and Prather and Hurst (2022) on the bidirectional link between sleep and circadian rhythms, Kocevska et al. (2021) expand the scope to a global scale, confirming the universality of the problem. The most striking difference is the scale of evidence presented, moving from specific physiological mechanisms to a large-scale meta-analysis, yet they all consistently point to the same problem: blue light is messing with our sleep and mental health. This collective understanding underscores the urgent need for interventions, especially in an increasingly digital world.

The synthesis of these studies, all viewed through the lens of the Circadian Rhythm Disruption Theory, consistently highlights how evening blue light exposure profoundly impacts sleep quality and circadian timing by suppressing melatonin. This physiological disruption then exacerbates mental health challenges like anxiety and depression, particularly in adolescents. The findings also reveal that adolescents are more sensitive to these effects than adults and that prolonged screen time leads to reduced sleep duration and quality, impacting cognitive performance. The proposed Pop-Up Break Platform directly addresses these issues by aiming to interrupt

continuous screen use, thereby allowing the body's natural circadian rhythm to resynchronize and mitigate these negative physiological and psychological outcomes. This approach leverages the understanding that controlling blue light exposure, especially in the evening, is key to improving sleep hygiene and overall well-being.

The Circadian Rhythm Disruption Theory, with its focus on how light impacts our internal clocks, shares a core similarity with Cognitive Load Theory (CLT) because both theories explain how external stimuli can overwhelm and disrupt internal biological or cognitive processes. While one focuses on sleep-wake cycles and physiological responses, the other zeroes in on how the brain processes information.

***Theory II**: Cognitive Load Theory: When Your Brain Says "Enough!"*

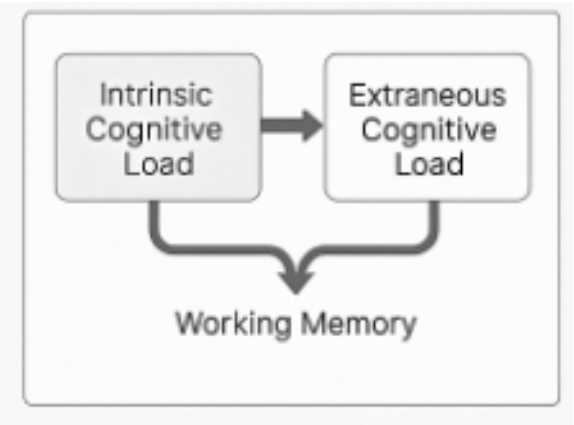


Figure 2.1: Cognitive Load Interaction Model based on CLT

John Sweller's Cognitive Load Theory (CLT), developed in the late 1980s, delves into how the human brain processes and stores information, particularly during complex learning. It posits that working memory has a limited capacity, akin to a small desk that can only accommodate a few papers at a time. If an excessive amount of distractions or overly difficult information, referred to by CLT as "extraneous cognitive load," is presented, working memory becomes overwhelmed, hindering learning and clear thinking. This theory is crucial because, in the digital age, prolonged screen time and constant blue light exposure exemplify factors that can induce this overload. The key feature of CLT that this paper considers for the research methodology is its emphasis on managing extraneous cognitive load to optimize learning and mental functioning. If students are to truly benefit from their digital learning environments, it is imperative to consider how to lessen the burden on their working memory.

When comparing Cognitive Load Theory with the Circadian Rhythm Disruption Theory, the most significant difference that truly sets them apart is their primary focus: Circadian Rhythm Disruption Theory centers on the physiological impact of external stimuli (like blue light) on internal biological rhythms, leading to sleep and mood disturbances, whereas Cognitive Load Theory focuses on the cognitive impact of information and environmental design on working memory, affecting learning and mental performance. One is fundamentally about the body's

clock, and the other is about the brain's processing capacity. However, they share a common goal of understanding how environmental factors can negatively affect well-being and performance.

This section highlights that Cognitive Load Theory offers a crucial lens through which to understand the cognitive toll of prolonged screen time. It emphasizes that when students are exposed to excessive digital stimuli, it creates an "extraneous cognitive load" that hampers their ability to learn, focus, and even regulate their emotions. By understanding this, it becomes clear that simply instructing students to "focus harder" will not resolve the issue if their working memory is already overwhelmed. The implication for this research methodology is clear: any intervention aimed at improving student well-being in a digital learning environment must consider strategies to reduce extraneous cognitive load, potentially through more efficient structuring of screen time or by encouraging regular mental breaks. The feature of Cognitive Load Theory that this paper will consider in writing the research methodology is the necessity of reducing extraneous cognitive load to improve student focus and well-being.

Theory III: *Digital Intervention Theory and Co-Design Principles*

Digital Intervention Theory advocates for behavioral tools that are engaging, personalized, and grounded in psychological science. Research by Ferrari et al. (2022)

and Wageningen University (2024) underscores the success of co-designed interventions, particularly among adolescents.

The proposed platform integrates these principles by allowing students to customize their reminders—from quirky pop-ups like a clown jumpscare to a baby chick hatching from an egg. These emotionally resonant and playful elements enhance engagement and disrupt long, unbroken screen sessions effectively.

As digital wellness tools evolve, selecting the right development framework is crucial for effectiveness and engagement. The proposed Pop-Up Break Platform will be built using Flutter, an open-source UI toolkit developed by Google, which allows for simultaneous deployment on Android, iOS, and Web using a single codebase written in Dart. This aligns with co-design principles discussed in Ferrari et al. (2022) and Wageningen University (2024), which emphasize accessibility, personalization, and emotional resonance in digital tools.

Flutter is well-suited for this intervention due to its support for highly interactive and animated user interfaces, features essential for capturing adolescent attention through pop-ups and reminders. Additionally, Firebase will be used to handle Google account authentication, cloud data storage, and website hosting, enabling seamless syncing of user preferences across both mobile and browser

platforms. This integration reflects current trends in adaptive, cross-platform mental wellness apps and reinforces the importance of designing technology that is both engaging and scientifically grounded.

Pop-Up Interventions Strategy: A Digital Lifeline for Students

Building on the understanding from both Circadian Rhythm Disruption Theory and Cognitive Load Theory, the idea of a Pop-Up Break Platform emerges as a promising intervention. This approach leverages the insights that both blue light and cognitive overload negatively impact students. The platform aims to actively interrupt prolonged screen sessions with customizable alarms, helpful notepads, and engaging "pop-up reminders", ranging from a cute chick hatching to a funny clown jumpscare (if that's what gets one's attention!). These digital nudges are designed to break the cycle of continuous exposure, allowing for physical and mental respite.

The development and implementation of such digital wellness interventions are gaining traction, especially for students exposed to prolonged screen time. Studies show that digital health interventions can be effective in improving psychological well-being among university students and that co-designing these tools with young people is crucial for their effectiveness. The aim here is to create a tool that is not just useful but also engaging and personalized enough that students will use it. The

specific feature this study considers is the use of interactive pop-up reminders to prompt scheduled breaks and encourage healthier digital habits. This directly addresses the problem of students neglecting breaks amidst demanding academic schedules.

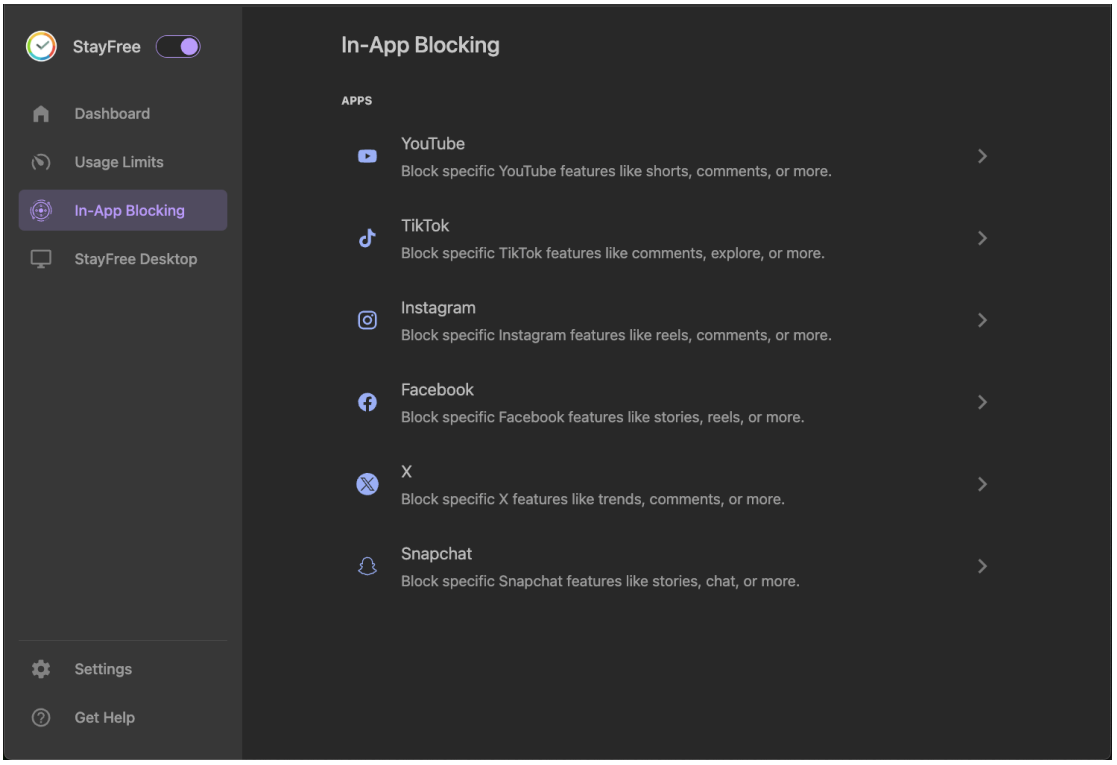


Figure 2.2. StayFree In-App Blocking Dashboard

Indeed, digital tools for managing screen time are already in use. Applications like StayFree and built-in features such as Android's Digital Wellbeing offer usage limits, focus modes, app blocking, and bedtime settings. For instance, the StayFree interface (as shown in the image) demonstrates a sophisticated in-app blocking

system, where users can target specific platform features—such as YouTube Shorts, Instagram Reels, or TikTok comments—rather than blocking entire apps. This granular control empowers users to curate their digital experience, reduce temptation, and stay focused without fully disconnecting.

Another widely adopted method for managing screen time and productivity is the Pomodoro Technique, popularized through browser-based tools such as Pomofocus (n.d.). This technique, originally devised by Francesco Cirillo, structures work into 25-minute "pomodoro" focus intervals, followed by short 5-minute breaks, with a longer rest after four cycles, supporting sustained attention and reducing mental fatigue. Pomofocus enhances this method by allowing users to list tasks with estimated pomodoros, then visually track their progress, all in a minimalist, customizable interface

Yet, while Pomofocus effectively supports productivity and encourages healthy focus-break rhythms, its primary aim is task management. In contrast, the proposed Pop-Up Break Platform builds on this time-boxing foundation but explicitly targets student wellness by delivering emotionally engaging, interactive pop-ups that interrupt prolonged screen use. These pop-ups are designed not merely to pause work, but to promote cognitive clarity, align with circadian health, and support psychological well-being in a more immersive, responsive way.

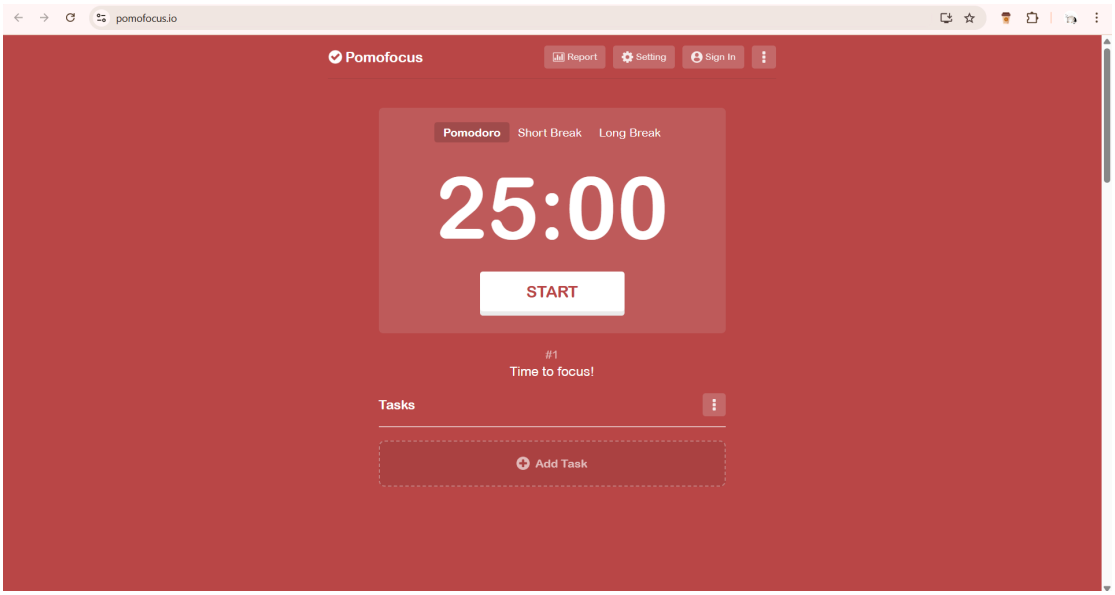


Figure 2.3. PomoFocus Pomodoro Setting/ Landing Page

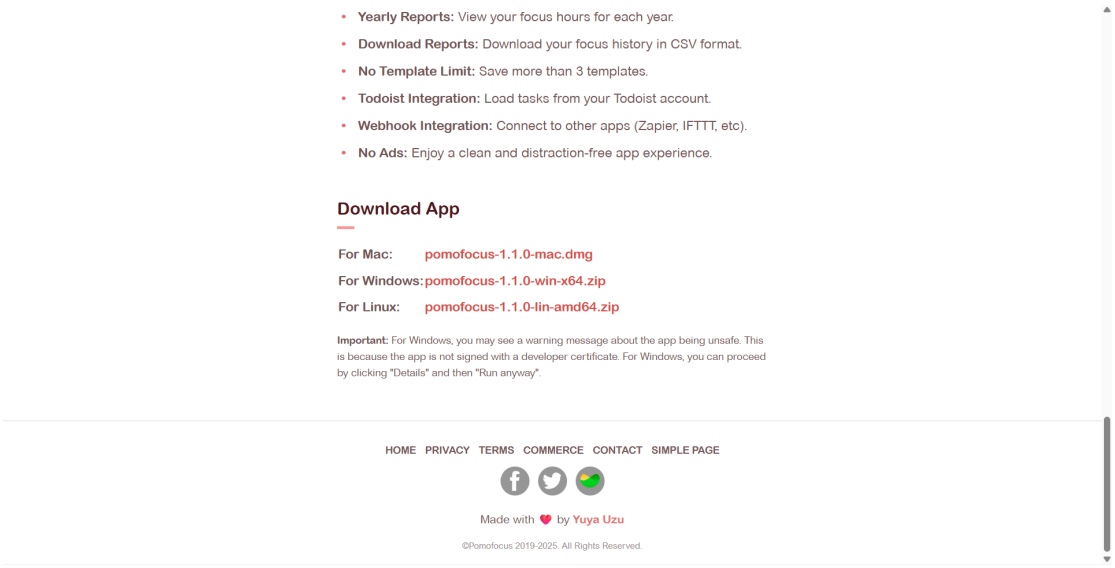


Figure 2.4. PomoFocus Pomodoro Instructions & Download attached files/ Landing Page

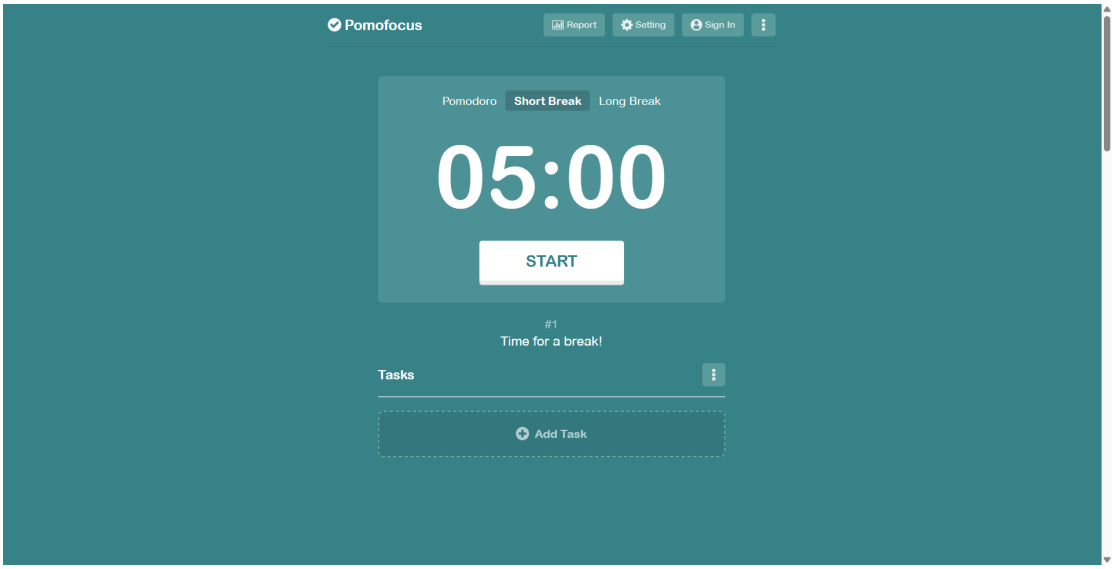


Figure 2.5. PomoFocus Pomodoro Short Break/ Landing Page

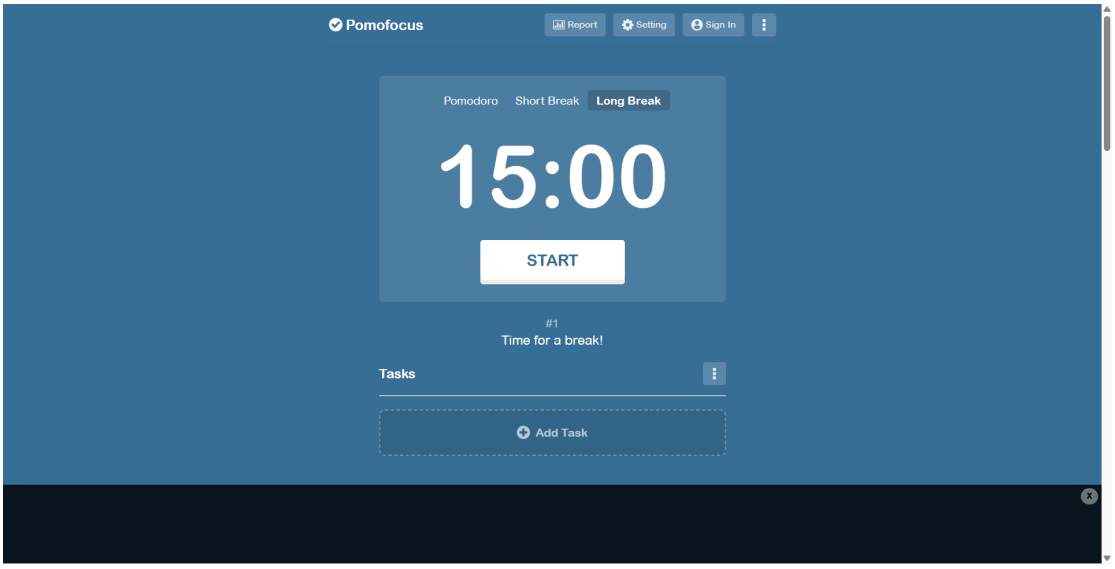


Figure 2.6. PomoFocus Pomodoro Long Break/ Landing Page

Similarly, Google's "Screen Time Reminders" subtly deploys prompts that nudge users after prolonged usage. These existing solutions illustrate that digital intervention for screen management is not only functional but evolving to be more

user-responsive. However, while most tools focus on passive tracking and restriction, the proposed Pop-Up Break Platform would go further—emphasizing customizable, emotionally engaging, and interactive pop-up reminders. These are designed to disrupt screen overuse while supporting cognitive clarity and circadian rhythm alignment, directly addressing the theoretical needs established in this study.

The previous studies on digital interventions for screen time management, including generalized applications like StayFree and system-level features such as Android's Digital Wellbeing, demonstrate that while the market offers tools for managing screen time, they often lack the personalized, engaging, and theoretically-driven features necessary for effective behavioral change in students. These general applications provide basic timers and usage reports, but they do not necessarily address the psychological or physiological mechanisms of blue light exposure or cognitive overload in a targeted manner. The crucial difference is that while existing apps offer passive monitoring and basic controls, the Pop-Up Break Platform aims for active, engaging, and customizable interventions directly linked to mitigating circadian disruption and cognitive overload through behavioral prompts. This distinction signifies a move beyond simple timers to a more dynamic, user-centered approach to digital wellness. The specific detail of the literature that the paper will consider is the necessity of interactive and user-friendly design in digital interventions to promote active engagement and behavioral change.

This section synthesizes that digital interventions offer a powerful way to mitigate the adverse effects of blue light and cognitive overload. The evidence suggests that while current screen time management tools exist, a truly effective solution needs to be designed with a deep understanding of how these digital habits impact circadian rhythms and cognitive processing. The key is to create a platform that does not just block usage but actively encourages healthier habits through engaging, customizable, and timely prompts. This direct application of theoretical insights into practical tools aims to foster more regulated digital habits, ultimately leading to improved sleep quality, reduced anxiety, and enhanced academic focus for students.

Complementary Tools: Notion and Coffeelings

To promote healthier journaling and emotional regulation habits among adolescents, this study also recognizes the supportive role of productivity tools such as Notion and the Coffeelings Chrome Extension. While the proposed Pop-Up Break Platform functions as an intervention tool to disrupt prolonged digital exposure and promote rest, complementary tools like Notion and Coffeelings serve to reinforce intentional self-tracking, digital reflection, and mental wellness routines in the daily lives of students.

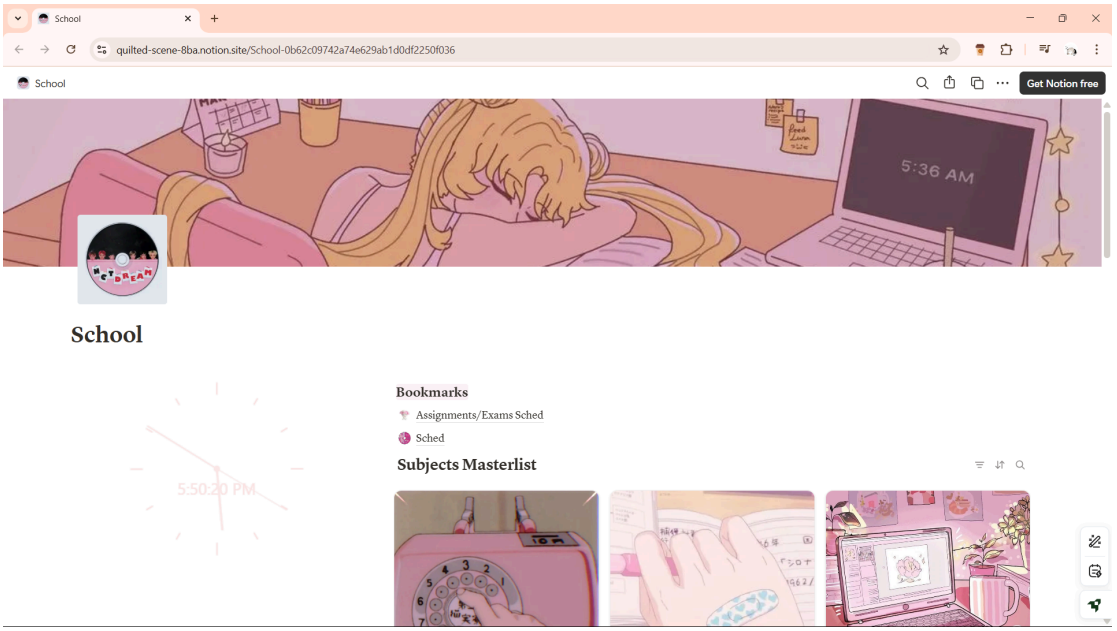
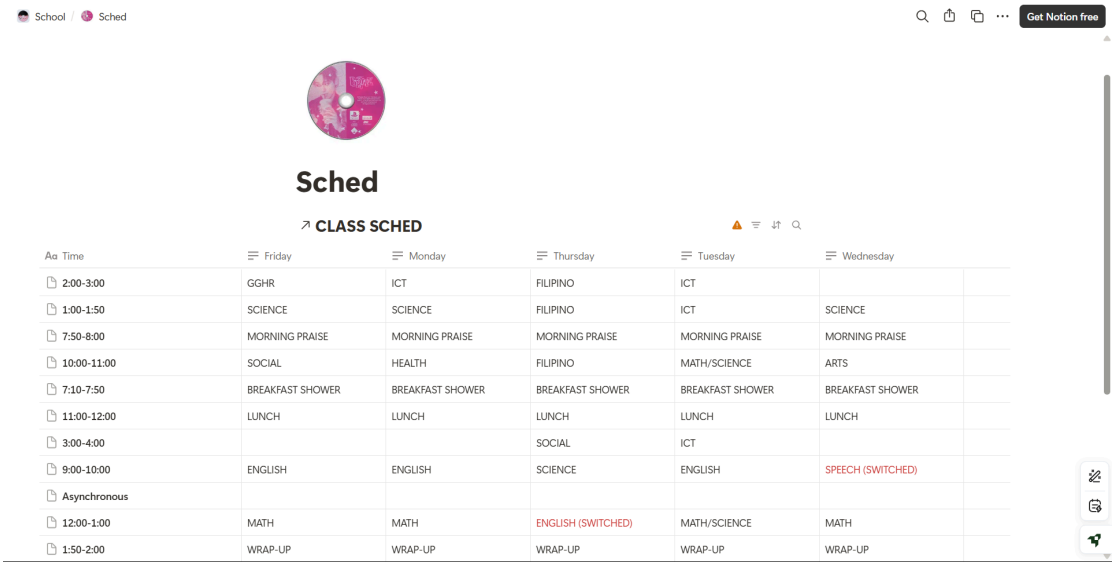


Figure 2.7. Notion Screenshot in School dashboard



Time	Friday	Monday	Thursday	Tuesday	Wednesday
2:00-3:00	GGHR	ICT	FILIPINO	ICT	
1:00-1:50	SCIENCE	SCIENCE	FILIPINO	ICT	SCIENCE
7:50-8:00	MORNING PRAISE	MORNING PRAISE	MORNING PRAISE	MORNING PRAISE	MORNING PRAISE
10:00-11:00	SOCIAL	HEALTH	FILIPINO	MATH/SCIENCE	ARTS
7:10-7:50	BREAKFAST SHOWER	BREAKFAST SHOWER	BREAKFAST SHOWER	BREAKFAST SHOWER	BREAKFAST SHOWER
11:00-12:00	LUNCH	LUNCH	LUNCH	LUNCH	LUNCH
3:00-4:00			SOCIAL	ICT	
9:00-10:00	ENGLISH	ENGLISH	SCIENCE	ENGLISH	SPEECH (SWITCHED)
Asynchronous					
12:00-1:00	MATH	MATH	ENGLISH (SWITCHED)	MATH/SCIENCE	MATH
1:50-2:00	WRAP-UP	WRAP-UP	WRAP-UP	WRAP-UP	WRAP-UP

Figure 2.8. Notion Screenshot in School dashboard Class schedule

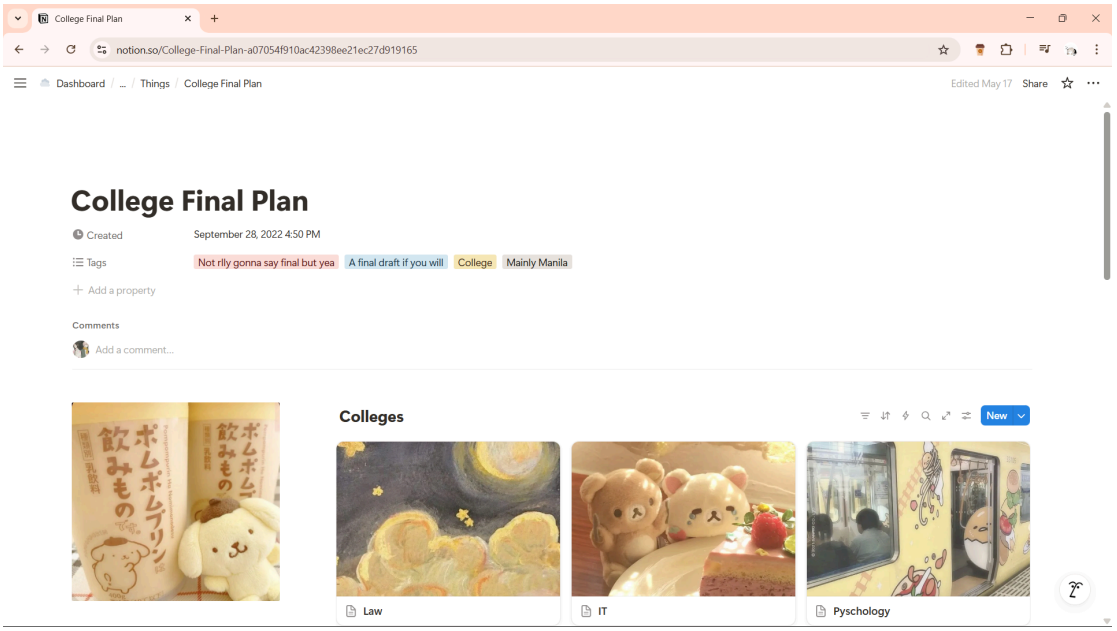


Figure 2.9. Notion Screenshot in College Final Plan

Notion is a flexible, all-in-one productivity workspace that allows users to build personalized dashboards, databases, habit trackers, and digital journals. In the screenshot from the researcher's Notion titled College Final Plan, one can observe visual layouts combining images, links, and content blocks categorized under fields like "Law" and "IT", demonstrating Notion's ability to manage both academic and emotional planning. This digital system helps users organize tasks and schedules, but more importantly, it allows them to track moods, study goals, and cognitive burdens through interactive journal entries and embedded media. With customizable pages for mental health check-ins, students can reflect on their stress levels, sleep quality, and anxieties—all variables central to this study—within the same platform where their

academic life is managed. This fosters a more integrative and mindful approach to digital planning.

Figure 2.10. Coffeelings choose today's roasts 'amazing' & 'ok' Screenshots

Figure 2.11. Coffeelings choose today's roasts 'tired' & 'sad' Screenshots

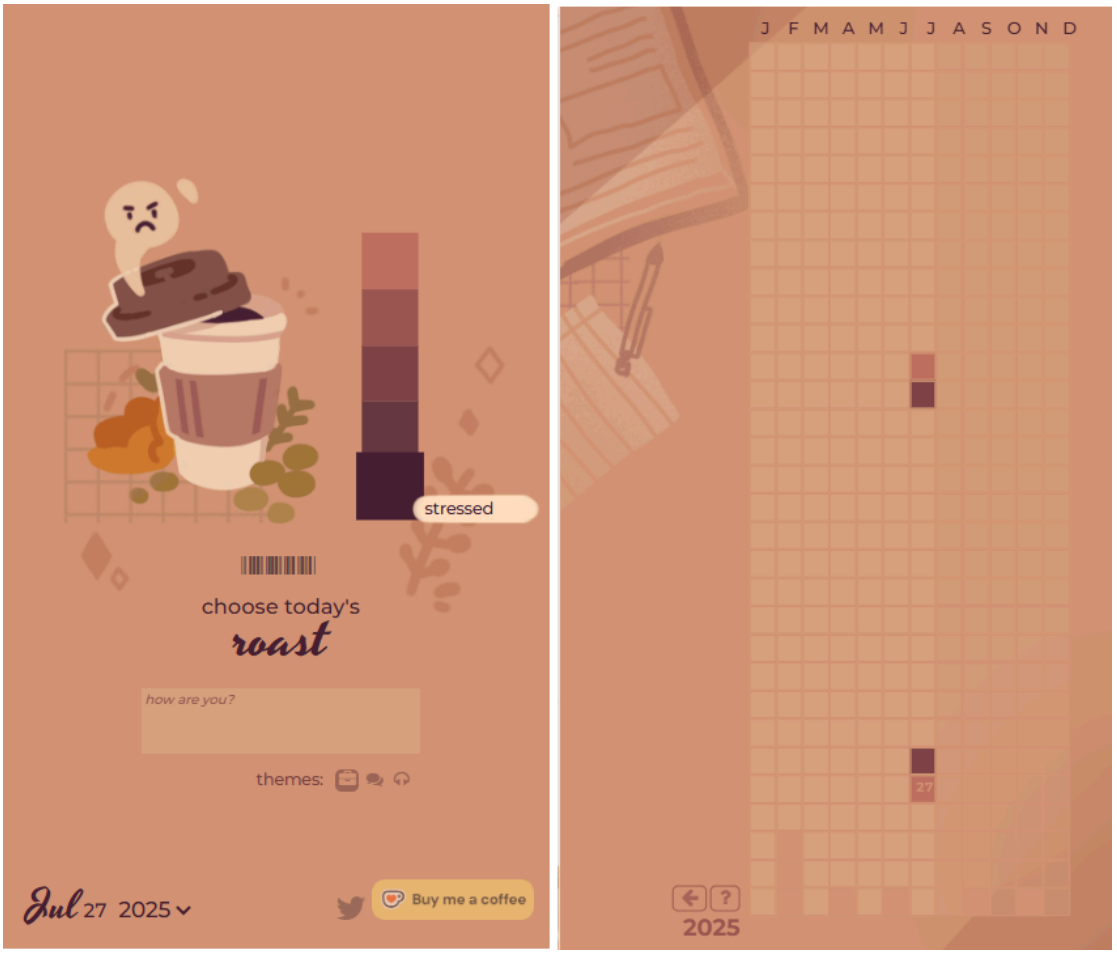


Figure 2.12. Coffeelings choose today's roasts 'stressed' & calendar Screenshots

Meanwhile, the Coffeelings Chrome Extension, shown in the accompanying screenshot, exemplifies a lightweight but emotionally rich tool that maps users' moods to daily "coffee roasts." Each roast represents a specific emotional tone (e.g., "sad" or "stressed"), and the visual calendar uses colored blocks to represent mood patterns over time. It also provides space for short reflective entries, all while maintaining an aesthetic and minimal design that reduces cognitive load. Unlike traditional mental

health tracking apps that can feel clinical or intrusive, Coffeelings gently encourages emotional awareness by embedding journaling into a calming interface. The extension's visual representation of mood over weeks and months aligns with this study's interest in long-term emotional trends and their correlation with screen time, anxiety, and sleep patterns. It becomes especially valuable as an intervention not by limiting digital usage, but by giving it emotional meaning.

Both tools address crucial components of digital wellness: Notion enhances structured self-monitoring and academic-life integration, while Coffeelings supports emotional regulation through non-invasive, aesthetically guided journaling. Their usage supports the framework laid out by the Circadian Rhythm Disruption Theory and the Cognitive Load Theory in this paper. While the Pop-Up Break Platform interrupts harmful digital cycles in real time, tools like Notion and Coffeelings encourage students to take ownership of their digital behaviors and mental states, bridging intervention with reflection.

Moreover, research has shown that co-designed digital interventions are more likely to succeed when users have autonomy, aesthetic affinity, and ease of use (Wageningen University & Research, 2024). These platforms provide evidence that students are more likely to engage with tools that are not only functional but also customizable and emotionally expressive—qualities embedded in both Notion and Coffeelings.

Ultimately, this research leverages the user-centered design principles observed in Notion and Coffeelings to reinforce the effectiveness of the proposed Pop-Up Break Platform, proving that emotionally aware, flexible, and visually engaging digital tools can positively influence adolescent habits around journaling, emotion regulation, and digital time management.

The First Main Variable

Blue Light Exposure is defined as a type of visible light with a wavelength between 380 to 500 nanometers (nm), possessing the shortest wavelength and highest energy of all visible light types. It affects human circadian rhythms and melatonin levels and originates from both natural sources, like sunlight, and artificial sources such as fluorescent lights, LED TVs, computer monitors, smartphones, and tablet screens.

This variable is highly relevant in light of the Circadian Rhythm Disruption Theory, which explains how blue light, particularly in the evening, functions as a powerful external cue that interferes with the body's natural 24-hour cycle. By actively suppressing melatonin production, blue light delays sleep onset and disrupts sleep quality, leading to a cascade of negative physiological and psychological effects. The theory underpins the understanding that controlling blue light exposure is

essential for maintaining healthy circadian rhythms and, consequently, improving sleep and mental health.

A study by Mudassir Alam, Kashif Abbas, Yusra Sharf, and Sarfaraz Khan (2024) specifically addressed the impacts of blue light exposure from electronic devices on circadian rhythm and sleep disruption in adolescent and young adult students. Their research highlighted the unique vulnerability of adolescents to these effects due to their developmental sleep needs. The study found that evening blue light exposure consistently suppresses melatonin secretion, leading to poor sleep duration and quality, and consequently, negative mental health outcomes. This paper will consider the detailed physiological grounding of melatonin suppression discussed in their research, along with the consistent evidence of blue light's detrimental effects on adolescent sleep, when formulating the intervention strategies for the Pop-Up Break Platform.

Similarly, a systematic review by Alicja Grzelak (2024) on managing blue light exposure and its impacts on sleep quality and circadian health echoed these findings, emphasizing the pervasive nature of blue light from digital devices in modern life. Both Grzelak's and Alam et al.'s studies share the commonality of highlighting the widespread impact of blue light on adolescent sleep and mental health. While Alam et al. specifically focused on the vulnerability of adolescent

students, Grzelak's review provided a broader overview of blue light's effects on circadian health. This common understanding of blue light as a significant disruptor of sleep and well-being forms a crucial basis for this study's proposed intervention.

I. Sleep Quality is measured by how well a person sleeps, encompassing duration, depth, and how they feel after waking up. It is intrinsically linked to blue light exposure and circadian rhythm disruption, as the suppression of melatonin by blue light directly impairs the ability to fall asleep and maintain restorative sleep. The Pittsburgh Sleep Quality Index (PSQI) is utilized as a standardized instrument to assess sleep quality over one month, providing a global score that reflects the severity of sleep disturbances.

The Circadian Rhythm Disruption Theory explicitly connects to sleep quality by demonstrating that when the body's internal clock is desynchronized by external factors like blue light, the natural sleep-wake cycle is disturbed, resulting in reduced sleep duration and poorer sleep efficiency. The theory posits that consistent blue light exposure, particularly at night, shifts the timing of melatonin release, making it harder for individuals to initiate and sustain sleep. Thus, poor sleep quality is a direct consequence of a disrupted circadian rhythm.

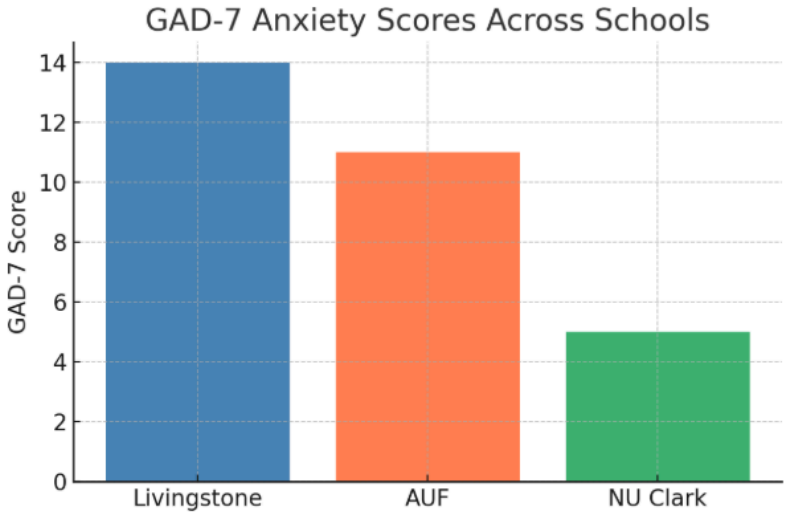


Table 2.2. Average GAD-7 scores among Grade 11 Students from Livingstone International School, National University of Clark, and Angeles University Foundation(Manuel, 2025)

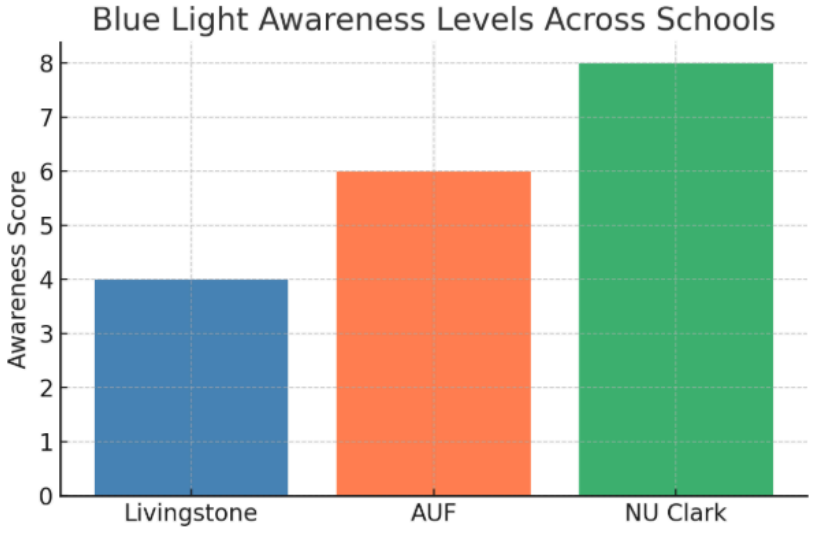


Table 2.3. Average Awareness scores among Grade 11 Students from Livingstone International School, National University of Clark, and Angeles University Foundation (Manuel, 2025)

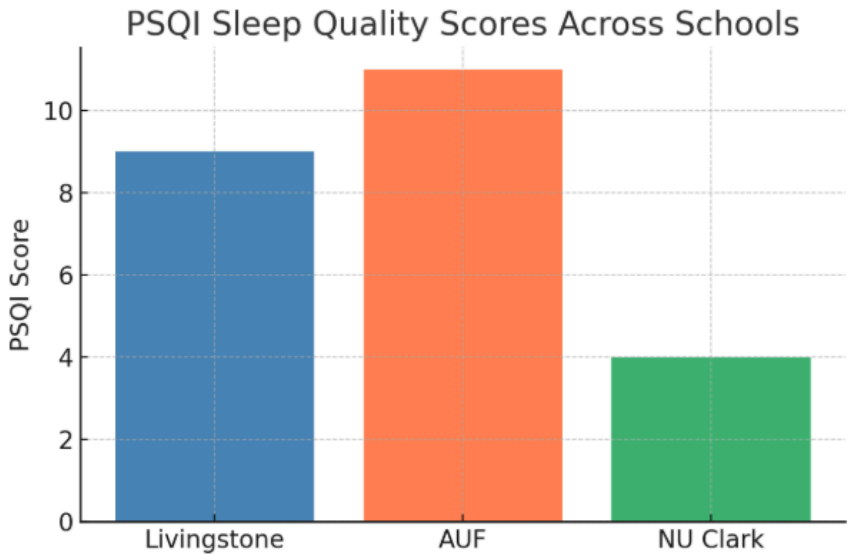


Table 2.4. Average PSQI scores among Grade 11 Students from Livingstone International School, National University of Clark, and Angeles University Foundation (Manuel, 2025)

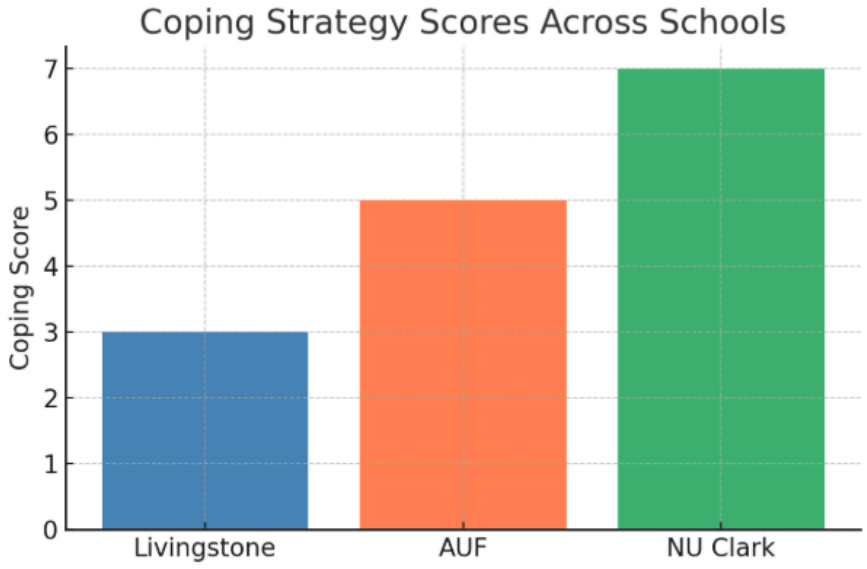


Table 2.5. Average coping strategy awareness among Grade 11 Students from Livingstone International School, National University of Clark, and Angeles University Foundation (Manuel, 2025)

The previous quantitative study by Manuel (2025), "Comparative Study on Blue Light Exposure in Grade 11 Students from Livingstone International School, National University of Clark, and Angeles University Foundation," utilized the PSQI to assess sleep quality among 66 Grade 11 students. The study found statistically significant differences in sleep quality between the schools, with Angeles University Foundation (AUF) students reporting the poorest sleep quality, while National University of Clark (NU) students reported significantly better sleep. The study considered the PSQI scores as a key indicator of the students' sleep health and found that academic environment and lifestyle factors played a significant role. This paper will consider the PSQI as the primary research instrument for measuring sleep quality and will leverage the comparative findings to understand specific school-related influences on students' sleep habits.

Another related article by Christopher Höhn and his peers (2024) investigated the effects of evening smartphone use on sleep and declarative memory consolidation in male adolescents and young adults. Their study supported the idea that blue light exposure suppresses melatonin and affects sleep, although it noted age-related differences in recovery. Höhn et al.'s findings similarly highlighted the negative impact of blue light on sleep, reinforcing the importance of addressing evening screen time. Both Höhn et al. and Manuel (2025) underline the critical need to address screen time before bed to improve sleep quality in adolescents. While Manuel (2025)

focused on comparative sleep quality across schools, Höhn et al. provided physiological insights into melatonin suppression and its recovery.

Synthesis

This section of the second chapter provides a list of all the theories and variables reviewed in the paper. This serves as the overall synthesis, which fuses in one summarizing paragraph the feature/s of relevant theories and the details of the related literature that the paper will consider in writing the research methodology.

II. Anxiety Levels represent the degree of worry, nervousness, or unease a person experiences, often in response to stress or perceived threats. Elevated anxiety levels can manifest as heightened emotional states, impaired cognitive function, and physical symptoms, often exacerbated by insufficient or poor-quality sleep. The General Anxiety Disorder-7 (GAD-7) scale is used as a diagnostic self-report tool for screening, diagnosing, and assessing the severity of anxiety disorder.

The Circadian Rhythm Disruption Theory implies that chronic sleep disturbances, often caused by blue light exposure disrupting the circadian rhythm, can significantly contribute to increased anxiety levels. The physiological stress resulting from a misaligned body clock and poor sleep can lead to emotional dysregulation, making individuals more susceptible to anxiety and depression. This theory provides a

framework for understanding the bidirectional relationship between sleep, circadian rhythms, and mental health, wherein disrupted sleep can both cause and exacerbate anxiety.

In the quantitative study conducted by Manuel (2025), both the Generalized Anxiety Disorder-7 (GAD-7) and the Pittsburgh Sleep Quality Index (PSQI) were employed to assess the mental health and sleep patterns of Grade 11 students across three institutions: Livingstone International School (LSIS), Angeles University Foundation (AUF), and National University (NU) Clark. The results revealed statistically significant differences in both anxiety levels and sleep quality among the schools. LSIS students reported the highest levels of severe anxiety, while NU students showed the lowest, suggesting that school-specific factors such as academic pressure, classroom competitiveness, and institutional support systems directly impact students' emotional well-being. In parallel, AUF and LSIS students also showed the poorest sleep quality, whereas NU students exhibited significantly better sleep patterns, as evidenced by lower PSQI scores.

These findings are especially critical because poor sleep quality and heightened anxiety often exist in a cyclical relationship, where increased stress impairs sleep, and disrupted sleep further amplifies psychological distress. The study suggests that lifestyle habits, extended screen time, and school culture—particularly surrounding expectations and workloads—may exacerbate this cycle. By highlighting

the prevalence and consequences of these issues in specific academic contexts, Manuel's research underscores the necessity of targeted mental health and wellness initiatives in high-pressure school environments.

This paper adopts both the GAD-7 and PSQI as core instruments for measuring anxiety and sleep quality, respectively. Anchoring this research in the data and conclusions drawn by Manuel (2025) strengthens the rationale for focusing on educational environments where students are more vulnerable to mental health challenges. The insights from this prior study not only provide a validated framework for data collection but also guide this paper's emphasis on exploring institutional influences on adolescent well-being, to inform more responsive and supportive school-based interventions.

An article by Andre Comiran Tonon and 18 other researchers (2022) examined the significant association between sleep disturbances, disruptions in circadian rhythm, and the prevalence of major depressive disorder among adolescents. Their findings indicated that artificial light at night (ALAN) exposure, particularly blue light, is linked to late bedtimes and a higher incidence of mood disorders due to melatonin suppression. Both Tonon et al. and Manuel (2025) provide compelling evidence for the link between blue light-induced sleep disturbances and mental health issues like anxiety and depression in adolescents. While Manuel (2025) identified specific school contexts with higher anxiety, Tonon et al. provided a broader

understanding of the physiological mechanisms linking ALAN to depressive disorders.

III. Digital Habits refer to the routine patterns and behaviors associated. With the use of electronic devices, particularly concerning screen time duration, timing of use (especially before bed), and the activities engaged in on these devices. These habits directly influence blue light exposure, which in turn impacts sleep quality and anxiety levels. Assessing digital habits helps to identify problematic usage patterns that contribute to adverse health outcomes.

Cognitive Load Theory is highly relevant to digital habits because prolonged and unstructured screen time, a common digital habit, can induce extraneous cognitive load. When students continuously engage with digital devices without breaks, their working memory becomes overwhelmed, impairing their ability to concentrate, learn, and emotionally regulate. This theory suggests that managing digital habits by incorporating regular breaks and reducing excessive stimuli is crucial for optimizing cognitive function and preventing burnout.

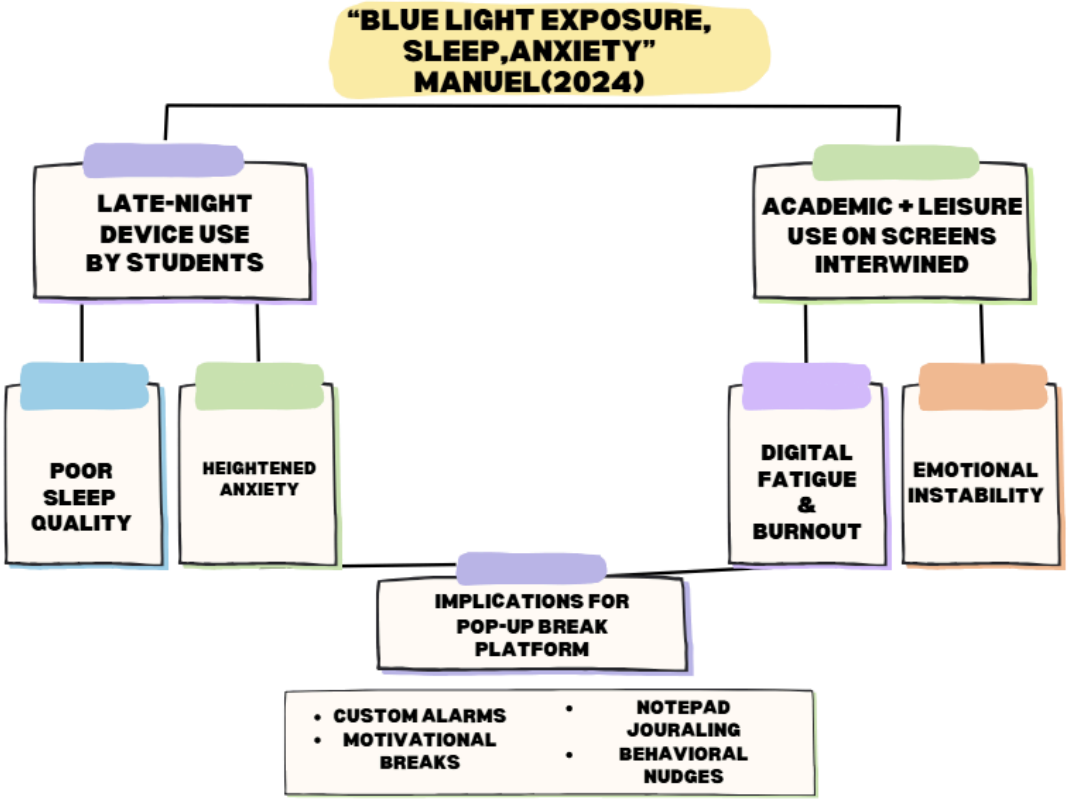


Figure 2.13. Conceptual Framework of Blue Light Exposure and Its Psychological and Behavioral Impacts on Grade 11 Students (Adapted from Manuel, 2024)

The previous qualitative study by Manuel (2024), "The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students aged 15 to 16," utilized semi-structured interviews and open-ended surveys to explore students' screen time habits. The study found that students often engaged in academic and recreational activities on their devices late into the night, contributing to poor sleep and heightened anxiety. This paper will consider the qualitative insights on students' reasons for

device use and their perceived effects, which will inform the design of the Pop-Up Break Platform's features to effectively address these habits.

A systematic review by Marcia Ines Silvani, Robert Werder, and Claudio Perret (2022) further emphasized the adverse effects of increasing blue light-emitting electronic device use on sleep and mental health, especially in adolescents. Their review suggested that interventions such as blue light filtering apps and limiting screen time before bed could be beneficial in managing digital habits. Both Manuel (2024) and Silvani, Werder, and Perret (2022) underscore the critical role of managing screen time to mitigate the negative impacts of blue light exposure on adolescent health. While Manuel (2024) provided specific insights into students' self-reported screen habits, Silvani, Werder, and Perret (2022) offered a broader scientific review of the physiological consequences of these habits.

Synthesis

This chapter has provided a comprehensive review of relevant theories and literature concerning the impact of blue light exposure and digital habits on student well-being. The synthesis of all relevant theories—Circadian Rhythm Disruption Theory, Cognitive Load Theory, and the principles of digital intervention design—reiterates that effective solutions for student well-being in a digital world

require a multifaceted approach. The paper will consider the understanding that prolonged evening blue light exposure disrupts circadian rhythms and suppresses melatonin, leading to poor sleep and increased anxiety. Concurrently, it acknowledges that continuous screen use without breaks can overwhelm working memory, causing cognitive overload and reducing focus. Therefore, the research methodology for the Pop-Up Break Platform will integrate these insights by focusing on developing customizable, engaging, and timely pop-up reminders and tools designed to interrupt digital overexposure, thereby allowing for circadian resynchronization and reduced cognitive strain. This holistic approach, informed by both physiological and cognitive principles, aims to promote regulated digital habits that lead to improved sleep quality, lower anxiety levels, and enhanced concentration in Grade 11 students, acknowledging the unique academic environments and their influence on screen time behaviors.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methodology used in the conduct of the study. The discussions include the research design and techniques, population and sample of the study, research instruments, data gathering procedure, and data processing and statistical treatment. The chapter reflects the actual implementation of the system, data collection, and platform deployment.

Research Design

This study utilized the Agile Development Model combined with an Experimental Research Design. The Agile approach allowed continuous iteration based on user and technical feedback throughout the development of the Pop-Up Break Platform, later deployed as the “Break The Light” website. The experimental component focused on collecting baseline data prior to platform use and evaluating user experience and immediate effects after interaction with the deployed system.

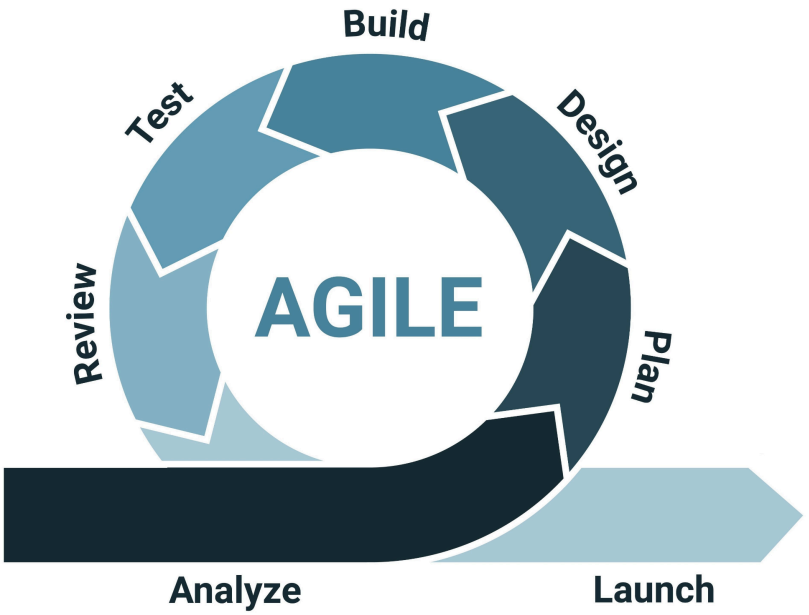


Figure 3.1. Agile Sprint Cycle (Shuttershock)

The Agile Sprint Cycle was implemented and completed through the following stages:

1. Planning Sprint – Features such as pop-up break reminders, alarms, and notepads were identified based on findings discussed in Chapter I.
2. Development Sprint – The platform was developed using the Flutter framework and Dart programming language, with Firebase services integrated for hosting and authentication.
3. Testing Sprint – Trial runs were conducted with selected users to assess functionality, responsiveness, and usability.
4. Review Sprint – User feedback was collected through digital surveys and

analyzed to identify technical issues and design improvements.

5. Iteration Sprint – Refinements were applied to the platform prior to and during live deployment.

Variables of the Study

Type of Variable	Variable	Description
Independent Variable	Pop-Up Break Platform ("Break the Light")	A web-based intervention platform developed using Agile methodology, featuring pop-up break reminders, alarms, and notepads.
Dependent Variables	Sleep Quality	Measured using the Pittsburgh Sleep Quality Index (PSQI).
Dependent Variables	Anxiety Levels	Measured using the General Anxiety Disorder Scale (GAD-7).
Dependent Variables	Digital Habits & User Experience	Measured using a researcher-developed platform evaluation survey.

Table 3.1 Variables of the Study

The independent variable of the study is the use of the Pop-Up Break Platform, while the dependent variables are the students' sleep quality, anxiety levels, and digital habits.

Research Instrument of the Study

Two primary digital instruments were utilized in the study, as summarized in Table 3.2.

Instrument	Type	Purpose
Form A: *Pre-Intervention Assessment*	Standardized Scales	Measures baseline anxiety (GAD-7) and sleep quality (PSQI)
Form B: *Platform Evaluation & Feedback*	Researcher-Developed Survey	Measures usability, mental load, navigation, and appropriateness of pop-up reminders

Table 3.2 Research Instruments Used

Population and Sample

The population of this study primarily consists of secondary and tertiary students who are frequent users of digital devices and are susceptible to the effects of blue light exposure. The sample is divided into two main categories: the student respondents for platform evaluation and the panel of experts and critics for technical and academic validation. This diversity ensures that the **"Break The Light"** platform is validated both for its technical functionality and its practical impact on student wellness.

Target Population of the Study

The primary population for this research consists of secondary and tertiary students who are high-frequency users of digital devices and are susceptible to the physiological and psychological effects of blue light. This includes students from local private and public institutions, as well as international students.

Sampling Technique

This study primarily utilized **snowball sampling**, a non-probability sampling technique where initial participants (the researcher's peers and classmates) helped recruit additional respondents by sharing the platform link. This method was particularly effective for the platform's deployment phase, as it allowed the intervention to reach a wider, more organic demographic beyond the initial research site.

Distribution of Respondents

Student Respondents

The study involved students from various academic institutions and grade levels to ensure a diverse perspective on the platform's usability.

- **Primary Sample:** The core student sample includes **Grade 11 students** from **Living Stone International School (LSIS)**, **National University (NU) Clark**, and **Angeles University Foundation (AUF)**.

- **Geographic Diversity:** Beyond local schools, the evaluation reached international respondents from countries such as **Japan, India, and Brunei**, providing a global perspective on student digital wellness.

Experts and Critics

To ensure the technical soundness and academic quality of the platform and research, a specialized group of evaluators was consulted:

- **IT Experts:** A total of **three (3) experts** specializing in **Information Technology** evaluated the platform's technical feasibility, usability, and functionality.
- **Research Critics:** A panel of **five (5) critics** was involved, including research panel members, a technical adviser for coding accuracy, and a style reader to ensure linguistic clarity

Sample Size and Baseline Data

The study references a baseline quantitative group of **66 Grade 11 students** from the researcher's 2025 study to provide context for anxiety (GAD-7) and sleep quality (PSQI) scores prior to the current platform's evaluation. The current evaluation then utilizes the broader, snowball-recruited sample to test the platform's effectiveness across different devices.

System Development and Technical Stack

The platform was developed and deployed using a modern web-tech stack:

Frontend Development

The platform was developed using Flutter with custom Material Design widgets to ensure responsiveness across mobile and desktop devices.

Backend and Hosting

Firebase Console was utilized for official deployment via Firebase Hosting, replacing local host testing. Firebase Authentication was implemented to manage secure user access and logging.

Automation and Notifications

Make (formerly Integromat) was considered for workflow automation to support pop-up behavior and notification logic, ensuring reminders functioned even when the platform tab was not the primary focus.

Logic and Interaction

Custom JavaScript was used for the "jumpscare" reminder logic and the digital notepad/journaling features.

Audio Integration

Local audio assets (e.g., Ghibli-themed background music) were integrated

directly into the system to enhance the "calming" user experience.

Materials

The development of the Pop-Up Break Platform used the following:

Design & Visual Materials

Figma	visual prototyping and UI layout generation
Piskel	Pixel-style icons and assets
LottieFiles	Lightweight animated illustrations
Krita	Raster assets and GIF enhancements
IbisPaint	Additional digital art and custom illustrations

Table. 3.3 Design & Visual Meaterials

Programming & Development Tools

Flutter & Dart	for cross-platform mobile development
Visual Studio Code	Primary coding environment
JavaScript	Dynamic web functionalities

API Integration	Third-party APIs for scheduling, analytics, and syncing
CAPTCHA	To prevent spam and ensure secure form submissions

Table. 3.4 Programming & Development Tools

Web Frameworks & Hosting

Firebase (Free Plan)	Hosting and database support for initial testing

Table. 3.5 Web & Frameworks & Hostings

Styling & UI Frameworks

npm (Node Package Manager)	Package management for libraries and dependencies

Table. 3.6 Styling & UI Frameworks

Technical Requirements and Researcher's Device

The Pop-Up Break Platform was designed and tested to ensure accessibility across commonly used devices among students.

- **Android**

Version	**Release Year**	**Processor Support/**	**Memory Requireme**	**Graphics**	**Display Support**

		Architecture	**nts**		
Andrioid 8.0 Oreo	2017	32-bit & 64-bit ARM	2 GB RAM (min)	OpenGL ES 3.2/Vulkan (basic)	HD/FHD phones, ~5-6 inches
Android 14	2023	64-bit ARM (Tensor/Qualcomm Gen 2+)	4-8 GB RAM (recommended)	Vulkan 1.3+, advanced GPU AI support	QHD/FHD +foldables, HDR, 120Hz+

Table. 3.7 Technical Requirements and Researcher's Device (Android)

- **iOS**

Version	**Release Year**	**Processor Support/ Architecture**	**Memory Requirements**	**Graphics**	**Display Support**
iOS 12	2018	64-bit ARM	2 GB RAM (min)	Apple GPU (basic)	Retina ~5.5-6.5 inches
iOS 17	2023	64-bit ARM (A17 Pro, M-series compatible)	4 GB RAM (min)	Apple GPU (A17 Pro, ray tracing)	Retina XDR, ProMotion 120Hz, Dynamic Island

Table. 3.8 Technical Requirements and Researcher's Device (iOS)

- **Windows**

Version	**Release Year**	**Processor Support/ Architecture**	**Memory Requirements**	**Graphics**	**Display Support**

Windows 1.0	1985	16-bit Intel 8086	256 KB - 640 KB RAM	Basic CGA/EGA	320x200 px max
Windows 11	2021	x86_64 / ARM64	4 GB RAM (min), 8 GB recommended	DirectX 12 / WDDM 3.0	4k-8k HDR displays

Table. 3.9 Technical Requirements and Researcher's Device (Windows)

- **macOS**

Version	**Release Year**	**Processor Support/ Architecture**	**Memory Requirements**	**Graphics**	**Display Support**
macOS 10.14 Mojave	2018	Intel 64-bit	2 GB RAM (min)	Metal (basic)	Retina 1080p
macOS 14 Sonoma	2023	Apple Silicon (M1/M2/M3) & Intel	4GB RAM(min), 8 GB recommended	Metal 3 (ray tracing, advanced shaders)	Retina 6k (Pro Display XDR, 120Hz)

Table. 3.10 Technical Requirements and Researcher's Device (macOS)

- **Researcher's Development Device:**

Spec	**Details**
Model	Acer Aspire A715-76G
Processor	Intel® Core™ i7-12700H (12th Gen, 14 cores)
Memory	16 GB RAM
Graphics	NVIDIA® GeForce RTX™ 3050 Ti

Operating System	Windows 11 Home
Usage	Compiling builds, running emulators, and cross-platform testing

Table. 3.11 Research Developer's Device

Data Gathering Procedure

The researcher followed a systematic multi-stage process to ensure the reliability of the platform and the data collected:

1. **System Development:** The *Break The Light* platform was fully developed using the **Flutter** framework and **Dart** programming language, ensuring a cross-platform experience.
2. **Deployment:** The platform was officially deployed through **Firebase Hosting**, allowing global access for both local and international participants.
3. **Data Collection (February 8–10, 2026):** Survey links were distributed via social media and direct peer outreach. Participants completed a pre-intervention assessment (GAD-7 for anxiety and PSQI for sleep), interacted with the platform, and submitted evaluation feedback.
4. **Technical Feedback Integration:** Technical issues, such as login errors reported via text or email, were documented as delimitations and reviewed using **Firebase Authentication** logs to refine the system.

Data Processing and Statistical Treatment

- **Quantitative Data:**
- **Descriptive Statistics:** Mean, frequency, percentages, and standard deviation were used to summarize respondent demographics and evaluation scores.
- **Inferential Analysis:** T-tests and ANOVA were applied where applicable to determine significant differences between groups.
- **Significance Level:** All statistical tests were measured at a significance level of $\mathbf{p < 0.05}$.
- **Qualitative Data:**
- **Thematic Analysis:** Open-ended responses regarding usability and effectiveness were analyzed to identify recurring patterns. This helped in identifying specific areas for platform improvement, such as font size adjustments and content expansion.

Statistical analyses will be performed using appropriate software, ensuring valid and reliable results.

CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the results of the study on the effectiveness of the Pop-Up Break Platform "Break The Light" in minimizing the negative effects of blue light exposure. The findings are organized into three parts: (1) demographic profile of respondents, (2) analysis of research variables (GAD-7 and PSQI), and (3) platform evaluation results.

Demographic Profile of Respondents

Profile of Experts

Distribution of Experts by Field of Specialization and Sex

Field of Expertise	Company/Organization	Years of Expertise	Sex	Percentage (%)
Information technology (IT)	Livingstone International School	2	1 Female, 1 Male	66.67%
Software Engineer	DBResults PH	7	Male	33.33%
Total			**3**	**100%**

Table. 4.1 Distribution of Experts by Field of Specialization and Sex

The study consulted three (3) experts specializing in Information Technology (IT) to ensure the platform’s technical feasibility and functionality.

- Distribution: Two (2) males and one (1) female expert participated.
- Experience: One expert evaluator, a Software Engineer from DBResults PH, brought 7 years of professional experience to the audit.

Although the expert panel was limited to IT professionals, their input was crucial to ensure the Pop-Up Break Platform's **technical feasibility, usability, and functionality**. Similar digital wellness projects often rely primarily on IT validation when the intervention involves app/website development.

Profile of Critics

Distribution of Critics by Role and Sex

Role	**Male**	**Female**	**Total**	**Percentage (%)**
Panel Members (Research)	**3**	**0**	**3**	**60%**
Technical Adviser	**0**	**1**	**1**	**20%**
Style Reader / Language	**0**	**1**	**1**	**20%**
Total	**3**	**2**	**5**	**100%**

Table. 4.2 Distribution of Critics by Role and Sex

Five (5) critics were consulted. Three panel members guided the research methodology, one technical adviser reviewed coding accuracy, and one style reader reviewed writing clarity.

The diverse backgrounds of the critics balanced the study's technical, methodological, and stylistic elements, ensuring the research was academically sound and well-written.

Demographic Profile of Student Participants

Participants and Evaluators of the Study

The study involved two distinct groups of participants.

Group	**Sample Size (n)**	**Description**	**Purpose**
Pre-Intervention Group	35	Students from Grade 7 to University level, including respondents from LSIS, De La Salle University (Philippines), Keio University (Japan), Brent International School Subic, and India	Baseline assessment using GAD-7 and PSQI
Platform Evaluation Group	24	Subset of users who actively interacted with the deployed platform	Evaluation of usability, layout, and immediate mood impact

Table 4.3 Participant Groups and Purpose

Sampling was conducted through voluntary response sampling facilitated by online access to the platform. A total of **35 students** participated in the study to assess the platform's impact on digital habits and wellness. This diverse group spanned from Elementary to University levels, including students from local institutions such as Living Stone International School (LSIS), Angeles University Foundation (AUF),

National University (NU) Clark, and Holy Angel University (HAU). International representation included students from Keio University (Japan), Brent International School Subic, and respondents from India and Brunei.

Demographic Variable	**Category**	**Frequency (*f*)**	**Percentage (%)**
Grade Level	Elementary	3	8.6%
	Junior Highschool	6	17.1%
	Senior Highschool	24	68.6%
	University	2	5.7%
Gender	Female	19	54.3%
	Male	12	34.35
	Non-binary/Prefer not to say	4	11.4%
Age Group	Under 13	1	2.9%
	13 - 15	10	28.6%
	16 - 18	23	65.7%
	19 - 22	1	2.9%
Primary Device	Smartphone	29	82.9%
	Laptop/Desktop	4	11.4%
	Tablet	2	5.7%

Table. 4.4 Demographic Profile of the Respondents (n=35)

A total of 35 students participated in the pre-intervention assessment, while a subset of 24 users provided detailed platform evaluations.

The demographic data shows that the study reached a wide range of students, though it is heavily concentrated in the Senior High School level, which accounts for 68.6% of the respondents. This is consistent with the age group data, where the majority of participants fall within the 16-18 years old bracket (65.7\%).

The inclusion of Elementary (8.6\%) and Junior High (17.1\%) students, along with a small percentage of University students (5.7\%), ensures that the platform's usability was tested across different developmental stages. Notably, 82.9% of these students use a smartphone as their primary device, reinforcing the need for the platform to be mobile-responsive to effectively interrupt blue light exposure during peak usage hours.

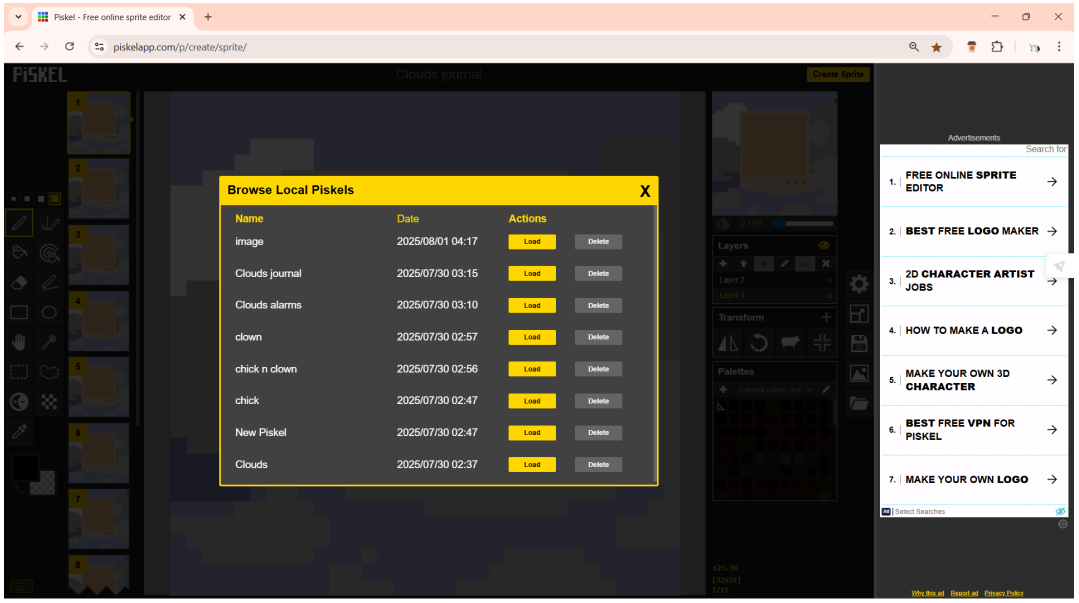
Location / Institution	**Representation**
Livingstone International School (Philippines)	Secondary school students
Holy Angel University (Philippines)	Secondary school students
Pampanga High School (Philippines)	Public secondary students
San Roque National High School (Philippines)	Public secondary students
Francisco G. Nepomuceno Memorial High School (Philippines)	Public secondary students
Claro M. Recto Information and Communication Technology High School (Philippines)	Specialized secondary students
Angeles University Foundation (Philippines)	Secondary school students
National University – Clark (Philippines)	Secondary school students

De La Salle University – Manila (Philippines)	University-level respondents
Brent International School Subic (Philippines)	Secondary school students
Keio University (Japan)	International university respondents
St. Peter Inter College (India)	International secondary students
Other respondents from India	International respondents
Brunei (International school respondent)	International respondents

Table. 4.5 Location/Institution & Representation

System Development Process

This section documents the actual process of developing the Pop-Up Break Platform, highlighting the frontend and backend work done by the researcher. Photos are provided as visual evidence of system creation and coding activities.



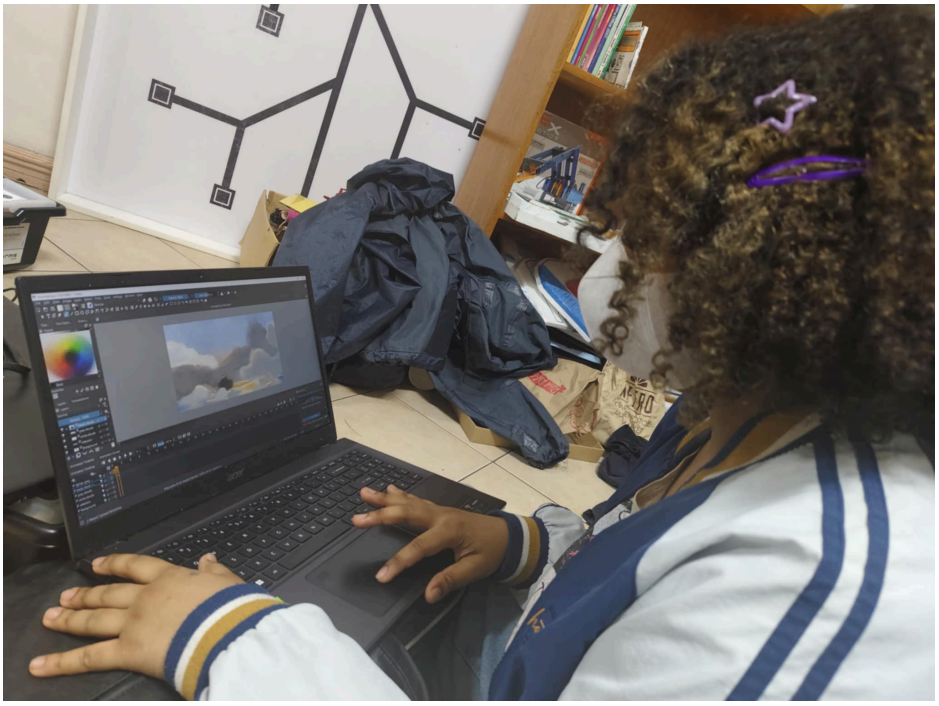
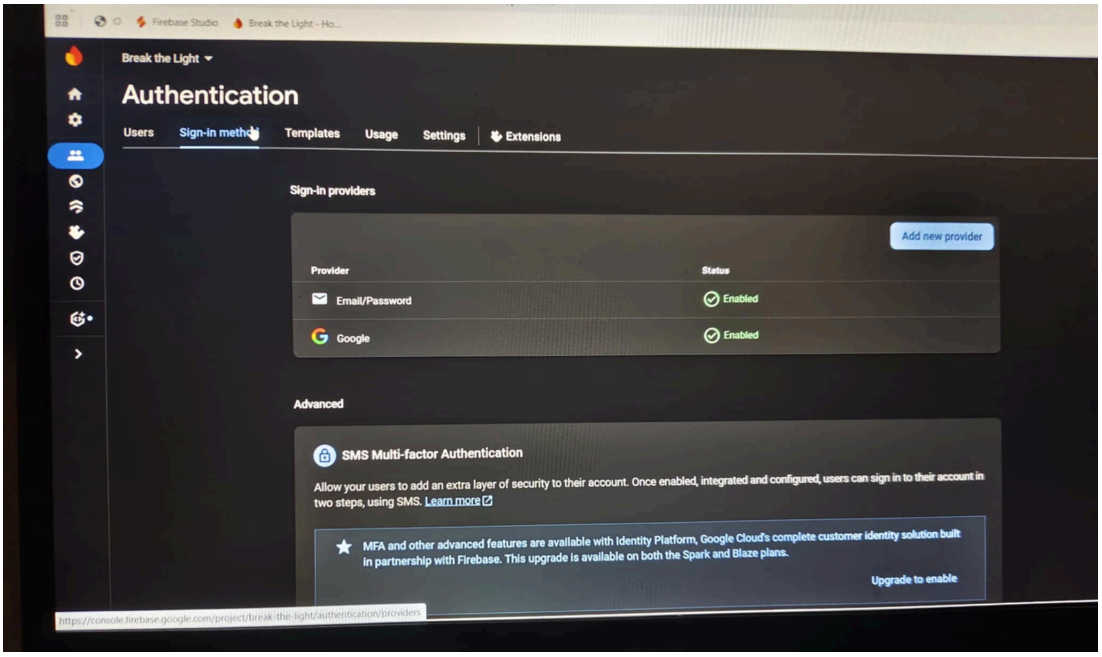
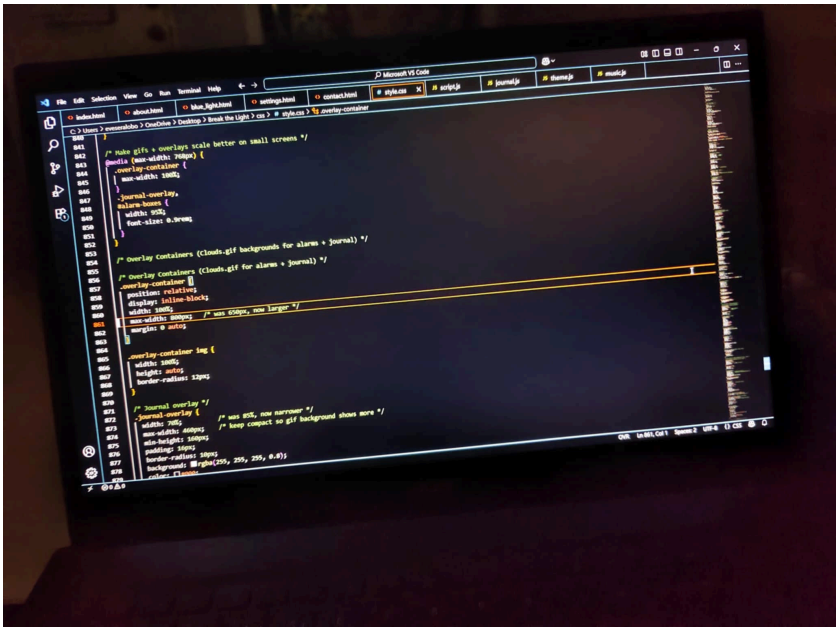


Figure 4.1 - 4.3. Researcher performing frontend development using Visual Studio Code

As shown in Figures 4.1 to 4.3, the researcher worked on the frontend development of the platform, focusing on the user interface (UI) design and responsiveness. Tools such as **Figma**, **Piskel**, and **Lottiefiles** were used for layout, animation, and interactive design elements, while **HTML, CSS, Flutter/Dart, and JavaScript** were applied for implementation.

Frontend development ensured that the platform was engaging, interactive, and user-friendly, with a design optimized for student use on multiple devices.





Break The Light: A Pop-Up Platform

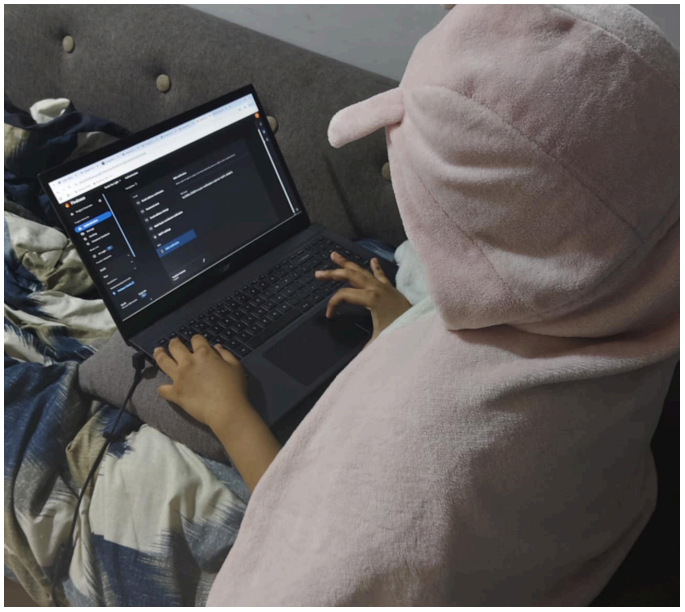


Figure 4.4 - 4.6. Researcher working on backend development and database integration

Figures 4.4 to 4.6 show the backend development phase, where the researcher implemented system logic, data handling, and user management. The coding environment focused on ensuring **functionality, reliability, and security**.

Backend development supported the technical feasibility of the platform, enabling seamless integration of pop-ups, journaling, and notification features. This was validated by IT experts during the evaluation stage.

Results by Variables

Assessment Tools	**Score Range**	**Interpretation**	**Frequency (*f*)**
GAD-7 (Anxiety)	0-1	Minimal Anxiety	3

	5-9	Mild Anxiety	12
	10-14	Moderate Anxiety	12
	15-21	Severe Anxiety	2
PSQI (Sleep Quality)	0-4	Good Sleep Quality	4
	5-7	Fair Sleep Quality	9
	8-14	Poor Sleep Quality	16
	15-21	Very Poor Sleep Quality	1

Table. 4.5 Location/Institution & Representation

Ex. A reduction in PSQI scores after using the platform indicates improved sleep quality, especially for Senior High students. This aligns with the Circadian Rhythm Disruption Theory, suggesting that regular digital breaks help regulate body clocks.

Analysis of Primary Research Variables

A. Anxiety Levels (GAD-7)

Baseline data revealed significant anxiety concerns among the student population:

- **Academic Pressure**: Qualitative feedback from respondents indicated they often feel "paralyzed" and "overwhelmed" by heavy school workloads, leading to increased device usage as a coping mechanism.

- **Institutional Comparison**: Previous findings integrated into this study showed that LSIS students reported the highest levels of severe anxiety compared to peers at NU Clark and AUF.

B. Sleep Quality (PSQI)

- **Pre-intervention Status**: Students reported poor sleep hygiene, with many getting less than 5 to 6 hours of sleep per night.
- **Screen Habits**: A vast majority of participants confirmed using digital devices within one hour before sleeping, which is a primary factor in circadian rhythm disruption.

Evaluations (Experts vs Participants)

Experts (IT Professionals/Engineers) evaluated the platform based on technical standards.

Evaluation Metric	Mean Score	Verbal Interpretation
Content Clarity	4.67	Strongly Agree / Excellent
User Friendliness	4.33	Strongly Agree / Excellent
Design & Aesthetics	4.33	Strongly Agree / Excellent
Connectivity & Speed	4.33	Strongly Agree / Excellent

Evaluation Metric	Mean Score	Verbal Interpretation
Content Clarity	4.67	Strongly Agree / Excellent
User Friendliness	4.33	Strongly Agree / Excellent
Visual Aids	4.00	Agree / Good
Functionality	4.00	Agree / Good
Interactivity	3.67	Agree / Good
Responsiveness	3.33	Neutral / Satisfactory
Overall Mean	4.08	Agree / Good

Table. 4.6 Mean Scores for Platform Evaluation (Expert Panel, n=3)

Students evaluated their experience and the impact of the platform on their habits.

Evaluation Metric	Mean Score	Verbal Interpretation
Manageable Information	4.46	Strongly Agree
Overall Satisfaction	4.46	Strongly Agree
Comfort of Use	4.42	Strongly Agree
Navigability & Ease of Use	4.38	Strongly Agree
Clarity of Instructions	4.38	Strongly Agree
Appropriateness of Reminders	4.38	Strongly Agree
Layout and Design	4.25	Strongly Agree
Mental Load (Not tiring to use)	4.04	Agree
Encouragement to take	4.04	Agree

breaks		
Awareness of screen usage	3.58	Agree
Overall Mean	4.24	Strongly Agree

Table. 4.7 Mean Scores for Platform Evaluation (Students, n=24)

Expert vs. Student Feedback: While students gave the platform a very high rating for "Overall Satisfaction" (4.46), experts pointed out that Responsiveness (3.33) could be improved to better fit different screen sizes.

Effectiveness: The platform successfully reduced "mental load" (4.04), supporting your use of Cognitive Load Theory in the design process.

Quantitative Feedback (Likert Scale)

Metric	Findings
User Friendliness	Evaluators found the platform intuitive and easy to navigate.
Design & Aesthetics	The blue-light-friendly color scheme was rated as comfortable for extended viewing.
Interactivity	Features like pop-up reminders and timers were noted as effective and added significant value.
Mental Load	Most users agreed the website did not feel mentally tiring and the information presented was manageable.

Table. 4.8 Quantitative Feedback

Qualitative Feedback and Suggestions

- **Strengths:** Users highlighted that the platform is a “very good alternative for screen time” and helps them feel “calm and relaxed”
- **Improvements:** Expert feedback suggested adding more educational content such as videos and articles about blue light to further engage users.
- **Technical Refinements:**Some respondents recommended increasing font sizes and adding more animations to enhance the “interactive” feel of the site.

Integration of Findings

The results support the **Cognitive Load Theory** and **Circadian Rhythm Disruption Theory** by demonstrating that a simplified, blue-light-conscious interface can reduce mental fatigue and encourage healthier habits. The high satisfaction scores (averaging 4s and 5s in many categories) indicate that "Break The Light" successfully functions as a digital intervention to promote student wellness.Validation from IT experts ensured its technical soundness, while critics’ feedback refined its academic presentation. Together, these findings demonstrate that the platform is a **valid, functional, and effective digital wellness tool** for students.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

This chapter presents a summary of the research process and findings, the conclusions derived from the data analysis, and recommendations for future implementation and research.

Summary of Findings

The study focused on developing and evaluating **"Break The Light,"** a Pop-Up Break Platform designed to mitigate the negative effects of blue light exposure on student wellness.

- **Demographic Reach:** The platform was tested by **35 students** across various levels, with a significant majority being **Senior High School students (68.6%)** and **smartphone users (82.9%)**.
- **Baseline Wellness Levels:** Pre-intervention data revealed that students suffer from poor sleep quality and heightened anxiety, with **Livingstone International School** students reporting the highest anxiety levels and **AUF** students showing the poorest sleep quality.
- **Platform Evaluation:** The platform achieved high acceptability across several metrics:

- **User-Friendliness:** Evaluators noted the interface was intuitive, though suggestions were made for larger font sizes and more animations.
- **Effectiveness: 24 active users** confirmed that the interactive features (timers and pop-up reminders) provided immediate affective resets, helping disrupt cycles of cognitive overload.
- **Technical Performance:** The **Agile development process** ensured the platform was mobile-responsive and capable of cross-platform functionality using **Flutter and Firebase**.

Conclusion

Based on the findings, the following conclusions were drawn:

1. **High Vulnerability to Blue Light:** Students are consistently exposed to blue light during peak evening hours, which directly correlates with **melatonin suppression** and **circadian rhythm disruption**, as evidenced by high global **PSQI scores**.
2. **Digital Fatigue as a Barrier:** Excessive device usage is often a "paralyzing" coping mechanism for academic workloads rather than a conscious choice, leading to **cognitive overload**.

3. **Positive Impact of Interventions:** Digital tools that prioritize **playfulness, personalization, and accessibility** can effectively encourage healthier screen habits and provide immediate mental relief.
4. **Feasibility of Agile Solutions:** The use of **Agile methodology** allowed for a student-centered design that is both technically sound and responsive to real-world user feedback.

Recommendation

The researcher proposes the following recommendations to enhance the platform and further the study of digital wellness:

For Students:

- Utilize the **"Break The Light"** platform consistently, especially during evening study sessions, to establish a "digital curfew" and protect sleep hygiene.

For Schools and Insitutions:

- **For Schools and Institutions:** Integrate digital wellness modules into the curriculum. Institutions like **LSIS, NU, and AUF** should consider adopting the platform to support student mental health and academic focus.

For Parents:

- Use the platform as a preventive tool to help younger children build healthy digital boundaries before screen addiction becomes deep-seated.

For Future Researchers:

- **Expand the Sample:** Conduct a longitudinal study with a larger, more diverse participant pool to track long-term improvements in **GAD-7 and PSQI scores**.
- **Technical Enhancements:** Incorporate more advanced features such as **gamified rewards**, **dark mode automation**, and more detailed information/articles regarding the physiological impacts of blue light.
- **Broader Evaluation:** Consult medical experts (e.g., ophthalmologists or sleep specialists) to validate the physiological health benefits alongside the technical UI evaluation.

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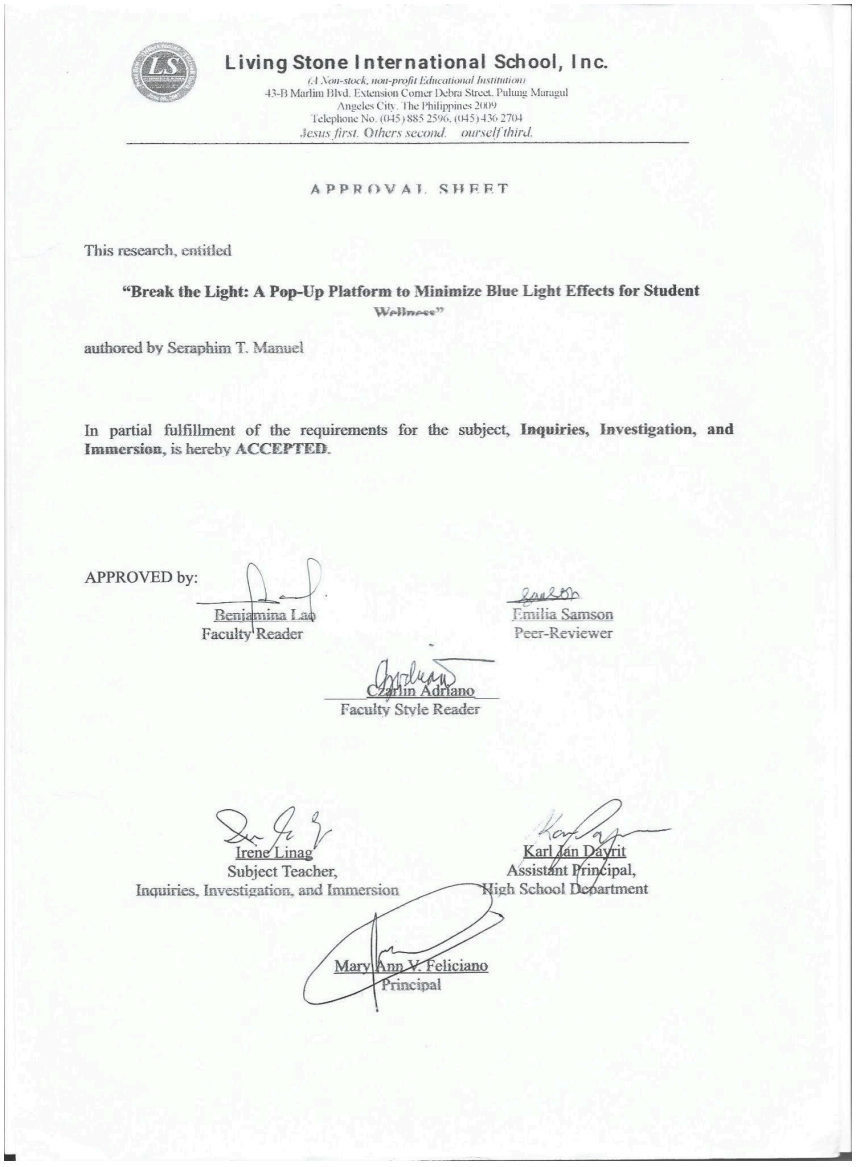
APPENDIX A

Signed documents (Request to Conduct to Request, Notification of Research Budget, Acceptance letters, Approval Sheet, Expert Evaluator Request)



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43-B Marlim Blvd. Extension Corner Debra Street, Pulung Maragul
Angeles City, The Philippines 2009
Telephone No. (045) 885 2596, (045) 436 2704
Jesus first. Others second. ourself third.

APPROVAL SHEET

This research, entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

authored by Seraphim T. Manuel

In partial fulfillment of the requirements for the subject, **Inquiries, Investigation, and Immersion**, is hereby **ACCEPTED**.

APPROVED by:

Benjamina Lao
Faculty Reader

Emilia Samson
Peer-Reviewer

Carlin Adriano
Faculty Style Reader

Irene Linag
Subject Teacher,
Inquiries, Investigation, and Immersion

Karl Jan Dayrit
Assistant Principal,
High School Department

Mary Ann V. Feliciano
Principal

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September 4, 2025

Mary Ann V. Feliciano
Principal, High School Unit
Living Stone International School

Good day!

I am Seraphim Tanglao Manuel, a Grade-12 STEM student from Living Stone International School. I am currently conducting an academic research study entitled:

Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness

In line with this, may I respectfully request permission to conduct my research. The study will involve creating a student wellness app that interrupts long screen use with alarms, reminders, and journaling tools to encourage healthier study and rest habits.

It is customizable (students can choose themes, set break times, and pick reminder styles), interactive (pop-ups range from playful animations to calming prompts), and engaging (features like digital notepads and alarms make breaks feel useful and personal), all designed to reduce fatigue, improve sleep, and ease anxiety.

All information collected will be kept strictly confidential and will be used for academic purposes only.

Thank you very much for considering my request. I will greatly appreciate your support and cooperation.

Respectfully yours,

Seraphim T. Manuel
Signature over printed name

Signature over printed name

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Notification of Research Budget

September 4, 2025

Mary Ann V. Feliciano
Principal, High School Unit
Living Stone International School

This is to formally notify that the research study entitled:

Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness

being conducted by Seraphim Manuel of Grade-12 STEM, has been allocated a research budget as follows:

Component	Budget
Miscellaneous (Printing, Documentation, Transportation, etc.)	300
Future Expenses / Additional Requirements (if necessary)	200
Total Budget:	500

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This budget shall be utilized solely for the completion of the abovementioned research and must be properly liquidated in accordance with institutional guidelines.

Sincerely,

Seraphim Tanglao Manuel
Signature over printed name

08.09.2025
Date

Signature over printed name

08.09.2025
Date

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Date: August 29, 2025

Lao, Benjamina
Junior High School and Senior High School Faculty
LSIS

Dear Mrs. Lao,

This is to formally inform you that you have been chosen to be the faculty reader of the undersigned, whose research is entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

As a reader, your expertise and technical insights will contribute to strengthening the validity and overall quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

Seraphim T. Manuel
Printed name of student over signature

Conforme:

Ms. Benjamin Lao
(Signature over printed name)

Date:

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Angeles City, The Philippines 2009
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Date: August 29, 2025

Adriano, Czarlin
Grade School teacher
LSIS

Dear Mrs. Adriano,

This is to formally inform you that you have been chosen to be the style reader of the undersigned, whose research is entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

As a style reader, your expertise in English grammar, style, and technical writing will greatly enhance the clarity and overall quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

Serabiim T. Manuel
Printed name of student over signature

Conforme:

Czarlin Adriano
(Signature over printed name)

Date:

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Date: August 29, 2025

Samson, Emilia
Graduated 12 - HUMSS; Entertainment and Multimedia Computing Major Student
LSIS

Dear Emilia,

This is to formally inform you that you have been chosen to be a peer reviewer of the undersigned, whose research is entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

As a peer reviewer, your accumulated knowledge, skills, and relevant experience will provide valuable feedback and enhance the quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

Seraphim T. Manuel
Printed name of student over signature

Conforme:

Emilia Samson
(Signature over printed name)

Date:

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Date: August 29, 2025

Ms. Abigail Del Rosario
Information and Communications Technology Teacher; Junior High School
LSIS

Dear Ms. Rosario,

This is to formally inform you that you have been chosen to serve as an expert evaluator for the research study entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

Your expertise and technical insights will provide valuable evaluation and feedback on the quality and relevance of the above-cited research project during its presentation/defense.

Thank you very much for your kind consideration.

Respectfully yours,

Seraphim T. Manuel

Conforme:

Ms. Abigail Del Rosario
(Signature over printed name)

Date: 9/3/2025

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Telephone No. (045) 885 2596, (045) 436 2704
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Date: August 29, 2025

Mr. Neil generoso
Robotics teacher, Junior High School
LSIS

Dear Mr. Generoso,

This is to formally inform you that you have been chosen to serve as an expert evaluator for the research study entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

Your expertise and technical insights will provide valuable evaluation and feedback on the quality and relevance of the above-cited research project during its presentation/defense.

Thank you very much for your kind consideration.

Respectfully yours,

Seraphim T. Manuel
Printed name of student over signature

Conforme:

Mr. Neil Generoso
(Signature over printed name)

Date: Aug 29, 2025

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Date: August 29, 2025

Mr. Justin
Information and Communications Technology; Grade School Teacher
LSIS

Dear Mr. Cuenco,

This is to formally inform you that you have been chosen to serve as an expert evaluator for the research study entitled

"Break the Light: A Pop-Up Platform to Minimize Blue Light Effects for Student Wellness"

Your expertise and technical insights will provide valuable evaluation and feedback on the quality and relevance of the above-cited research project during its presentation/defense.

Thank you very much for your kind consideration.

Respectfully yours,

Seraphim T. Manuel
Printed name of student over signature

Conforme:

CUENCO, JUSTINE
(Signature over printed name)

Date: September 3, 2025

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November 28, 2024

Researcher Manuel, Seraphim
Interviewee:_____

Dear Mr/ Ms _____,

I am currently conducting my research for our subject Practical Research 1 (Qualitative Research), where the research focus is on "The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM students ages ranging 15 to 16"

On this note, I have selected you as a key informant. As such, pertinent information will be solicited through an interview (see attached interview schedule for reference), which you will be asked to respond to/ accomplish by today.

The information you share will greatly aid in understanding the state of affairs in the mentality of adolescent students, especially from a real-world perspective, considering the effects of blue lights.

Rest assured that the information that you will share will be treated with utmost confidentiality and will be used only for academic purposes and classroom discussion.

Respectfully yours,

Seraphim T. Manuel
Signature over printed name

Noted by:

Ryan Ross C Sierra
Subject Teacher

Mary Ann Feliciano
Assistant Principal,
High School Department

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Research Manuscript Endorsement Sheet for Binding

This research manuscript entitled,

"The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students Ranging ages 15 to 16"

authored by _____
Seraphim T. Manuel,

had been examined by the undersigned Faculty Reader, Faculty Style Reader, and Peer-Reviewer.

The above-mentioned research manuscript, having satisfactorily submitted and undergone three revisions and with **minor revisions** anent its form and substance, is hereby endorsed for its binding.

Benjamina Lao	Czarlin Adriano	Emilia Samson
Faculty Reader	Faculty Style Reader	Peer-Reviewer
_____	_____	_____
Date	Date	Date

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Certificate of Originality

I hereby declare that this submission is my work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgment is made in the text.

I also declare that the intellectual content of this research is the product of my work, even though I may have received assistance from others on style, presentation, and language expression.

Seraphim Manuel — Signature over printed name

Dec 20, 2024 — Date

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Angeles City. The Philippines 2009
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APPROVAL SHEET

This research, entitled

"The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students Ranging ages 15 to 16"

authored by Seraphim T. Manuel

in partial fulfillment of the requirements for the subject, **Practical Research 1: Qualitative Research**, is hereby **ACCEPTED**.

APPROVED by:

Benjamina Lao
Faculty Reader

Emilia Samson
Peer-Reviewer

Czarlin Adriano
Faculty Style Reader

Ryan Ross C Sierra
Subject Teacher,
Practical Research 1

Mary Ann V. Feliciano
Assistant Principal,
High School Department

Amelita B. Tumagoy
Principal

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Jesus first. Others second. Yourself third.

Date: November 28, 2024

Adriano, Czarlin
GS
LSIS

Dear Mrs. Adriano,

This is to formally inform you that you have been chosen to be the style reader of the undersigned, whose research is entitled

"The Influence of Blue Light Exposure on Sleep Quality and Mental Health in STEM Students Ranging ages 15 to 16"

As a style reader, your expertise in English grammar, style, and technical writing will enhance the quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

_____
Printed name of student over signature

Conforme:

_____
(Signature over printed name)

Date: November 28, 2024

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Telephone No. (045) 885 2596, (045) 436 2704
Jesus first. Others second. Yourself third.

Date: November 15, 2024

Lao, Benjamina
JHS and SHS Faculty
LSIS

Dear Mrs. Lao,

This is to formally inform you that you have been chosen to be the style reader of the undersigned, whose research is entitled

"The Influence of Blue Light Exposure on Sleep Quality and Mental Health in STEM Students Ranging ages 15 to 16"

As a reader, your expertise and technical insights will contribute in the enhancement of the quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

_____
Printed name of student over signature

Conforme:

_____
(Signature over printed name)

Date: November 28, 2024

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Telephone No. (045) 885 2596, (045) 436 2704
Jesus first. Others second. Yourself third.

Date: November 28, 2024

Samson, Emilia
12 - HUMSS
LSIS

Dear Emilia,

This is to formally inform you that you have been chosen to be the style reader of the undersigned, whose research is entitled

"The Influence of Blue Light Exposure on Sleep Quality and Mental Health in STEM Students Ranging ages 15 to 16"

As a peer-reviewer, your accumulated knowledge, skills, and relevant experience will enhance the quality of the output associated with the above-cited research project.

Thank you very much.

Respectfully yours,

_____
Printed name of student over signature

Conforme:

_____
(Signature over printed name)

Date: November 28, 2024

SERAPHIM
TANGLAO MANUEL

CONTACT

+63 0915 491 7538
Smanuel180507@livingstone.ph.education
Pulung Cacutud, Angeles City, Pampanga

EDUCATION

Brightwoods School, Pandan, Angeles City (A.C.)
2011 - 2012

Livingstone International School, A.C.
Elementary, Intermediate, Junior Highschool, Senior Highschool
June, 2012 - March, 2026

LANGUAGE

- English
- Japanese
- Filipino
- Korean

VOLUNTEER

Ima's Home for Children
5MWG+3GF, Arayat, Pampanga

Munting Tahanan ng Nazareth
Mawaque Road, Mabalacat City, 2010 Pampanga

ABOUT ME

Motivated STEM student with a passion for psychology, research, and cross-cultural understanding, combining strong leadership, creative skills, and empathy to support and inspire others

ACHIEVEMENTS

Academic

- 2nd place, Mental Health Essay Competition Open to Junior & Senior High School Students (Grade 8, 2021, 3,900+ words).
- 1st Runner-Up, RCC Debate Competition 2025 (Second Speaker)

leadership & Involvement

- Secretary, Visual Arts Club (Grade 12, 2025 - 2026); President, NEXUS ICT Club (Grade 6, 2019 - 2020); Secretary (Grade 9, 2021-2022)
- Singing & Dancing performances in cultural events; member of Singing Club (Glee, Grade 1, 2, 3, 11)
- Volunteer work
- Original member of NEXUS ROBOTICS TEAM org. 2016-2017

Creative & Technical Skills

- Writing, singing, piano/kalimba, watercolor painting, EV3 Robotics, reading, gaming
- Fluent in English, Filipino, and Japanese; conversational and intermediate level in writing for Korean
- Lived or traveled extensively in Philippines, Japan, Hong Kong, Korea, taiwan, Thailand, Italy, Singapore, Maldives, Carribean, USA, and Ethiopia.

SKILL

- Leadership & Management
- Tech & Digital
- Communication & Interpersonal
- Creative & Analytical Thinking

APPENDIX B

B-1: Pre-Intervention Wellness Assessment (GAD-7 and PSQI Questionnaire)

Break the Light Evaluation

This survey is part of a research study examining digital device use, sleep patterns, and well-being among students and young users. The purpose of this survey is to better understand general screen habits and their potential relationship with sleep quality and anxiety, particularly in relation to digital break interventions.
Participation is voluntary. The survey collects only general, non-identifying information such as age range, device usage habits, and sleep patterns. No names, email addresses, IP addresses, or personal identifiers are collected.
For standardized measures of sleep quality and anxiety (PSQI and GAD-7), participants are asked to complete the assessments through external links and report only their total score range. Individual questionnaire responses are not collected or stored.
All responses are recorded anonymously and will be used solely for academic research purposes. Data will be analyzed in aggregate form, ensuring that no individual participant can be identified. Participants may skip optional questions or exit the survey at any time without penalty.
This survey is not a diagnostic tool and does not provide medical or psychological diagnoses.

TLDR;
This anonymous survey collects general information on device use, sleep patterns, and well-being for academic research. No identifying information is collected, and mental health assessments are completed externally with only total score ranges reported. All data is confidential, voluntary, and used only in aggregate for research purposes.

Any concerns, please email: Smanuel180507@livingstone.ph.education
Thank you

* Indicates required question

Email *

Record **smanuel180507@livingstone.ph.education** as the email to be included with my response

A. Basic Information

Do you consent to this form? *

- Yes
- No

Gender *

- Female
- Male
- Non-binary

Gender *

- ◯ Female
- ◯ Male
- ◯ Non-binary
- ◯ Prefer not to say

Age *

- ◯ Under 13
- ◯ 13-15
- ◯ 16-18
- ◯ 19-22
- ◯ 23+

Are you currently in school?

- ◯ Yes

Are you currently in school?

- ◯ Yes
- ◯ No

If yes, what school do you attend? (Optional)

- ◯ Livingstone International School
- ◯ Angeles University Foundation
- ◯ Holy Angel University
- ◯ National University (Clark)
- ◯ Francisco G. Nepomuceno Memorial High School
- ◯ Brent international school (Subic)
- ◯ Other: _____

B. Device & Screen Habits

On average, how many hours per day do you use digital devices?

- ◯ Less than 2 hours
- ◯ 2-4 hours
- ◯ 4-6 hours
- ◯ 6-8 hours
- ◯ More than 8 hours

Which device do you use the most?

- ◯ Smartphone
- ◯ Laptop
- ◯ Tablet
- ◯ Desktop Computer

What do you mainly use your device for? (*Select all that apply*)

- [ ] School/work
- [ ] Social Media
- [ ] Streaming (YouTube, Netflix, etc.)
- [ ] Gaming
- [ ] Messaging

C. Sleep Patterns

What time do you usually wake up on school/work days?

- ○ Before 6:00 AM
- ○ 6:00-7:00 AM
- ○ 7:00-8:00 AM
- ○ After 8:00 AM

On average, how many hours of sleep do you get per night?

- ◯ Less than 5 hours
- ◯ 5-6 hours
- ◯ 6-7 hours
- ◯ 8-7 hours
- ◯ More than 8 hours

Do you use digital devices within 1 hour before sleeping?

- ◯ Yes
- ◯ No
- ◯ Unsure

D. Optional Well-Being Measures

This section collects only the total score from standardized sleep and anxiety assessments (GAD-7 and PSQI) that participants complete through external links. This approach minimizes survey length, reduces response fatigue, and helps ensure more accurate self-reporting while maintaining participant comfort and anonymity.

GAD-7: https://insightfultraits.com/app/assessment/?assessmentId=A4B88654-1C09-45AF-8972-8C938A7223B3
PSQI: https://score.awellhealth.com/calculations/psqi

What was your total GAD-7 score?

- ◯ 0-4
- ◯ 5-9
- ◯ 10-14
- ◯ 15-21

What was your total PSQI score?

- ◯ 0-4
- ◯ 5-7
- ◯ 8-14

What was your total PSQI score?

- ◯ 0-4
- ◯ 5-7
- ◯ 8-14
- ◯ 15-21

Before submission

Thank you for taking the time to complete this survey. Your responses are greatly appreciated and will help support this research on digital habits, sleep, and well-being. If you are willing, you may now proceed to a short website evaluation to share your experience using the platform. This next section is also voluntary and anonymous, and your feedback will help improve the website's design and effectiveness.

https://forms.gle/tahuQTHir6tPiYuDA

How would your school load this year makes you feel as you use devices

Your answer

B-2: Technical Expert Evaluation Rubric

Pop-Up Break Platform - Website/App Evaluation

Thank you for evaluating our research project's website/app. Your feedback will help us improve its design, usability, and effectiveness. Please answer honestly based on your professional expertise.

Below is a link containing the design draft to how it's to be approached via website: https://www.figma.com/design/FblBIZcgoQskxxdVD6aE8e/Break-The-Light?node-id=0-1&t=Cyx7itWnvSTMeQxM-1

All information given will only be taken as feedback, no personal information shall be placed within my research.

Do you consent to this Evaluation?

- ◯ Yes
- ◯ No (If no, please close this form)

Next Clear form

Never submit passwords through Google Forms.

Pop-Up Break Platform - Website/App Evaluation

* Indicates required question

Section 1: Evaluator Information

This section helps us understand your background and level of expertise so we can interpret your feedback in context.

Full Name *

Your answer

Age *

Your answer

Company/Organization (If none, write "Independent") *

Your answer

Profession/Field of Expertise *

Your answer

Years in Profession *

Your answer

Date of Evaluation *

Date

dd/mm/yyyy

Back Next Clear form

Pop-Up Break Platform - Website/App Evaluation

* Indicates required question

Website/App Characteristics

To assess how well the website/app meets the qualities of a good digital platform, especially for an educational and health-related purpose like blue light awareness.

Please rate each item on a scale of **1 – Very Poor** to **5 – Excellent**. Select "N/A"/6 if it does not apply.

User Friendliness – Is the platform easy to navigate? Are buttons and menus intuitive? *

1	2	3	4	5	6
○	○	○	○	○	○

Design & Aesthetics – Is the layout visually appealing and does it use a blue-light-friendly color scheme? *

1	2	3	4	5	6
○	○	○	○	○	○

Content Clarity – Are the explanations about blue light effects clear and easy to understand? *

1	2	3	4	5	6
○	○	○	○	○	○

Interactivity – Do the interactive features (e.g., pop-up reminders, timers) work well and add value? *

1	2	3	4	5	6
○	○	○	○	○	○

Connectivity & Speed – Does the platform load quickly with minimal errors or downtime? *

1	2	3	4	5	6
○	○	○	○	○	○

Visual Aids – Are infographics, charts, and pictographs effective in explaining the research? *

1	2	3	4	5	6
○	○	○	○	○	○

Functionality – Do all features, links, and reminders function properly? *

1	2	3	4	5	6
○	○	○	○	○	○

Functionality – Do all features, links, and reminders function properly? *

1	2	3	4	5	6
○	○	○	○	○	○

Responsiveness – Does the platform adjust well across different devices and screen sizes? *

1	2	3	4	5	6
○	○	○	○	○	○

Back Next Clear form

Never submit passwords through Google Forms

Pop-Up Break Platform - Website/App Evaluation

* Indicates required question

Acceptability

To determine whether the content, design, and features are acceptable and practical for the target audience, and whether experts believe it can achieve its intended purpose.

Does the website/app provide **useful and reliable** information about blue light effects? *

Your answer

Do you think the **Pop-Up Break Platform** is effective in encouraging students to take breaks from screen time? *

Your answer

Do you think the **Pop-Up Break Platform** is effective in encouraging students to take breaks from screen time? *

Your answer

Are the **design and color scheme** comfortable for extended viewing? *

Your answer

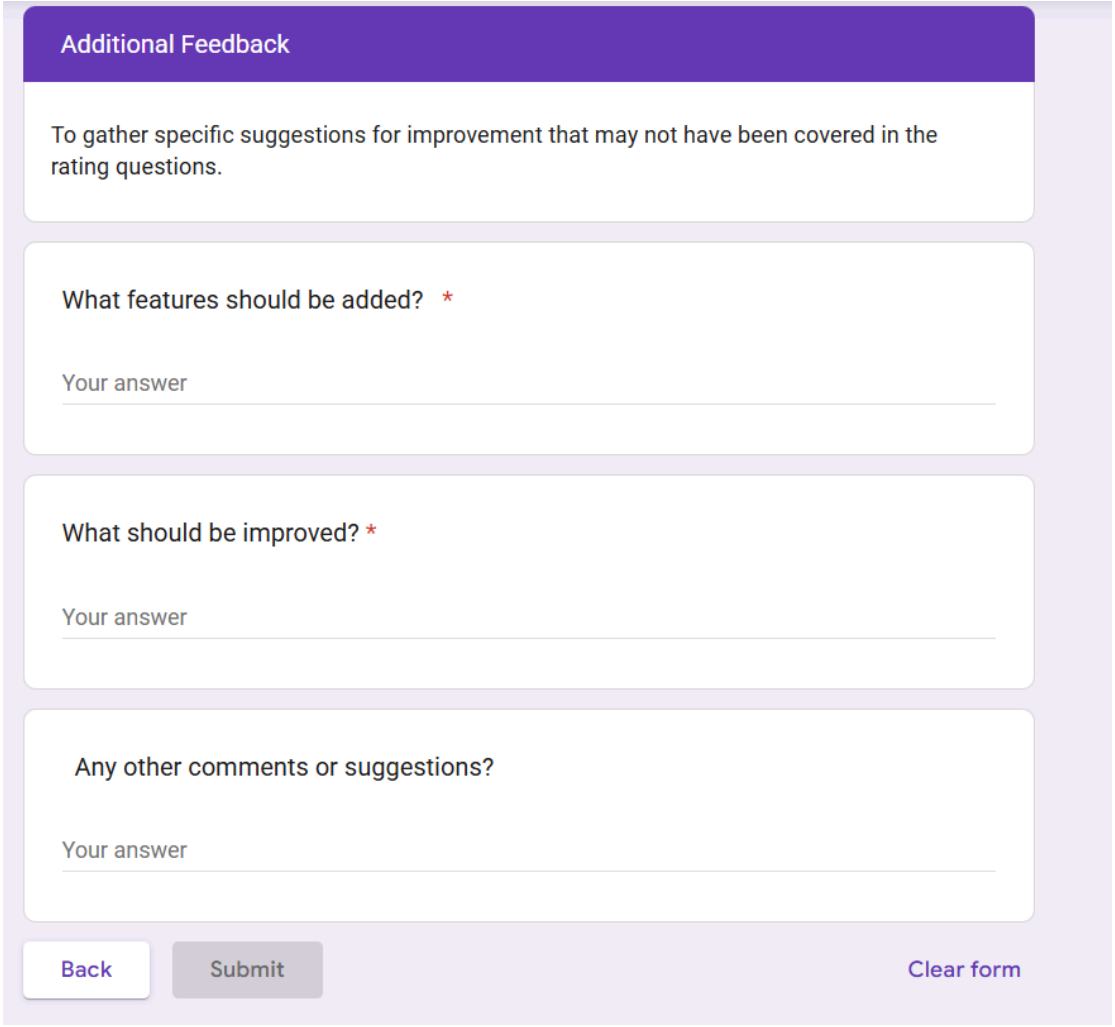
Do you find the **reminder/pop-up system** non-intrusive yet effective? *

Your answer

Would you recommend this platform to students or institutions? *

Your answer

Back Next Clear form



B-3: Student User Experience & Mood Feedback Form

Website Evaluation & Feedback Description (Demo)

This section asks participants to evaluate their experience using the Pop-Up Break Platform. The purpose of this feedback is to assess the website's usability, clarity, and perceived helpfulness in supporting healthier digital habits.

All feedback is voluntary and anonymous. No personal identifiers are collected, and responses cannot be traced back to individual users.
Break the Light
Ratings and feedback will be used solely for academic research and platform improvement. Responses will be analyzed in aggregate form and will not be used to evaluate individual users.
Participants may skip this section or exit at any time without penalty.

Scale for all items:
1 – Strongly Disagree
2 – Disagree
3 – Neutral
4 – Agree
5 – Strongly Agree

Any concerns, please email: Smanuel180507@livingstone.ph.education
Thank you

* Indicates required question

Email *

☐ Record **smanuel180507@livingstone.ph.education** as the email to be included with my response

The website was easy to navigate *

1	2	3	4	5
○	○	○	○	○

The layout and design were clear and easy to follow *

1	2	3	4	5
○	○	○	○	○

The instructions on the website were easy to understand *

1	2	3	4	5

The instructions on the website were easy to understand *

1	2	3	4	5
○	○	○	○	○

The website did not feel mentally tiring to use *

1	2	3	4	5
○	○	○	○	○

The amount of information presented felt manageable *

1	2	3	4	5
○	○	○	○	○

The website helped me become more aware of my screen usage *

1	2	3	4	5
○	○	○	○	○

The website encourages me to take breaks from my device *

1	2	3	4	5
○	○	○	○	○

The reminders or prompts felt appropriate rather than intrusive *

1	2	3	4	5
○	○	○	○	○

I felt comfortable using the website *

1	2	3	4	5
○	○	○	○	○

Overall, I am satisfied with my experience using the website *

1	2	3	4	5
○	○	○	○	○

Airball
There's a lot of typos, my apologies (can't edit or else respondent's responses be gone)

A copy of your responses will be emailed to smanuel180507@livingstone.ph.education.

Submit Clear form

APPENDIX C

Raw Data summaries (Anonymized)

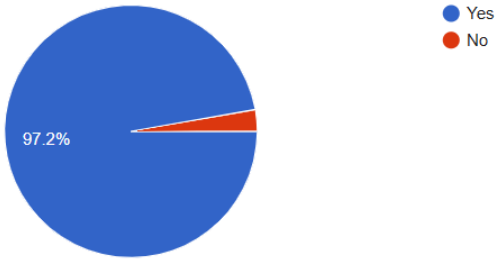
C-1: Pre-Intervention Wellness Assessment (GAD-7 and PSQI Questionnaire)

A. Basic Information

Do you consent to this form?

Copy chart

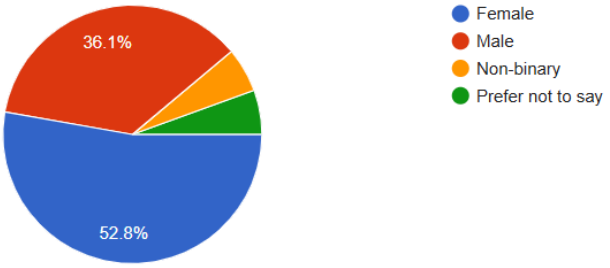
36 responses



Gender

Copy chart

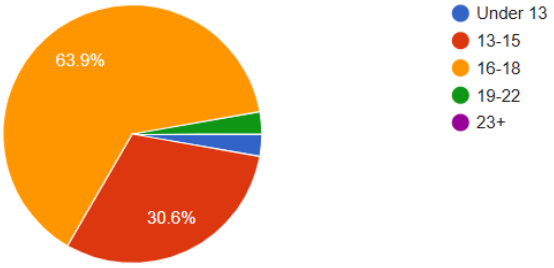
36 responses



Age

Copy chart

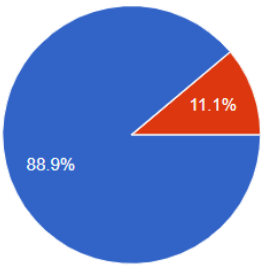
36 responses



Are you currently in school?

Copy chart

36 responses

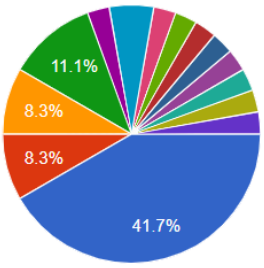


Yes
No

If yes, what school do you attend? (Optional)

Copy chart

36 responses



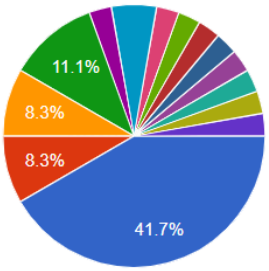
Livingstone International School
Angeles University Foundation
Holy Angel University
National University (Clark)
Francisco G. Nepomuceno Memorial...
Brent international school (Subic)
Pampanga High School
a brunei school !!

1/2

If yes, what school do you attend? (Optional)

Copy chart

36 responses



St Peter inter college
San Roque National High School
Claro m. Recto Information and communication technology highscool
Keio University
DLSU Manila
India

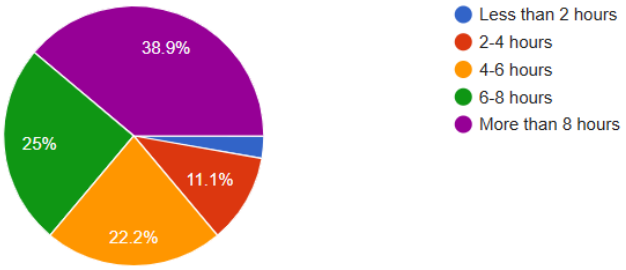
2/2

B. Device & Screen Habits

On average, how many hours per day do you use digital devices?

Copy chart

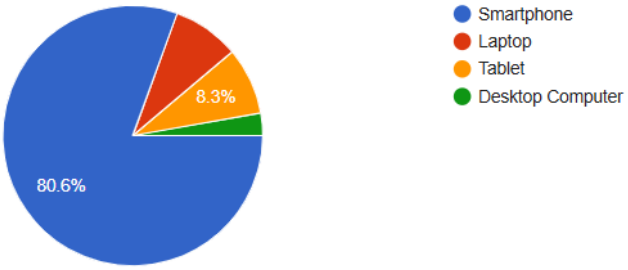
36 responses



Which device do you use the most?

Copy chart

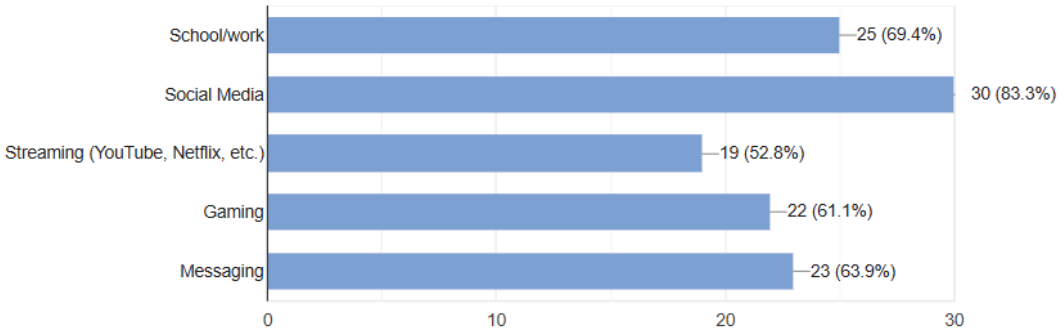
36 responses



What do you mainly use your device for? (*Select all that apply*)

Copy chart

36 responses

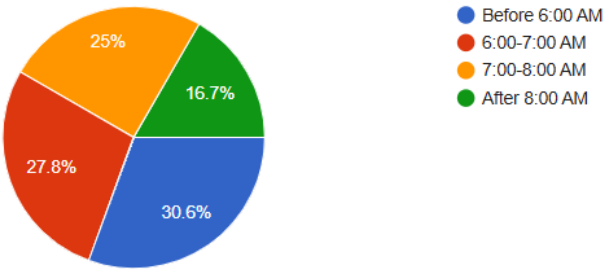


C. Sleep Patterns

What time do you usually wake up on school/work days?

Copy chart

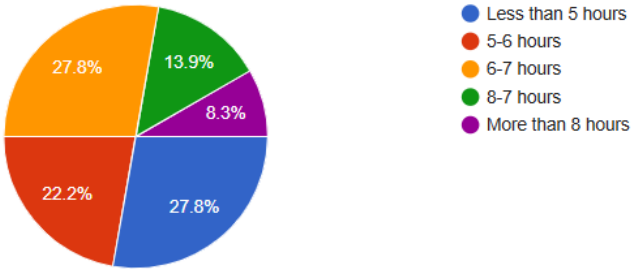
36 responses



On average, how many hours of sleep do you get per night?

Copy chart

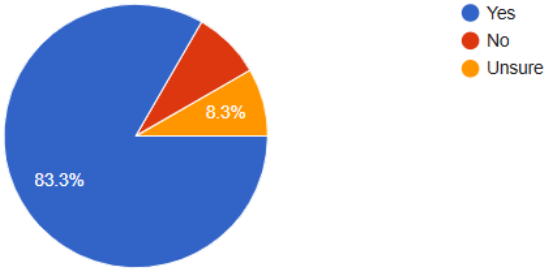
36 responses



Do you use digital devices within 1 hour before sleeping?

Copy chart

36 responses

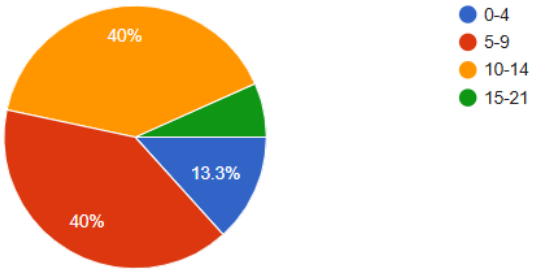


D. Optional Well-Being Measures

What was your total GAD-7 score?

Copy chart

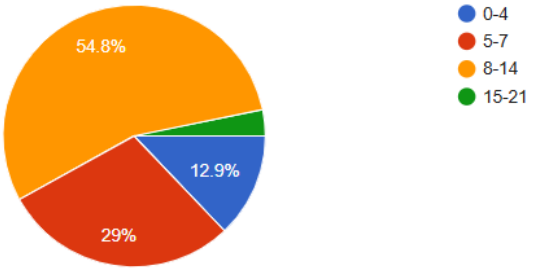
30 responses



What was your total PSQI score?

Copy chart

31 responses



Before submission

How would your school load this year makes you feel as you use devices

20 responses

I would say it affects me a lot. AS IN A LOT, not just in my academics but also my lifestyle.

I have to say that revenge bedtime procrastination is getting me as of now... also, hate that I have to pay for one of the optional surveys.

I would feel mixed as there is a lot of school load but there are also some research that can help with that load.

It feels very overwhelming as there is so much to do while having distractions on your phone other than

Before submission

How would your school load this year makes you feel as you use devices

20 responses

- It feels very overwhelming as there is so much to do while having distractions on your phone other than doing actual schoolworks. It makes you feel indecisive as to what will you do first.
- Bad
- It somehow makes me feel relieved
- It makes me feel like I need to use it more because of projects and other stuff.
- it'd make me feel decent
- Careful
- Burnt out
- A lot.
- A lot(not sure)but some doesn't require the use phones

Before submission

How would your school load this year makes you feel as you use devices

20 responses

Unbalanced, tend to forget to shut it down because I sleep in between

I feel okay although I juggle a lot of things for my schedule

Lots of online stuff to do in school so I have been using my device for more than I should probably 😵

I feel ok with devices, they help me get things done.

horrible

I use my smartphone because I'm overwhelmed with school tasks if that makes sense. Too much stuff to do and I feel paralysed so I reach for my phone instead. In simple words I feel overwhelmed as hell

I don't use my phone every school day, I use laptop for just every Tuesday to Wednesday, I have to increased focus on listening instead of using just device.

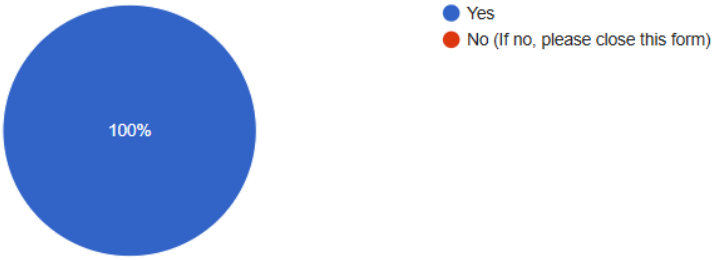
they dont really affect myself only if the school work is overloaded

C-2: Technical Expert Evaluation Rubric

Do you consent to this Evaluation?

Copy chart

3 responses



Section 1: Evaluator Information

Full Name

3 responses

- Neil Jason T. Generoso
- Ian Gregorio
- Abigail G. Del rosario

Age

3 responses

24

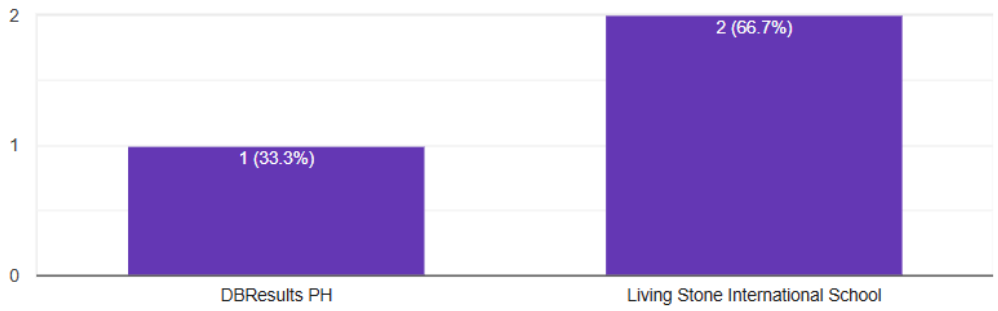
29

25

Company/Organization (If none, write "Independent")

Copy chart

3 responses



Profession/Field of Expertise

3 responses

Computer Engineer

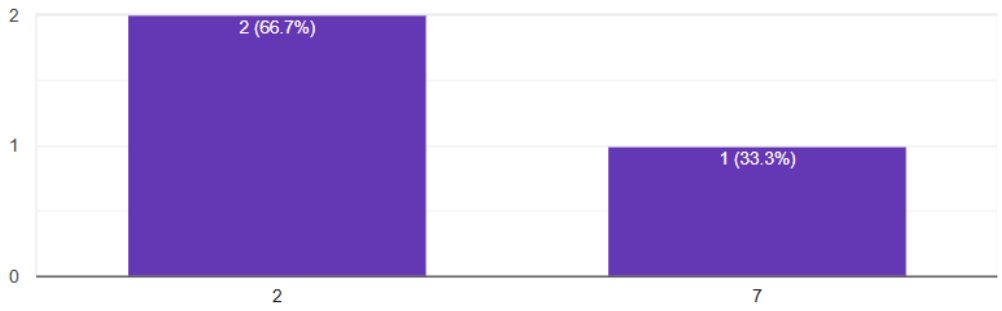
Software Engineer

Faculty Member

Years in Profession

Copy chart

3 responses



Date of Evaluation

3 responses

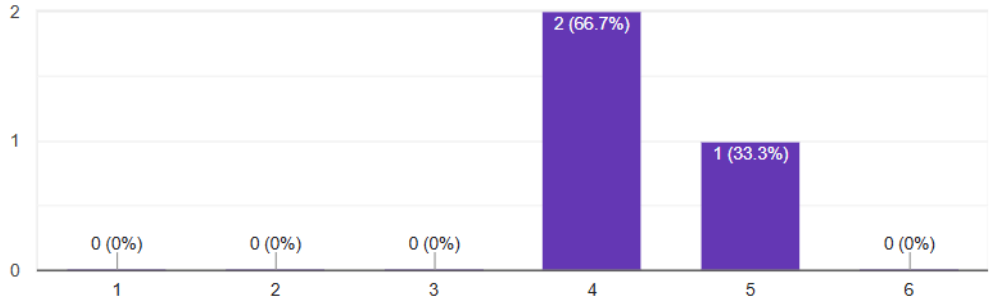
Aug 2025	31
Sep 2025	3
Feb 2026	10

Website/App Characteristics

Copy chart

User Friendliness – Is the platform easy to navigate? Are buttons and menus intuitive?

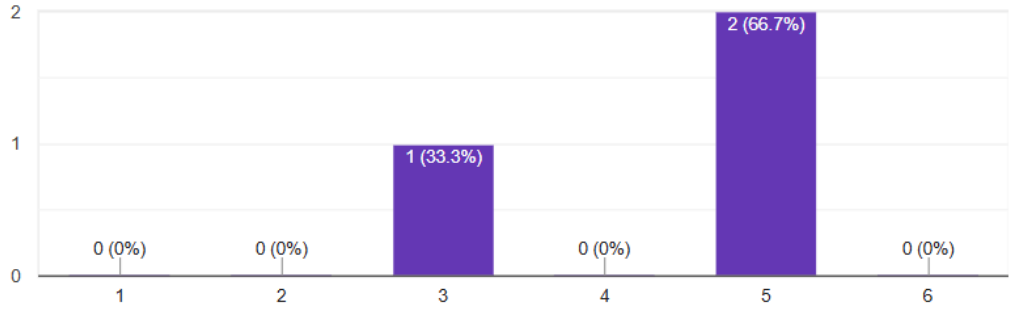
3 responses



Copy chart

Design & Aesthetics – Is the layout visually appealing and does it use a blue-light-friendly color scheme?

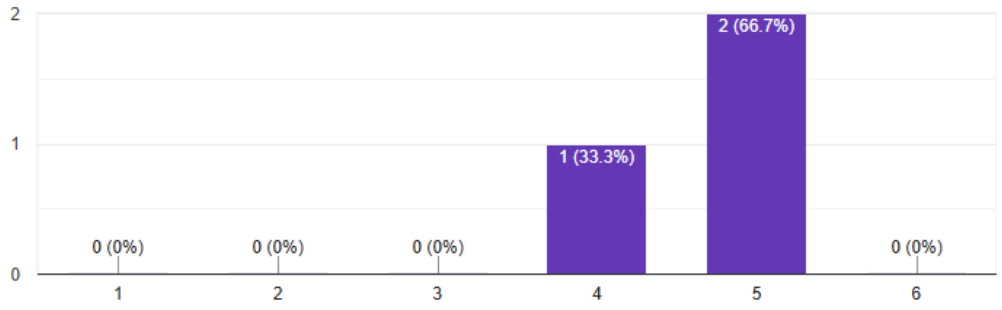
3 responses



Copy chart

Content Clarity – Are the explanations about blue light effects clear and easy to understand?

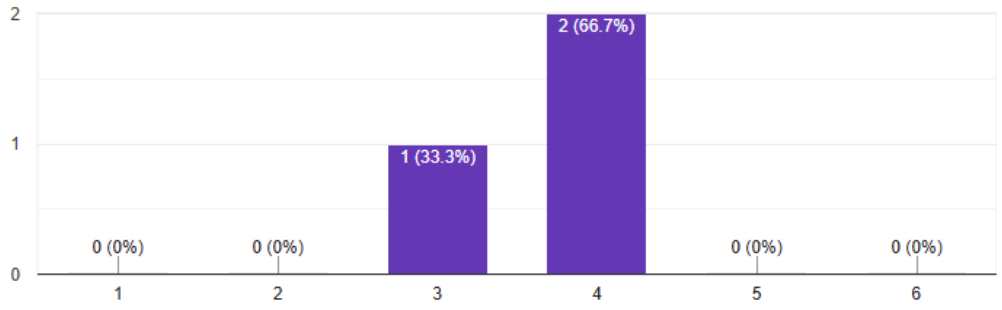
3 responses



Copy chart

Interactivity – Do the interactive features (e.g., pop-up reminders, timers) work well and add value?

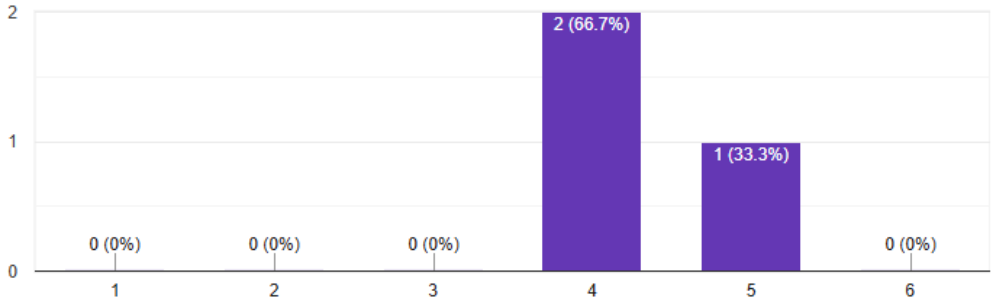
3 responses



Copy chart

Connectivity & Speed – Does the platform load quickly with minimal errors or downtime?

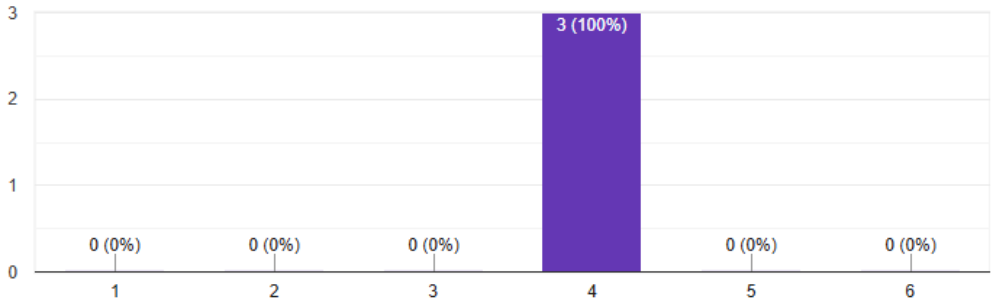
3 responses



Copy chart

Visual Aids – Are infographics, charts, and pictographs effective in explaining the research?

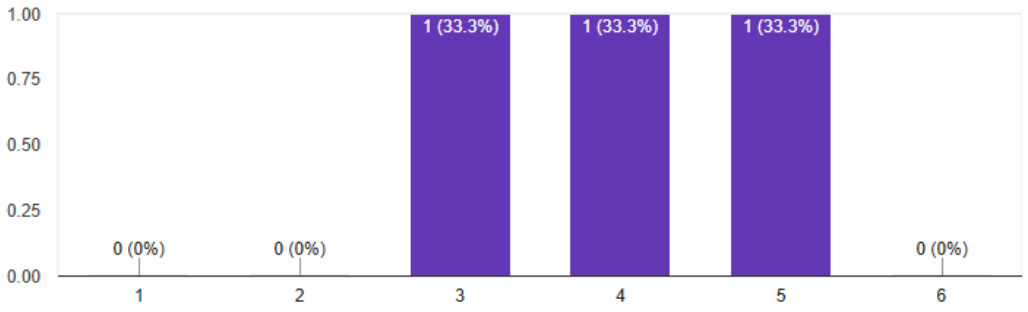
3 responses



Copy chart

Functionality – Do all features, links, and reminders function properly?

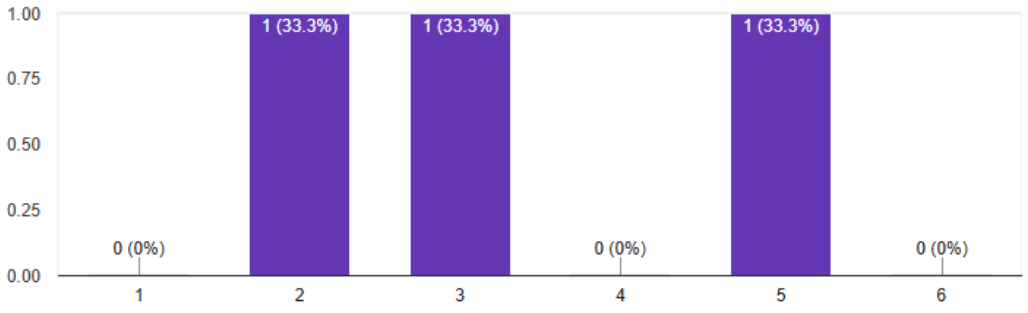
3 responses



Copy chart

Responsiveness – Does the platform adjust well across different devices and screen sizes?

3 responses



Acceptability

Does the website/app provide **useful and reliable** information about blue light effects?

3 responses

- Yes, the website/app provides useful and reliable information about blue light effects. The details are clear and easy to understand.
- Yes - it is very informative
- Yes, but i suggest add more.

Do you think the **Pop-Up Break Platform** is effective in encouraging students to take breaks from screen time?

3 responses

- Yes, Aside to the exposure for the blue light it can also help the user to reduce the eye strain
- Yes, its a very good alternative for screen time and helps the user be calm and relaxed
- Yes, since students can be distracted too much, having a reminder would be very useful.

Are the **design and color scheme** comfortable for extended viewing?

3 responses

Yes, the design and color scheme are comfortable to look at. The colors aren't too harsh, and it feels easy on the eyes even after a longer time.

Yes good choices of color, in line with the application function

They're fine, i think since the developer wanna promote the negative impact of blue light, it sort of limit the color, I suggest search more colors cheme appropriate for the website, that would both sustain both objective for the blue light, yet enough to make the website still look engaging.

Do you find the **reminder/pop-up system** non-intrusive yet effective?

3 responses

Yes

Yes, the reminder/pop-up system is effective without being distracting. It gets the point across but doesn't feel annoying.

Would you recommend this platform to students or institutions?

3 responses

Yes

Yes, to limit the students or the children from the exposure of blue light that may affect their health.

Additional Feedback

What features should be added?

3 responses

- A time bound game
- N/a
- Put more informations, videos, articles, anything that can inform the user about Blue light.

What should be improved?

3 responses

- The Navigation bar should also change or follow the same color of the background of the application or software
- N/a
- The content, quality of the pictures, more labels and text, font size, and alaso add more animations.

Any other comments or suggestions?

3 responses

- None
- N/a
- Although having a website that is interactive and engaging, i think we should also highlight the reason why we

Any other comments or suggestions?

3 responses

None

N/a

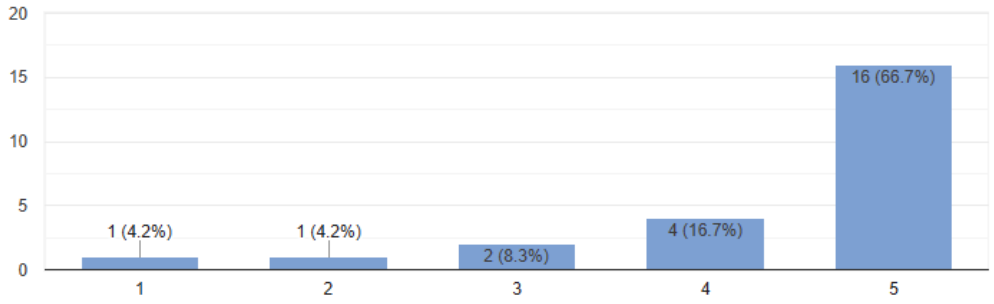
Although having a website that is interactive and engaging, i think we should also highlight the reason why we made it in the first place. Since its a type of websites that promotes and possibly control the risk of Blue Light, then the focus should also be about the Blue light. I think not everyone is aware what is, giving a detailed information will definitely engaged the user more about the website.

C-3: Student User Experience & Mood Feedback Form

The website was easy to navigate

Copy chart

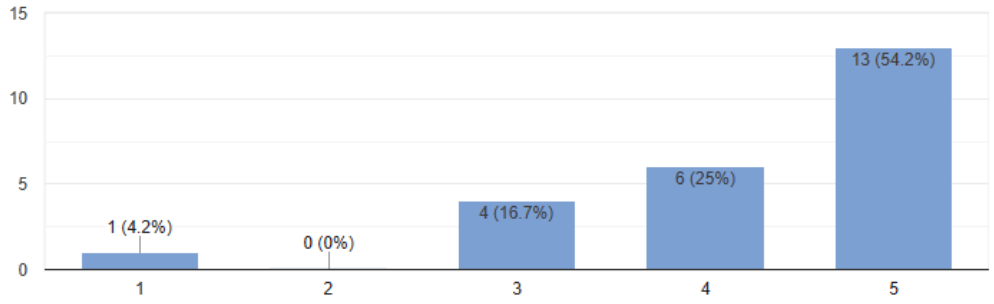
24 responses



The layout and design were clear and easy to follow

Copy chart

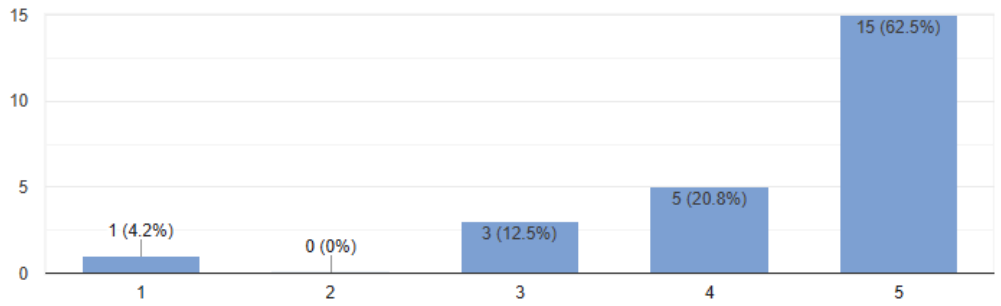
24 responses



The instructions on the website were easy to understand

Copy chart

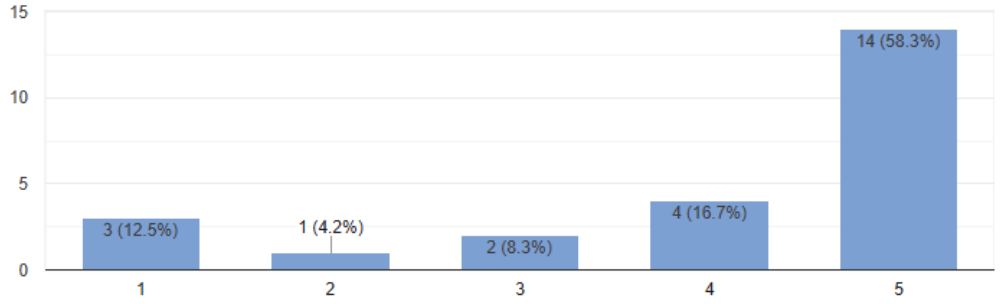
24 responses



The website did not feel mentally tiring to use

Copy chart

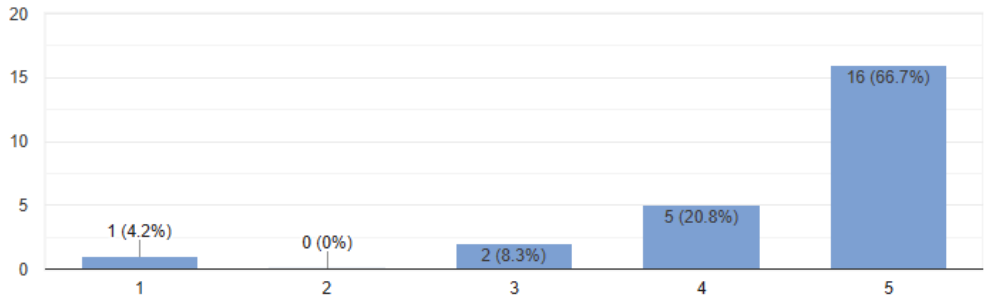
24 responses



The amount of information presented felt manageable

Copy chart

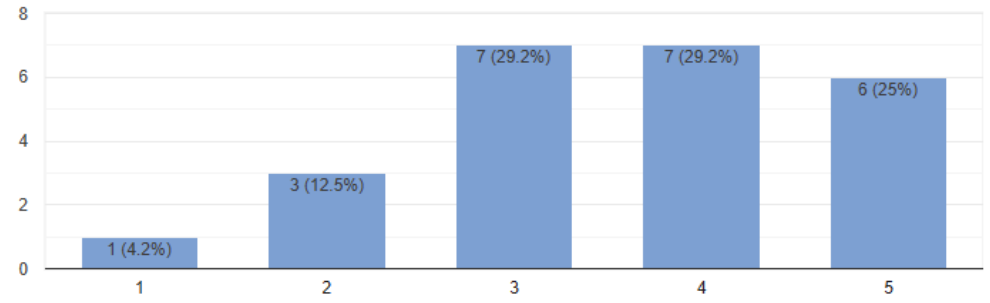
24 responses



The website helped me become more aware of my screen usage

Copy chart

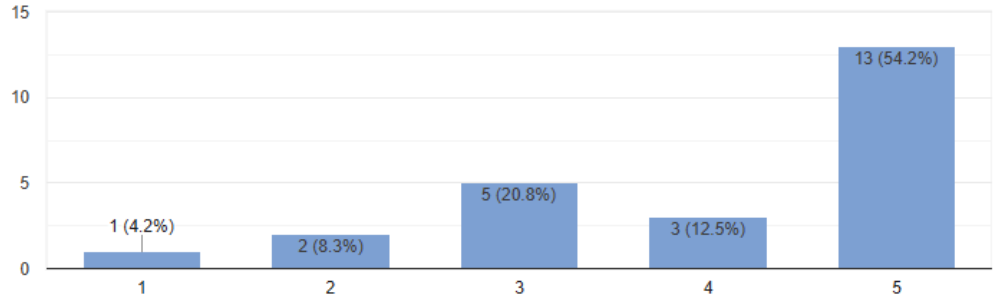
24 responses



The website encourages me to take breaks from my device

Copy chart

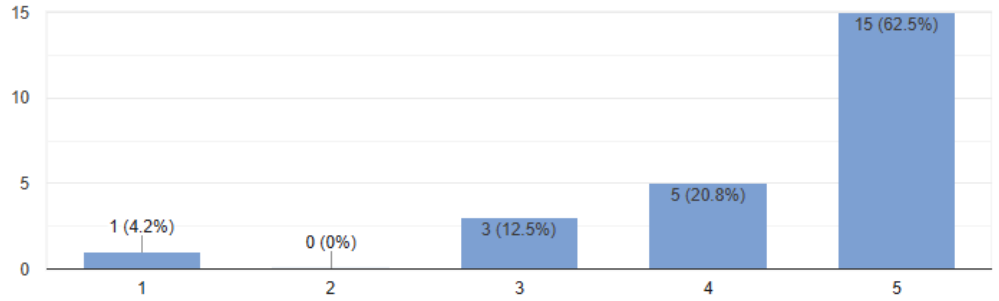
24 responses

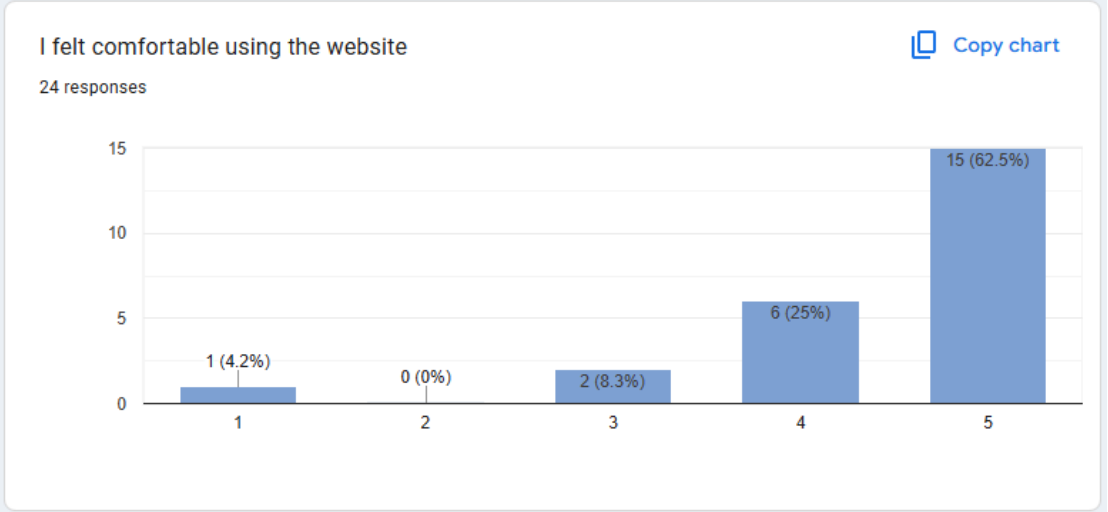


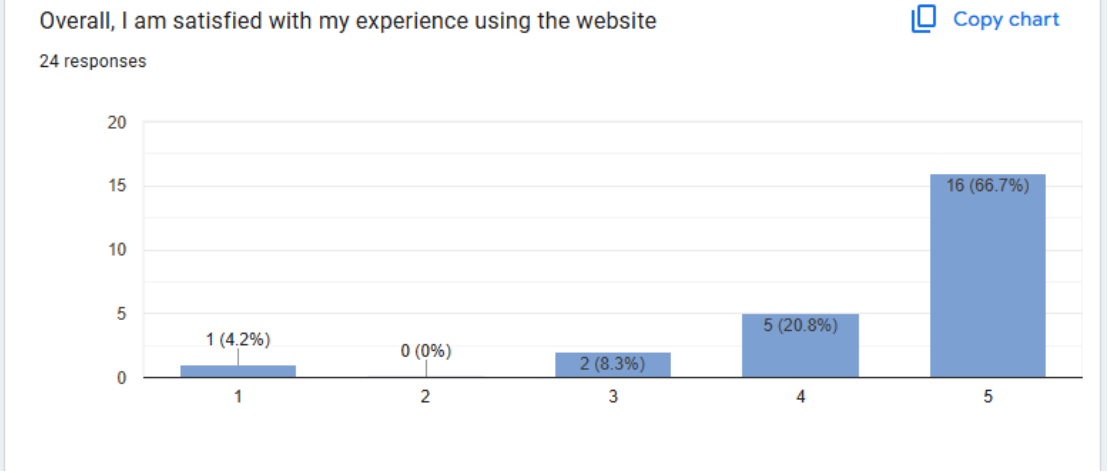
The reminders or prompts felt appropriate rather than intrusive

Copy chart

24 responses







APPENDIX D

APPENDIX D: System Source Code (Snippets)

Code is 170+ pages

- **D-1: `index.html` structure.**

<!DOCTYPE html>

```
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Break the Light - Home</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body class="home-page">
  <audio id="bg-music" src="assets/audio/Ghibli Music.mp3" loop></audio>

  <header>
    <h1>Break the Light</h1>
    <nav>
      <a href="index.html" class="active">Home</a>
      <a href="about.html">About</a>
      <a href="blue_light.html">Blue Light</a>
      <a href="settings.html">Settings</a>
      <a href="contact.html">Contact</a>
  <button id="theme-toggle" data-theme-toggle class="nav-button"
aria-label="Toggle theme">🌙</button>
    </nav>
  </header>

  <main>
    <section class="hero">
      <h2 class="hero-title">Break the Light</h2>
      <p class="hero-sub">Website to reduce blue light effects! One screen at a
time!</p>
      <a href="blue_light.html" class="button start-btn">Start!</a>
    </section>

    <section class="section">
      <h3>Features</h3>
      <div class="features-row">
        <div class="card">
```

```
        <img src="assets/images/chick n clown.gif" alt="Chick and Clown"
class="interactive" onclick="window.location.href='blue_light.html'">
        <div class="caption">
          <strong>Various reminders!</strong>
          <p>Be able to take breaks! Reminded by a chick or clown.</p>
        </div>
      </div>

      <div class="card">
        <img src="assets/images/Clouds alarms.gif" alt="Alarms"
class="interactive" onclick="window.location.href='blue_light.html'">
        <div class="caption">
          <strong>Adjustable alarms!</strong>
          <p>Have something important to do? Save it here.</p>
        </div>
      </div>

      <div class="card">
        <img src="assets/images/Clouds journal.gif" alt="Journal"
class="interactive" onclick="window.location.href='blue_light.html'">
        <div class="caption">
          <strong>Take a break Journal</strong>
          <p>Don't always overwork — express yourself with some rest time.</p>
        </div>
      </div>
    </div>
  </section>

  <section class="section">
    <h3>Future Features</h3>
    <div class="features-row two-cols">
      <div class="card centered">
        <img src="assets/images/chick.gif" alt="Chick Mentor"
class="interactive" onclick="window.location.href='blue_light.html'">
        <div class="caption">
          <strong>Chick Mentor</strong>
```

```
          <p>Don't know what to do off your device? Watch this cute chick teach
you!</p>
        </div>
      </div>
    </div>
  </section>
 </main>

 <footer>
  <div class="footer-inner">
   <span>Break the Light © Seraphim Manuel 2026</span>
  </div>
 </footer>

 <!-- Firebase SDK (compat builds to support existing compat-style code) -->
 <script
src="https://www.gstatic.com/firebasejs/10.7.0/firebase-app-compat.js"></script
>
 <script
src="https://www.gstatic.com/firebasejs/10.7.0/firebase-auth-compat.js"></script
>
 <script
src="https://www.gstatic.com/firebasejs/10.7.0/firebase-firestore-compat.js"></sc
ript>
 <script
src="https://www.gstatic.com/firebasejs/10.7.0/firebase-app-check-compat.js"></
script>

 <!-- Firebase Configuration -->
 <script src="js/firebase-config.js"></script>

 <!-- Modular App Check initializer (uses window.FIREBASE_CONFIG) -->
 <!-- modular App Check intentionally omitted (using compat App Check) -->

 <script src="js/script.js"></script>
 <script src="js/quickAccessMenu.js"></script>
```

```
 <script src="js/auto-responsive.js"></script>
</body>
</html>
```

- **D-2: Firebase Configuration and Authentication Logic**

  (`firebase-config.js`).

```
// Firebase Configuration
// For Firebase JS SDK v7.20.0 and later, measurementId is optional
window.FIREBASE_CONFIG = {
  apiKey: "AIzaSyBnJePbWah8Bqs_FEEo0IH24dPzW-5HHLM",
  authDomain: "break-the-light.firebaseapp.com",
  projectId: "break-the-light",
  storageBucket: "break-the-light.firebasestorage.app",
  messagingSenderId: "832176987679",
  appId: "1:832176987679:web:e717732424fcbf8d7fbf2a",
  measurementId: "G-73TNRKD26Y"
};

console.info('Firebase config loaded ✓');

// Initialize Firebase (compat-style if global 'firebase' present)
try {
  if (window.FIREBASE_CONFIG && window.firebase &&
firebase.initializeApp) {
    if (!firebase.apps || !firebase.apps.length) {
      firebase.initializeApp(window.FIREBASE_CONFIG);
      console.info('Firebase initialized (compat)');
    } else {
      console.info('Firebase already initialized');
    }

    // Initialize App Check using reCAPTCHA Enterprise (if available)
    try {
      if (firebase.appCheck) {
```

```
      // Try a few provider name variants for compatibility
      const Provider = (firebase.appCheck.RecaptchaEnterpriseProvider ||
firebase.appCheck.ReCaptchaEnterpriseProvider ||
(firebase.recaptchaEnterprise &&
firebase.recaptchaEnterprise.RecaptchaEnterpriseProvider));
      if (Provider) {
        const appCheckInstance = firebase.appCheck();
        appCheckInstance.activate(new
Provider('6LfLkWIsAAAAAKwKrsnT3GEqTeE-JhR2Ro9QH_oY'), true);
        console.info('App Check (reCAPTCHA Enterprise) activated (compat)');
      } else {
        console.warn('App Check provider not found. Ensure
firebase-app-check.js is loaded and compatible.');
      }
    }
  } catch (e) {
    console.warn('App Check init error (compat):', e && e.message ? e.message :
e);
  }
 } else {
  console.info('Firebase SDK not present (global). App will rely on modular
initialization in other scripts.');
 }
} catch (e) {
 console.error('Firebase auto-init failed:', e);
}
```

- **D-3: Ghibli Music Audio Integration code.**

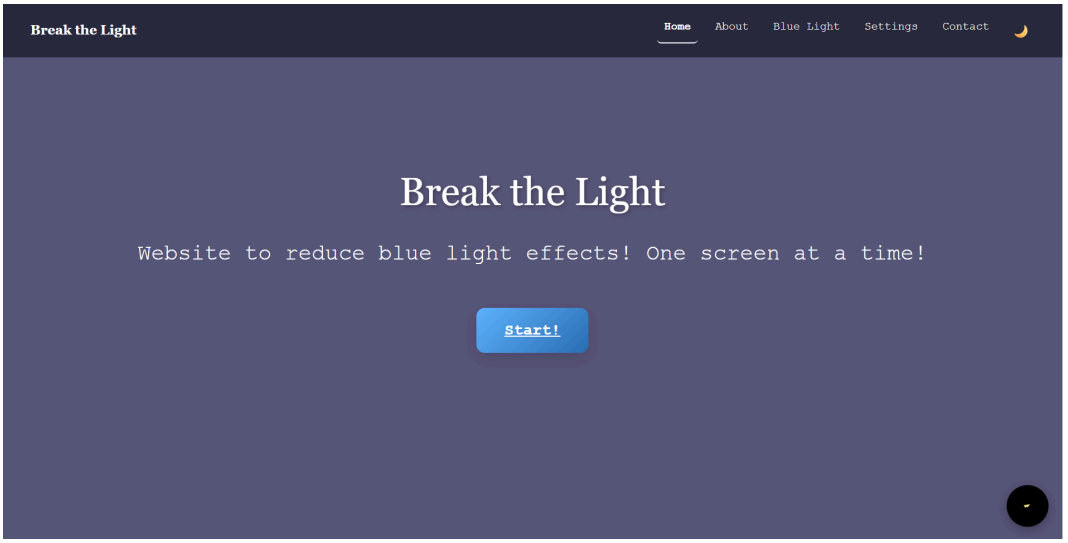
```
const audio = document.getElementById("bg-music");
const musicToggle = document.getElementById("music-toggle");

// Restore saved state
window.addEventListener("load", () => {
 const playing = localStorage.getItem("music-playing") === "true";
 const time = parseFloat(localStorage.getItem("music-time")) || 0;
```

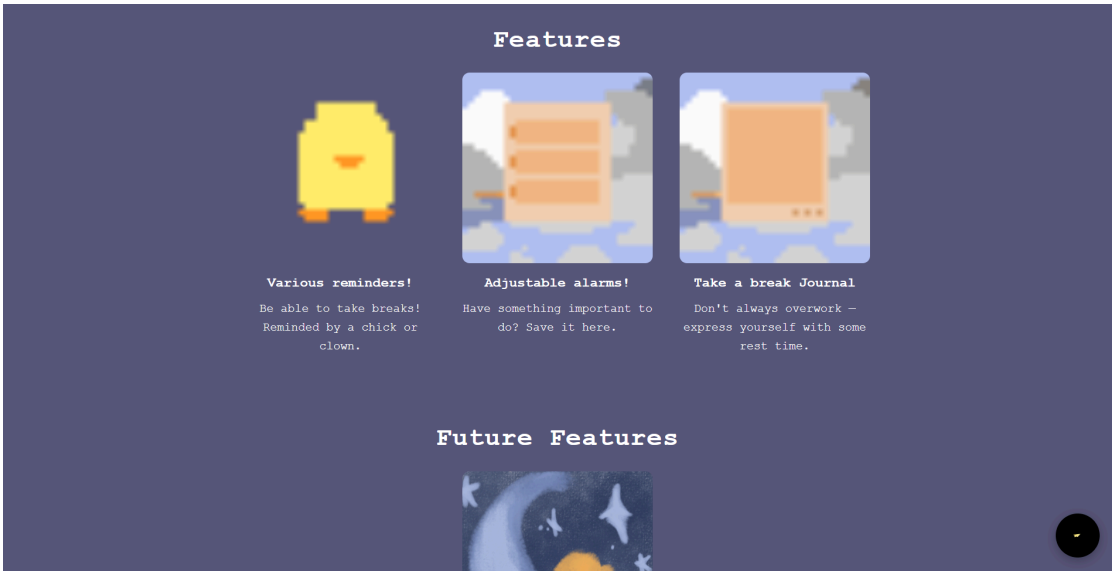
```
  audio.currentTime = time;
  if (playing) audio.play().catch(()=>{});
  updateBtn();
});

// Save state before leaving
window.addEventListener("beforeunload", () => {
  localStorage.setItem("music-playing", !audio.paused);
  localStorage.setItem("music-time", audio.currentTime);
});

// Toggle music
function updateBtn() {
  musicToggle.textContent = audio.paused ? "🎵" : "⏸️";
}
musicToggle.addEventListener("click", () => {
  if (audio.paused) {
    audio.play();
    localStorage.setItem("music-playing", "true");
  } else {
    audio.pause();
    localStorage.setItem("music-playing", "false");
  }
  updateBtn();
});
```

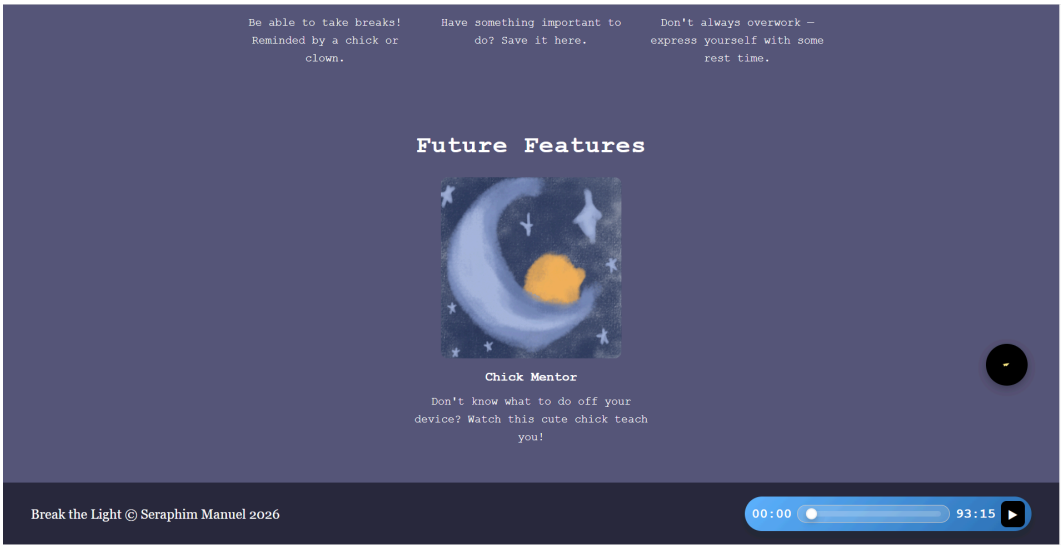
```
=== Deploying to 'break-the-light'...

i  deploying hosting
i  hosting[break-the-light]: beginning deploy...
i  hosting[break-the-light]: found 157 files in public
+  hosting[break-the-light]: file upload complete
i  hosting[break-the-light]: finalizing version...
+  hosting[break-the-light]: version finalized
i  hosting[break-the-light]: releasing new version...
+  hosting[break-the-light]: release complete

+  Deploy complete!

Project Console: https://console.firebase.google.com/project/break-the-light/overview
Hosting URL: https://break-the-light.web.app
```

APPENDIX E

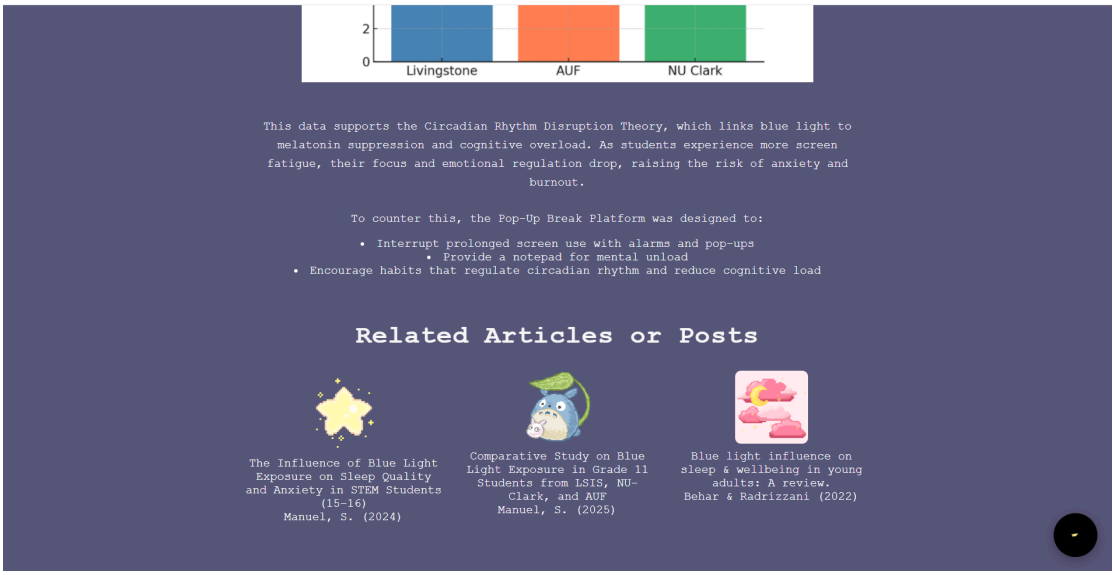
The website/App

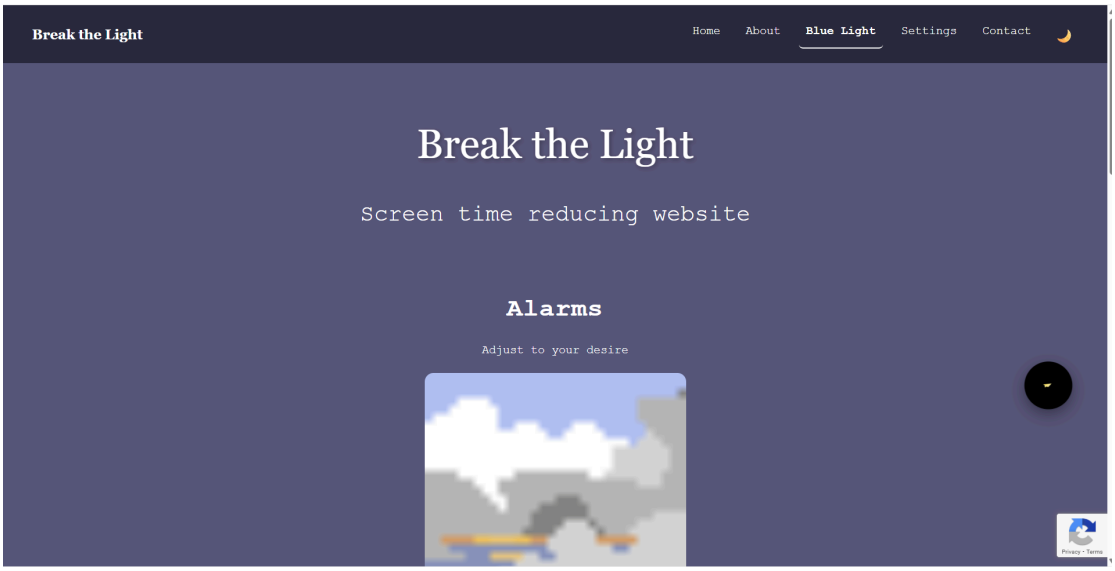


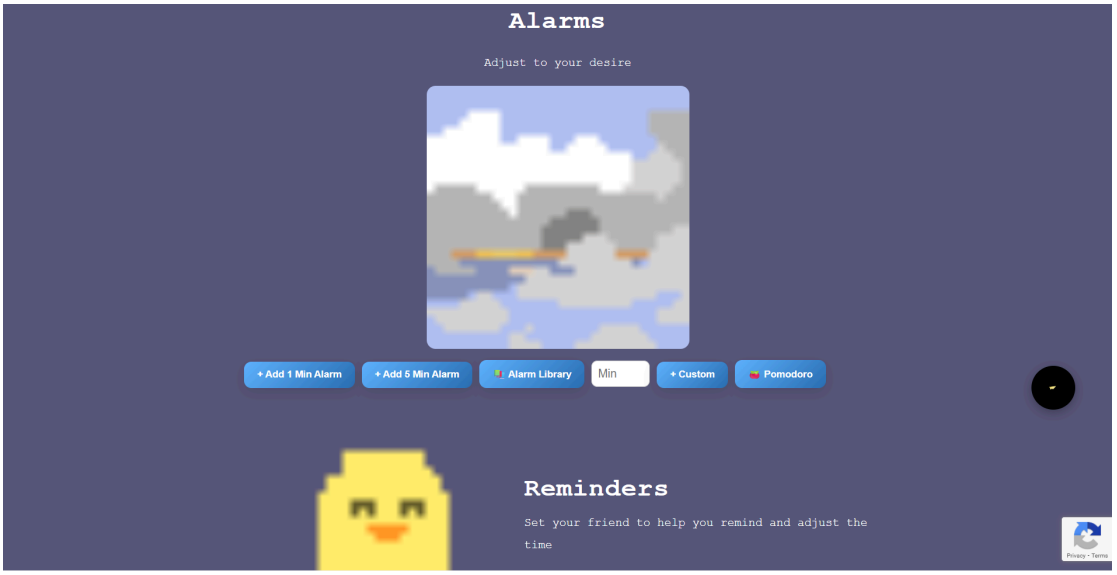



Be able to take breaks! Reminded by a chick or clown.
Have something important to do? Save it here.
Don't always overwork — express yourself with some rest time.
Future Features
Chick Mentor
Don't know what to do off your device? Watch this cute chick teach you!
Break the Light © Seraphim Manuel 2026
00:00
93:15

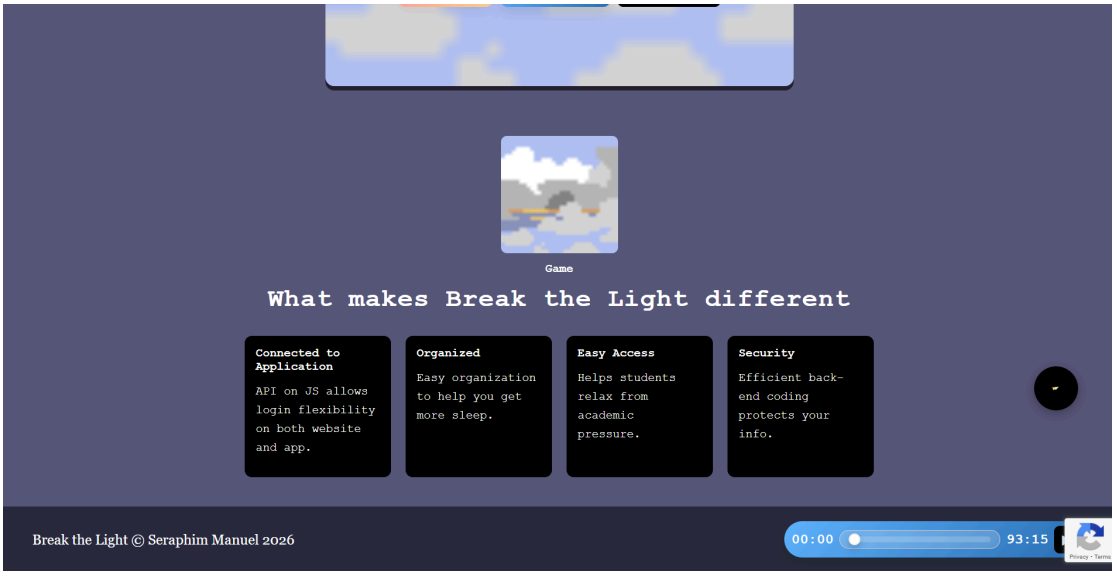
Break the Light
Home
About
Blue Light
Settings
Contact
About Blue Light
Figure out the importance to why we should lessen our screen time
Games-Howell Post-Hoc Test Mean Differences
Studies have shown that exposure to high-energy blue light in the evening delays sleep, shortens duration, and reduces melatonin production. This can lead to poor academic performance, fatigue, anxiety, and depression, especially in adolescents.

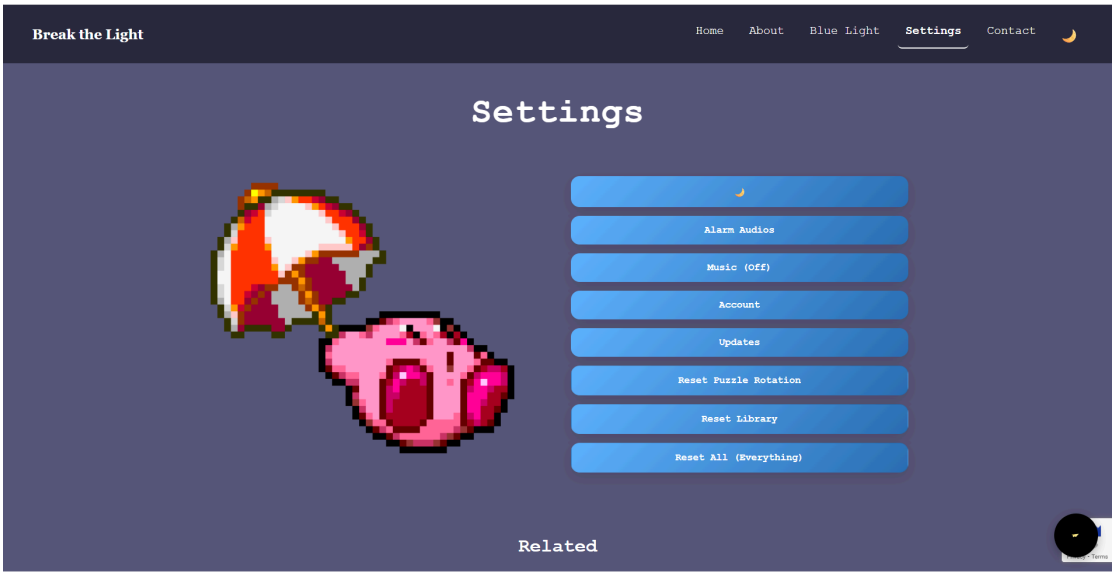

2
0
Livingstone
AUF
NU Clark
This data supports the Circadian Rhythm Disruption Theory, which links blue light to melatonin suppression and cognitive overload. As students experience more screen fatigue, their focus and emotional regulation drop, raising the risk of anxiety and burnout.
To counter this, the Pop-Up Break Platform was designed to:
• Interrupt prolonged screen use with alarms and pop-ups
• Provide a notepad for mental unload
• Encourage habits that regulate circadian rhythm and reduce cognitive load
Related Articles or Posts
The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students (15-16)
Manuel, S. (2024)
Comparative Study on Blue Light Exposure in Grade 11 Students from LSIS, NU-Clark, and AUF
Manuel, S. (2025)
Blue light influence on sleep & wellbeing in young adults: A review.
Behar & Radrizzani (2022)

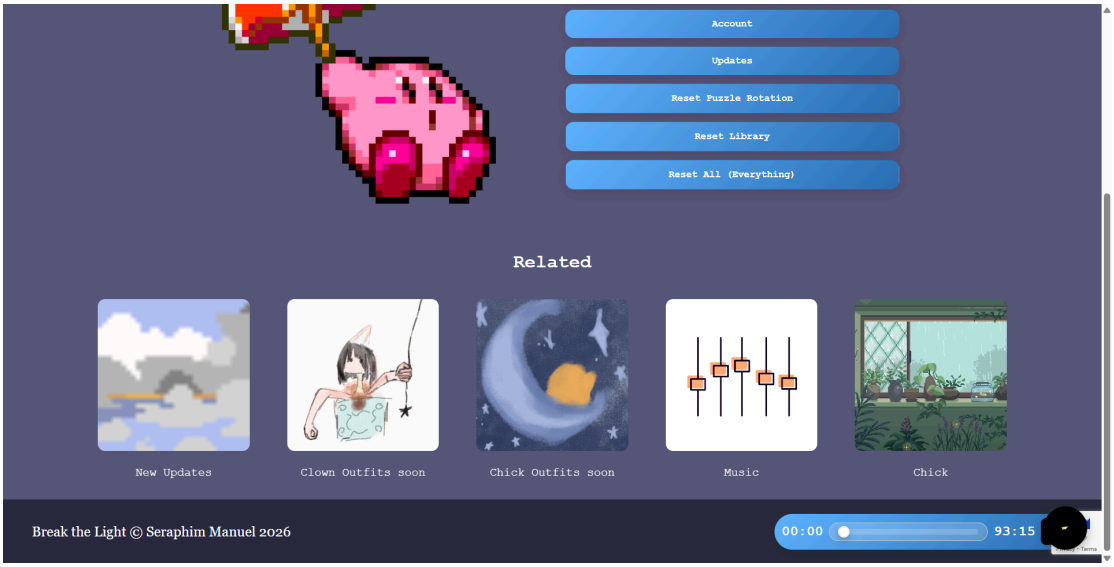

Break the Light
Home
About
Blue Light
Settings
Contact
Break the Light
Screen time reducing website
Alarms
Adjust to your desire

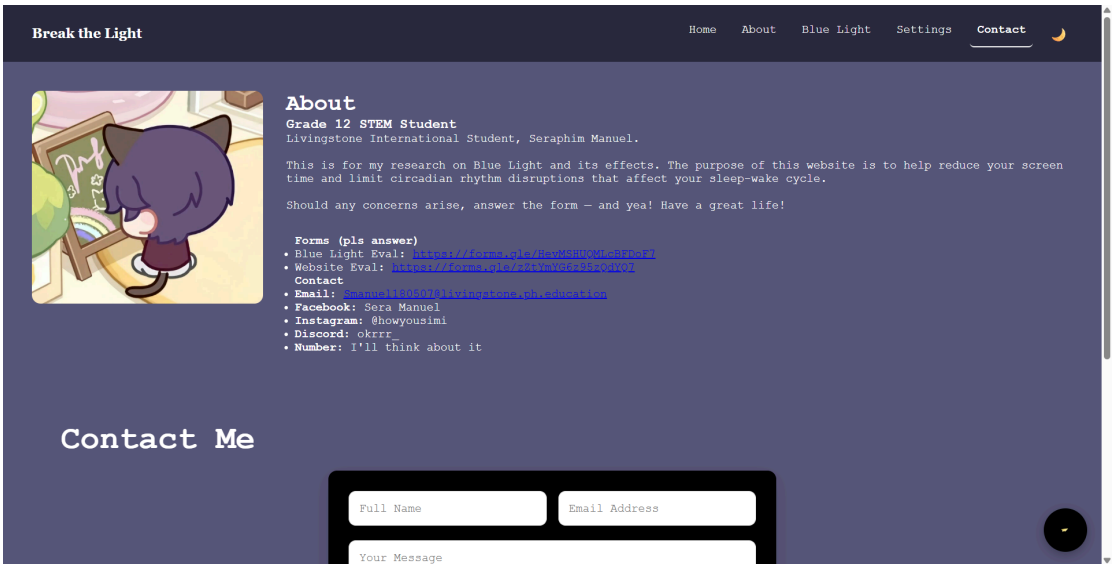

Alarms
Adjust to your desire
+ Add 1 Min Alarm
+ Add 5 Min Alarm
Alarm Library
Min
+ Custom
Pomodoro
Reminders
Set your friend to help you remind and adjust the time

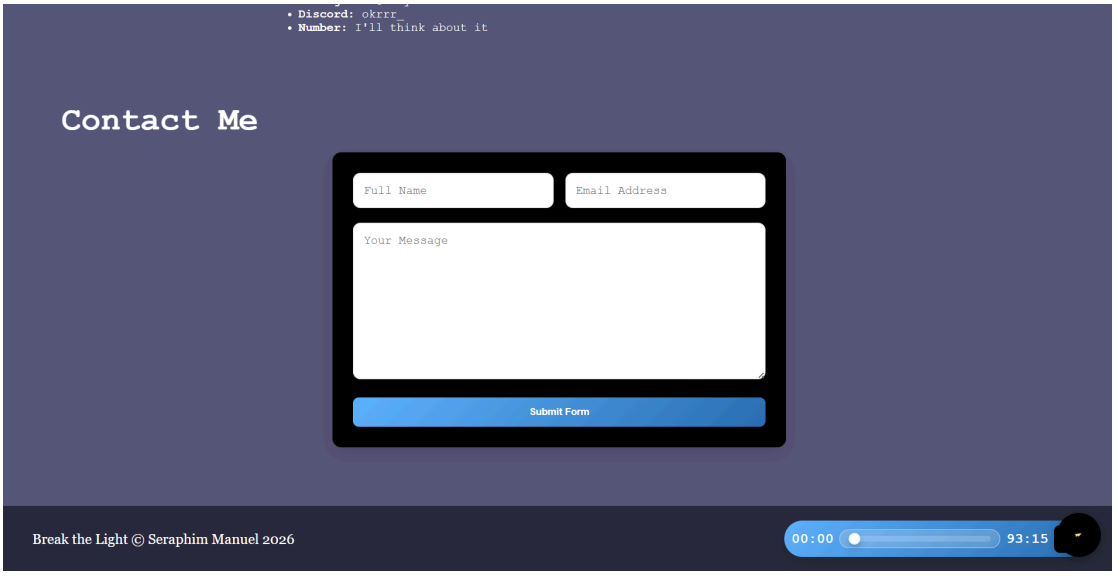
Write anything here...
Clear Journal
Save to Library
View Library

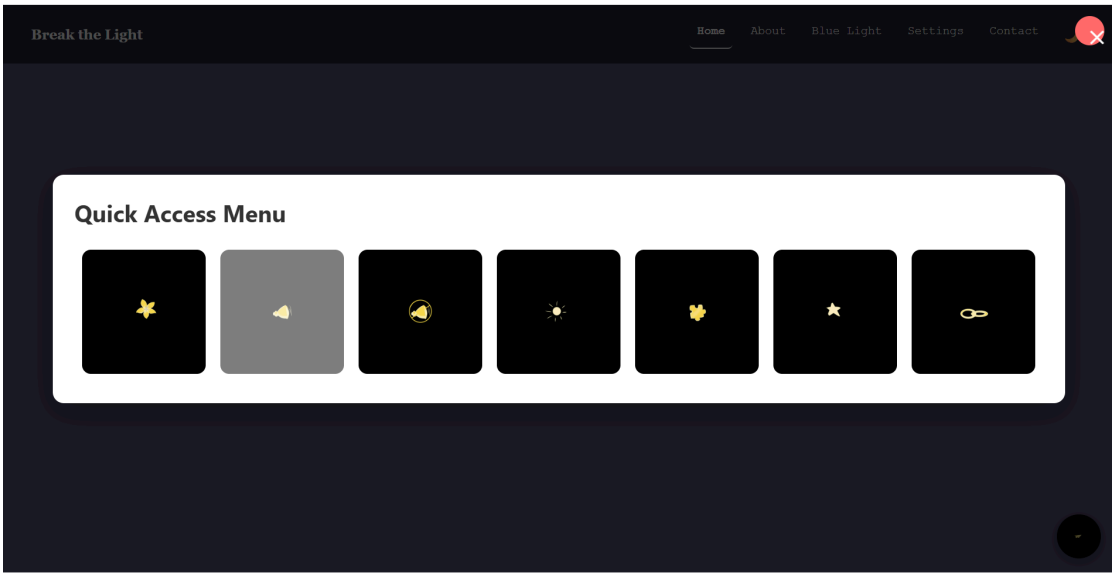

Game
What makes Break the Light different
Connected to Application
API on JS allows login flexibility on both website and app.
Organized
Easy organization to help you get more sleep.
Easy Access
Helps students relax from academic pressure.
Security
Efficient back-end coding protects your info.
Break the Light © Seraphim Manuel 2026
00:00
93:15

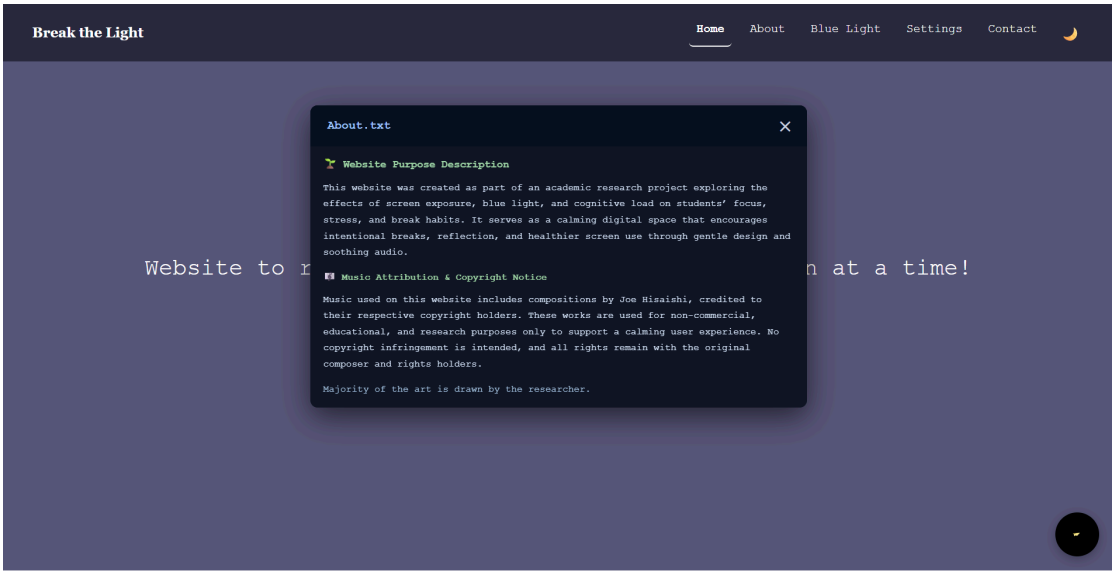

Break the Light
Home
About
Blue Light
Settings
Contact
Settings
Alarm Audios
Music (Off)
Account
Updates
Reset Puzzle Rotation
Reset Library
Reset All (Everything)
Related

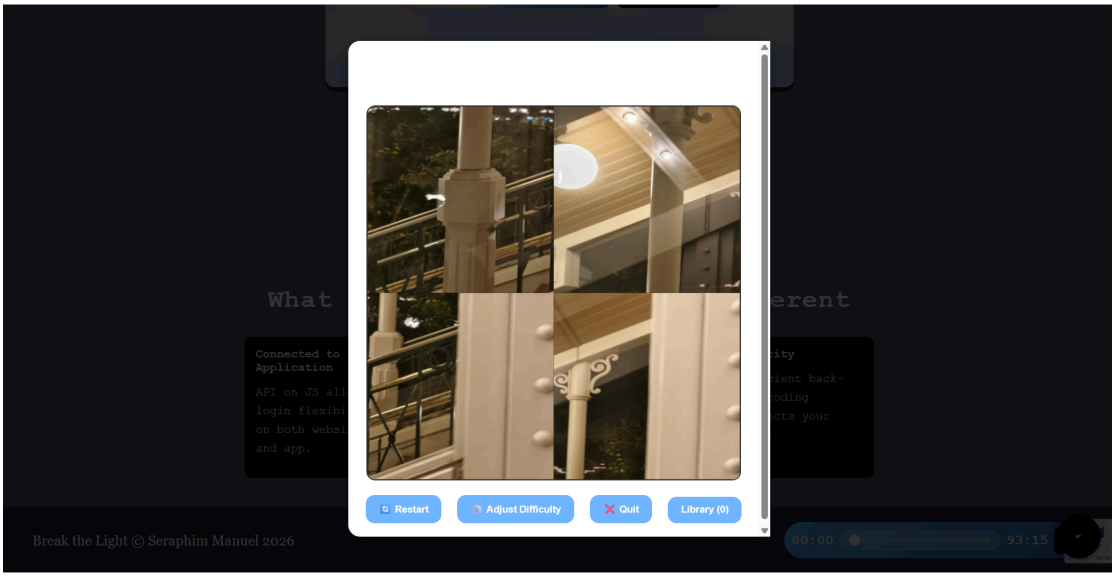

Account
Updates
Reset Puzzle Rotation
Reset Library
Reset All (Everything)
Related
New Updates
Clown Outfits soon
Chick Outfits soon
Music
Chick
Break the Light © Seraphim Manuel 2026
00:00
93:15


Break the Light
Home
About
Blue Light
Settings
Contact
About
Grade 12 STEM Student
Livingstone International Student, Seraphim Manuel.
This is for my research on Blue Light and its effects. The purpose of this website is to help reduce your screen time and limit circadian rhythm disruptions that affect your sleep-wake cycle.
Should any concerns arise, answer the form — and yea! Have a great life!
Forms (pls answer)
Blue Light Eval: https://forms.gle/HevMSHUQMLcBFDoF7
Website Eval: https://forms.gle/zZtYmYG6z95zQdXQ7
Contact
Email: Smanuel180507@livingstone.ph.education
Facebook: Sera Manuel
Instagram: @howyousimi
Discord: okrrr_
Number: I'll think about it
Contact Me
Full Name
Email Address
Your Message

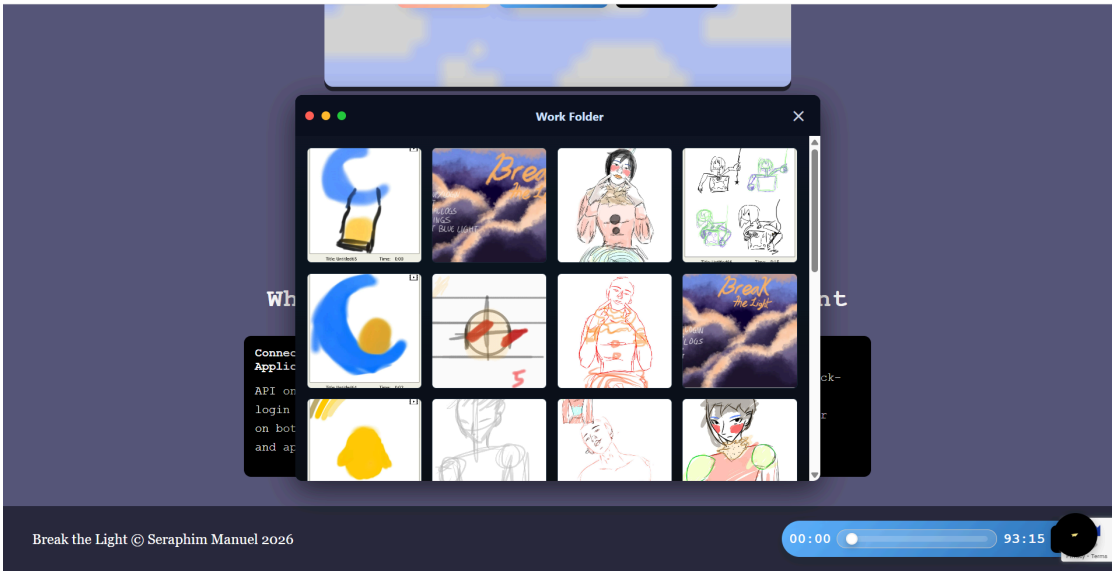

• Discord: okrrr_
• Number: I'll think about it
Contact Me
Full Name
Email Address
Your Message
Submit Form
Break the Light © Seraphim Manuel 2026
00:00
93:15

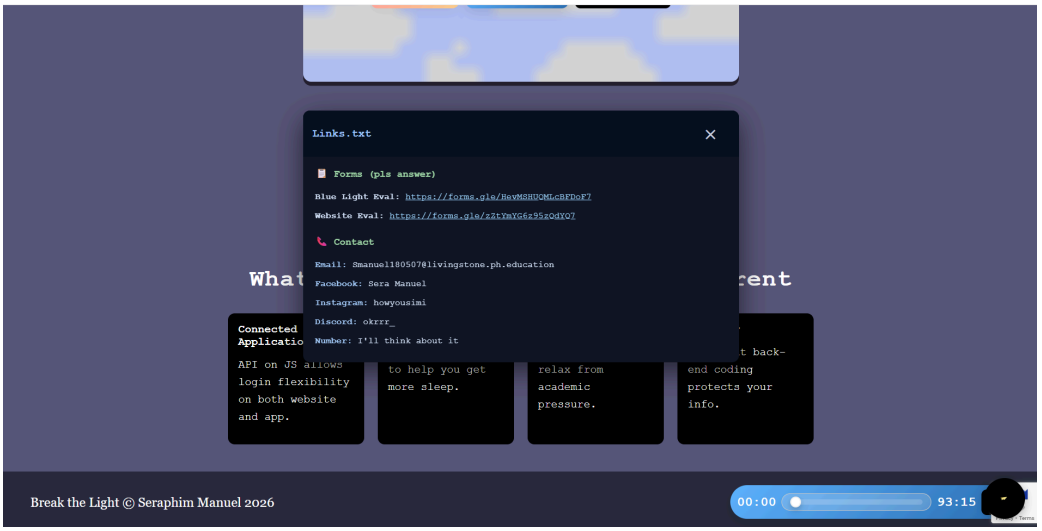

Break the Light
Home
About
Blue Light
Settings
Contact
Quick Access Menu

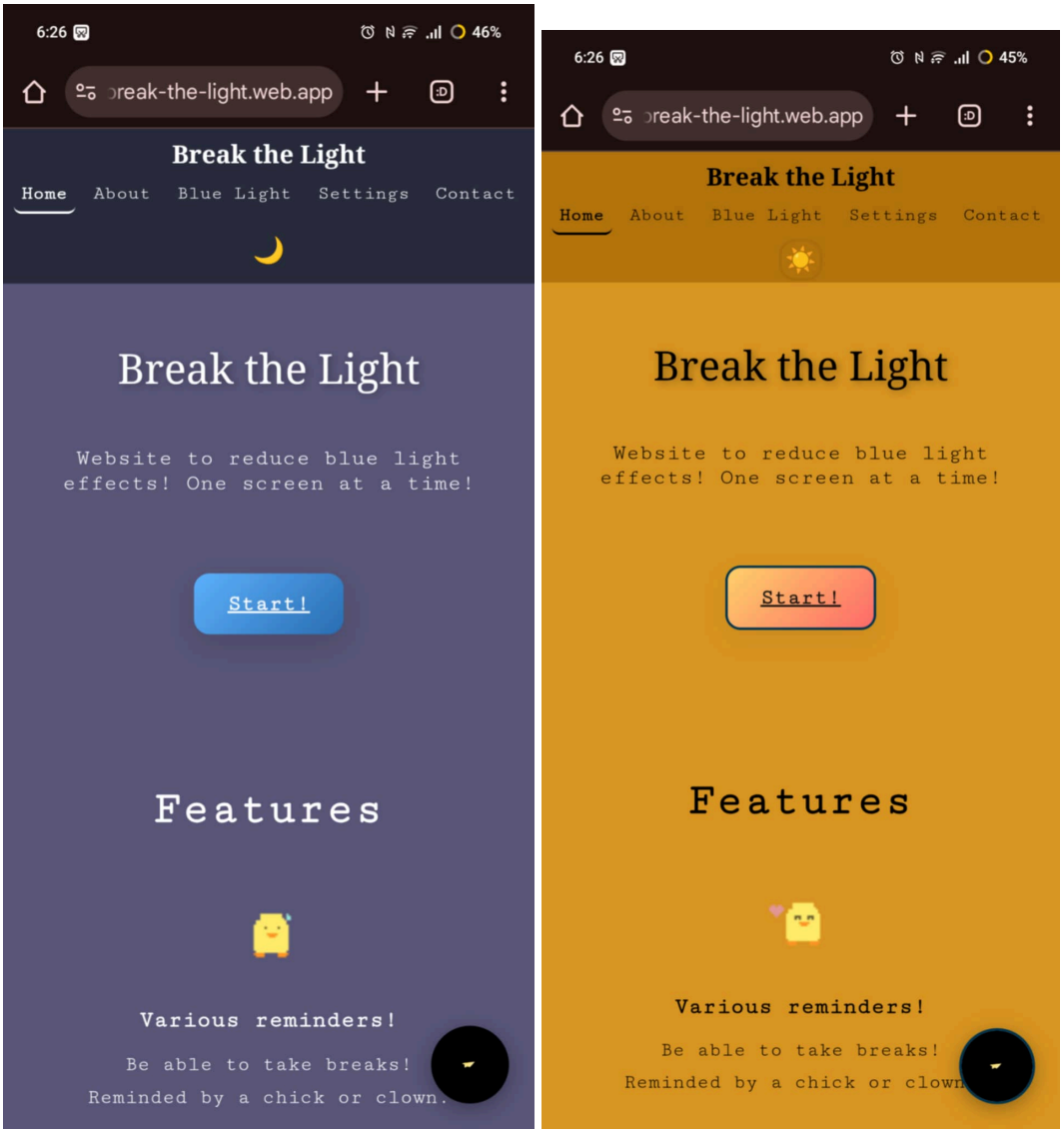


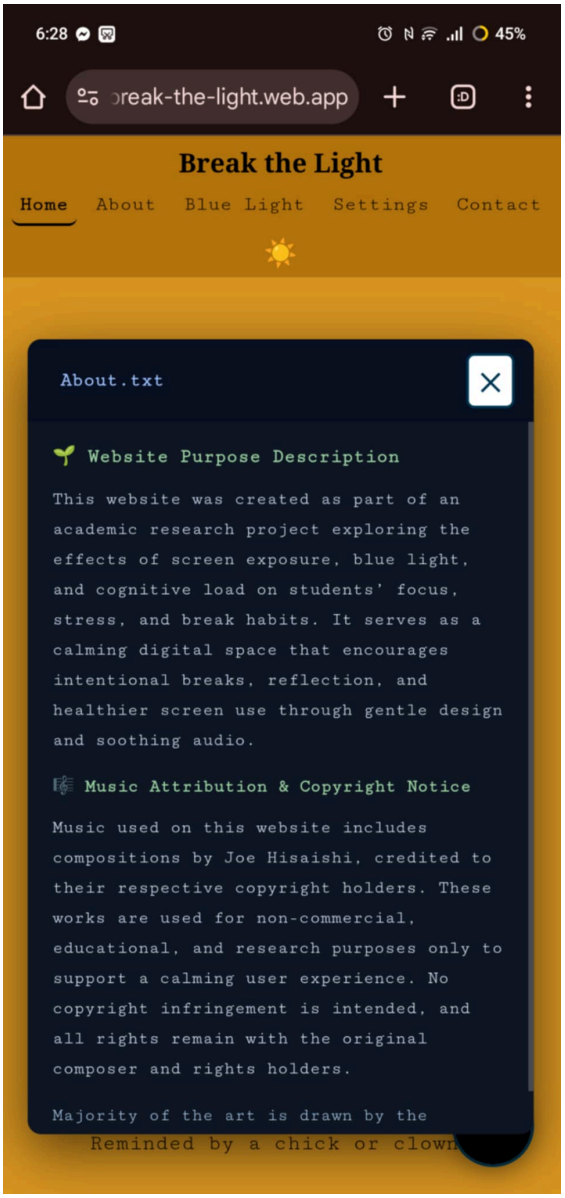


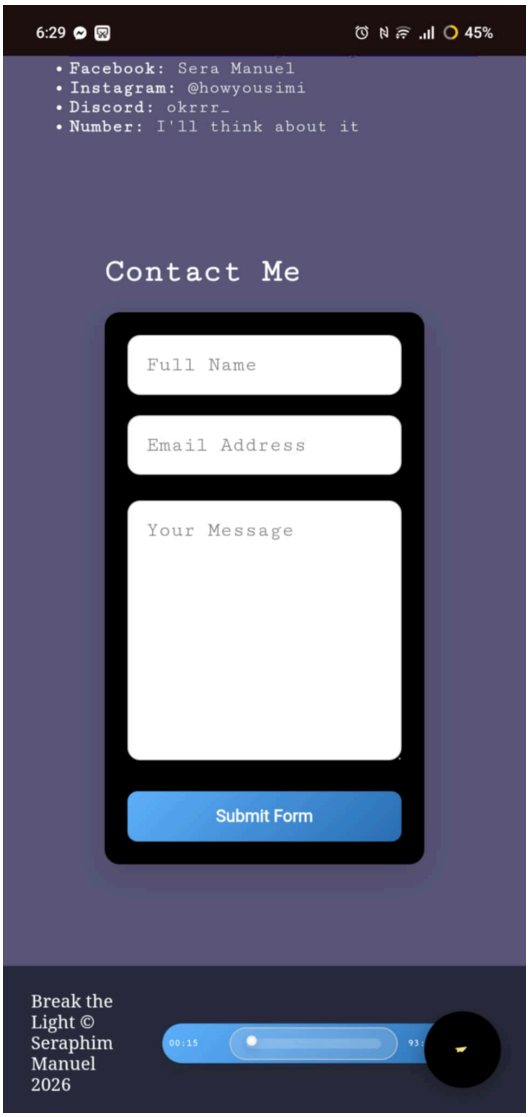
Break The Light: A Pop-Up Platform


Work Folder
Break the Light © Seraphim Manuel 2026
00:00
93:15


Links.txt
Forms (pls answer)
Blue Light Eval: https://forms.gle/HeyMSHUQMLcBFDoF7
Website Eval: https://forms.gle/zZtYmYG6z95zOdYO7
Contact
Email: Smanuel180507@livingstone.ph.education
Facebook: Sera Manuel
Instagram: howyousimi
Discord: okrrr_
Number: I'll think about it
Connected Applicatio
API on JS allows login flexibility on both website and app.
to help you get more sleep.
relax from academic pressure.
end coding protects your info.
Break the Light © Seraphim Manuel 2026
00:00
93:15


6:26
46%
reak-the-light.web.app
Break the Light
Home
About
Blue Light
Settings
Contact
Break the Light
Website to reduce blue light effects! One screen at a time!
Start!
Features
Various reminders!
Be able to take breaks!
Reminded by a chick or clown.
6:26
45%
reak-the-light.web.app
Break the Light
Home
About
Blue Light
Settings
Contact
Break the Light
Website to reduce blue light effects! One screen at a time!
Start!
Features
Various reminders!
Be able to take breaks!


6:28
45%
break-the-light.web.app
Break the Light
Home
About
Blue Light
Settings
Contact
About.txt
Website Purpose Description
This website was created as part of an academic research project exploring the effects of screen exposure, blue light, and cognitive load on students' focus, stress, and break habits. It serves as a calming digital space that encourages intentional breaks, reflection, and healthier screen use through gentle design and soothing audio.
Music Attribution & Copyright Notice
Music used on this website includes compositions by Joe Hisaishi, credited to their respective copyright holders. These works are used for non-commercial, educational, and research purposes only to support a calming user experience. No copyright infringement is intended, and all rights remain with the original composer and rights holders.
Majority of the art is drawn by the
Reminded by a chick or clown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